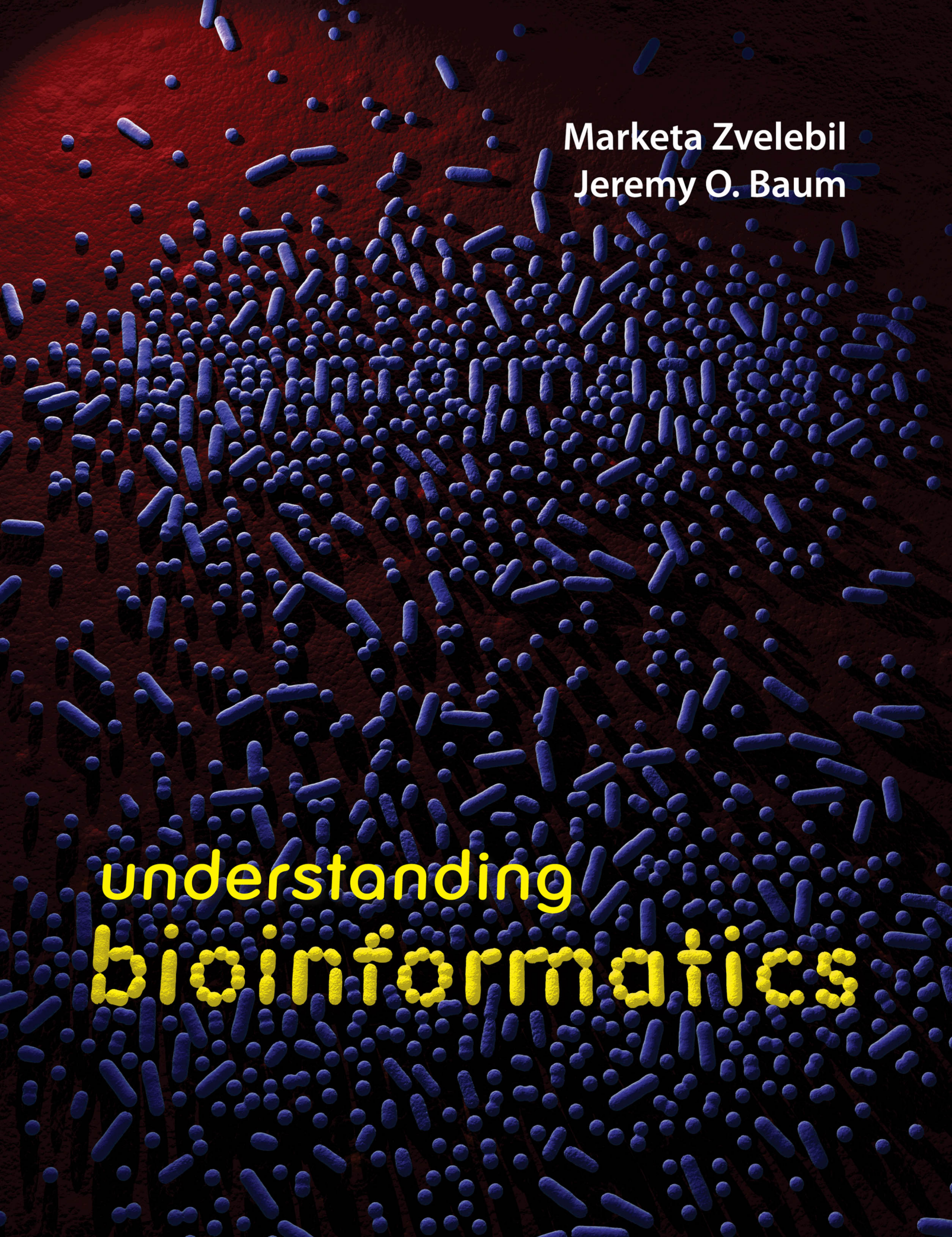


The background of the entire cover is a high-magnification micrograph showing a dense population of blue, rod-shaped bacteria. These bacteria are scattered across a dark red, textured surface that resembles a biological membrane or a petri dish. The lighting creates a sense of depth, with some bacteria appearing more prominent than others.

Marketa Zvelebil
Jeremy O. Baum

understanding
bioinformatics

Understanding Bioinformatics

In memory of Arno Siegmund Baum ז"ל

Understanding Bioinformatics

Marketa Zvelebil & Jeremy O. Baum

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Typesetting: Georgina Lucas
Cover design: Matthew McClements, Blink Studio Limited
Production Manager: Tracey Scarlett
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10-digit ISBN 0-8153-4024-9 (paperback)
13-digit ISBN 978-0-8153-4024-9 (paperback)

Library of Congress Cataloging-in-Publication Data

Zvelebil, Marketa J.

Understanding bioinformatics / Marketa Zvelebil & Jeremy O. Baum.
p. ; cm.

Includes bibliographical references and index.

ISBN-13: 978-0-8153-4024-9 (pbk.)

ISBN-10: 0-8153-4024-9 (pbk.)

1. Bioinformatics.

[DNLM: 1. Computational Biology--methods. QU 26.5 Z96u 2008] I. Baum, Jeremy O. II. Title.

QH324.2.Z84 2008

572.80285--dc22

2007027514

Published by Garland Science, Taylor & Francis Group, LLC, an informa business
270 Madison Avenue, New York, NY 10016, USA, and
2 Park Square, Milton Park, Abingdon, OX14 4RN, UK.

Printed in India.

15 14 13 12 11 10 9 8 7 6 5 4

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PREFACE

The analysis of data arising from biomedical research has undergone a revolution over the last 15 years, brought about by the combined impact of the Internet and the development of increasingly sophisticated and accurate bioinformatics techniques. All research workers in the areas of biomolecular science and biomedicine are now expected to be competent in several areas of sequence analysis and often, additionally, in protein structure analysis and other more advanced bioinformatics techniques.

When we began our research careers in the early 1980s all of the techniques that now comprise bioinformatics were restricted to specialists, as databases and user-friendly applications were not readily available and had to be installed on laboratory computers. By the mid-1990s many datasets and analysis programs had become available on the Internet, and the scientists who produced sequences began to take on tasks such as sequence alignment themselves. However, there was a delay in providing comprehensive training in these techniques. At the end of the 1990s we started to expand our teaching of bioinformatics at both undergraduate and postgraduate level. We soon realized that there was a need for a textbook that bridged the gap between the simplistic introductions available, which concentrated on results almost to the exclusion of the underlying science, and the very detailed monographs, which presented the theoretical underpinnings of a restricted set of techniques. This textbook is our attempt to fill that gap.

Therefore on the one hand we wanted to include material explaining the program methods, because we believe that to perform a proper analysis it is not sufficient to understand how to use a program and the kind of results (and errors!) it can produce. It is also necessary to have some understanding of the technique used by the program and the science on which it is based. But on the other hand, we wanted this book to be accessible to the bioinformatics beginner, and we recognized that even the more advanced students occasionally just want a quick reminder of what an application does, without having to read through the theory behind it.

From this apparent dilemma was born the division into Applications and Theory Chapters. Throughout the book, we wrote dedicated Applications Chapters to provide a working knowledge of bioinformatics applications, quick and easy to grasp. In most places, an Applications Chapter is then followed by a Theory Chapter, which explains the program methods and the science behind them. Inevitably, we found this created a small amount of duplication between some chapters, but to us this was a small sacrifice if it left the reader free to choose at what level they could engage with the subject of bioinformatics.

We have created a book that will serve as a comfortable introduction to any new student of bioinformatics, but which they can continue to use into their postgraduate studies. The book assumes a certain level of understanding of the background biology, for example gene and protein structure, where it is important to appreciate the variety that exists and not only know the canonical examples of first-year textbooks. In addition, to describe the techniques in detail a level of mathematics is



required which is more appropriate for more advanced students. We are aware that many postgraduate students of bioinformatics have a background in areas such as computer science and mathematics. They will find many familiar algorithmic approaches presented, but will see their application in unfamiliar territory. As they read the book they will also appreciate that to become truly competent at bioinformatics they will require knowledge of biomedical science.

There is a certain amount of frustration inherent in producing any book, as the writing process seems often to be as much about what cannot be included as what can. Bioinformatics as a subject has already expanded to such an extent, and we had to be careful not to diminish the book's teaching value by trying to squeeze every possible topic into it. We have tried to include as broad a range of subjects as possible, but some have been omitted. For example, we do not deal with the methods of constructing a nucleotide sequence from the individual reads, nor with a number of more specialized aspects of genome annotation.

The final chapter is an introduction to the even-faster-moving subject of systems biology. Again, we had to balance the desire to say more against the practical constraints of space. But we hope this chapter gives readers a flavor of what the subject covers and the questions it is trying to answer. The chapter will not answer every reader's every query about systems biology, but if it prompts more of them to inquire further, that is already an achievement.

We wish to acknowledge many people who have helped us with this project. We would almost certainly not have got here without the enthusiasm and support of Matthew Day who guided us through the process of getting a first draft. Getting from there to the finished book was made possible by the invaluable advice and encouragement from Chris Dixon, Dom Holdsworth, Jackie Harbor, and others from Garland Science. We also wish to thank Eleanor Lawrence for her skills in massaging our text into shape, and Nigel Orme for producing the wonderful illustrations. We received inspiration and encouragement from many others, too many to name here, but including our students and those who read our draft chapters.

Finally, we wish to thank the many friends and family members who have had to suffer while we wrote this book. In particular JB wishes to thank his wife Hilary for her encouragement and perseverance. MZ wishes to specially thank her parents, Martin Scurr, Nick Lee, and her colleagues at work.

Marketa Zvelebil

Jeremy O. Baum

May 2007

A NOTE TO THE READER

Organization of this Book

Applications and Theory Chapters

Careful thought has gone into the organization of this book. The chapters are grouped in two ways. Firstly, the chapters are organized into seven parts according to topic. Within the parts, there is a second, less traditional, level of organization: most chapters are designated as either Applications or Theory Chapters. This book is designed to be accessible both to students who wish to obtain a working knowledge of the bioinformatics applications, as well as to students who want to know how the applications work and maybe write their own. So at the start of most parts, there are dedicated Applications Chapters, which deal with the more practical aspects of the particular research area, and are intended to act as a useful hands-on introduction. Following this are Theory Chapters, which explain the science, theory, and techniques employed in generally available applications. These are more demanding and should preferably be read after having gained a little experience of running the programs. In order to become truly proficient in the techniques you need to read and understand these more technical aspects. On the opening page of each chapter, and in the Table of Contents, it is clearly indicated whether it is an Applications or a Theory Chapter.

Part 1: Background Basics

Background Basics provides three introductory chapters to key knowledge that will be assumed throughout the remainder of the book. The first two chapters contain material that should be well-known to readers with a background in biomedical science. The first chapter describes the structure of nucleic acids and some of the roles played by them in living systems, including a brief description of how the genomic DNA is transcribed into mRNA and then translated into protein. The second chapter describes the structure and organization of proteins. Both of these chapters present only the most basic information required, and should not in any way be regarded as an adequate grounding in these topics for serious work. The intention is to provide enough information to make this book self-sufficient. The third chapter in this part describes databases, again at a very introductory level. Many biomedical research workers have large datasets to analyze, and these need to be stored in a convenient and practical way. Databases can provide a complete solution to this problem.

Part 2: Sequence Alignments

Sequence Alignments contains three chapters that deal with a variety of analyses of sequences, all relating to identifying similarities. Chapter 4 is a practical introduction to the area, following some examples through different analyses and showing some potential problems as well as successful results. Chapters 5 and 6 deal with several of the many different techniques used in sequence analysis. Chapter 5 focuses on the general aspects of aligning two sequences and the specific methods employed in database searches. A number of techniques are described in detail, including dynamic programming, suffix trees, hashing, and chaining. Chapter 6 deals with methods involving many sequences, defining commonly occurring patterns, defining the profile of a family of related proteins, and constructing a multiple alignment. A key technique presented in this chapter is that of hidden Markov models (HMMs).

Part 3: Evolutionary Processes

Evolutionary Processes presents the methods used to obtain phylogenetic trees from a sequence dataset. These trees are reconstructions of the evolutionary history of the sequences, assuming that they share a common ancestor. Chapter 7 explains some of the basic concepts involved, and then shows how the different methods can be applied to two different scientific problems. In Chapter 8 details are given of the techniques involved and how they relate to the assumptions made about the evolutionary processes.

Part 4: Genome Characteristics

Genome Characteristics deals with the analysis required to interpret raw genome sequence data. Although by the time a genome sequence is published in the research journals some preliminary analysis will have been carried out, often the unanalyzed sequence is available before then. This part describes some of the techniques that can be used to try to locate genes in the sequence. Chapter 9 describes some of the range of programs available, and shows how complex their output can be and illustrates some of the possible pitfalls. Chapter 10 presents a survey of the techniques used, especially different Markov models and how models of whole genes can be built up from models of individual components such as ribosome-binding sites.

Part 5: Secondary Structures

Secondary Structures provides two chapters on methods of predicting secondary structures based on sequence (or primary structure). Chapter 11 introduces the methods of secondary structure prediction and discusses the various techniques and ways to interpret the results. Later sections of the chapter deal with prediction of more specialized secondary structure such as protein transmembrane regions, coiled coil and leucine zipper structures, and RNA secondary structures. Chapter 12 presents the underlying principles and details of the prediction methods from basic concepts to in-depth understanding of techniques such as neural networks and Markov models applied to this problem.

Part 6: Tertiary Structures

Tertiary Structures extends the material in Part 5 to enable the prediction and modeling of protein tertiary and quaternary structure. Chapter 13 introduces the reader to the concepts of energy functions, minimization, and *ab initio* prediction. It deals in more detail with the method of threading and focuses on homology modeling of protein structures, taking the student in a stepwise fashion through the process. The chapter ends with example studies to illustrate the techniques. Chapter 14 contains methods and techniques for further analysis of structural information and describes the importance of structure and function relationships. This chapter deals with how fold prediction can help to identify function, as well as giving an introduction to ligand docking and drug design.

Part 7: Cells and Organisms

Cells and Organisms consists of two chapters that deal in some detail with expression analysis and an introductory chapter on systems biology. Chapter 15 introduces the techniques available to analyze protein and gene expression data. It shows the reader the information that can be learned from these experimental techniques as well as how the information could be used for further analysis. Chapter 16 presents some of the clustering techniques and statistics that are touched upon in Chapter 15 and are commonly used in gene and protein expression analysis. Chapter 17 is a standalone chapter dealing with the modeling of systems processes. It introduces the reader to the basic concepts of systems biology, and shows what this exciting and rapidly growing field may achieve in the future.

Appendices

Three appendices are provided that expand on some of the concepts mentioned in the main part of this book. These are useful for the more inquisitive and advanced reader. Appendix A deals with probability and Bayesian analysis, Appendix B is mainly associated with Part 6 and deals with molecular energy functions, while Appendix C describes function optimization techniques.

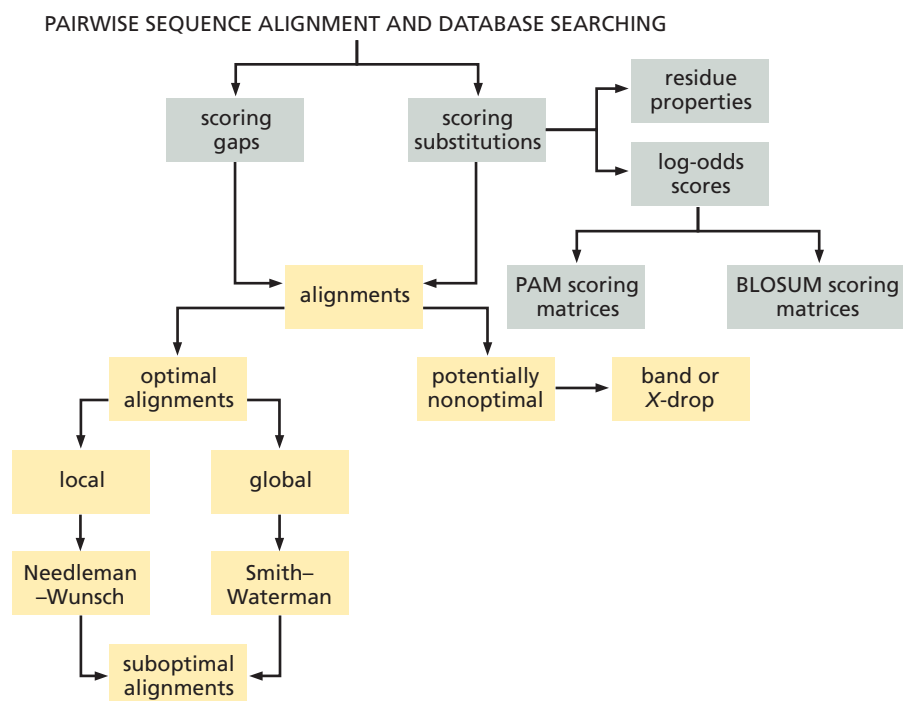
Organization of the Chapters

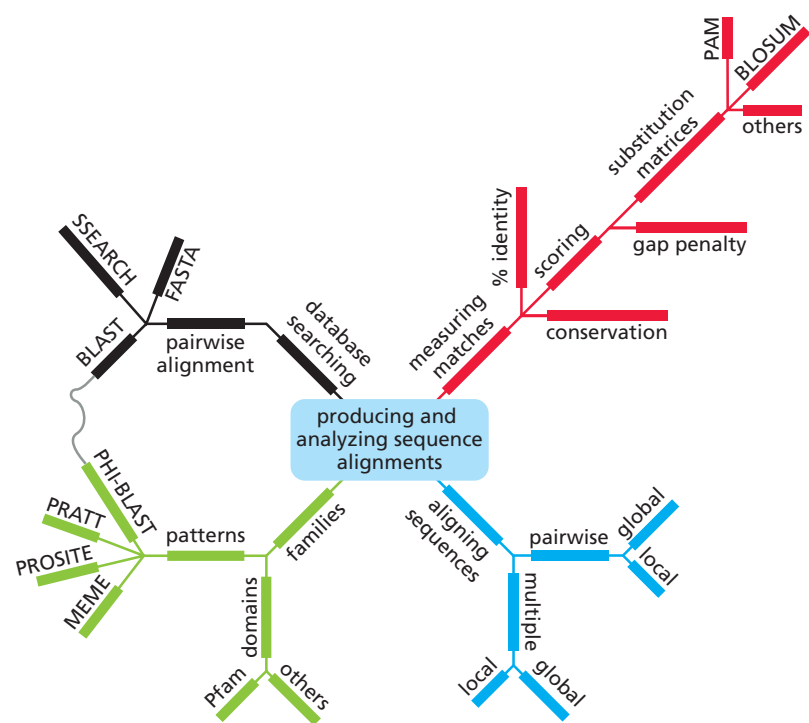
Learning Outcomes

Each chapter opens with a list of learning outcomes which summarize the topics to be covered and act as a revision checklist.

Flow Diagrams

Within each chapter every section is introduced with a flow diagram to help the student to visualize and remember the topics covered in that section. A flow diagram from Chapter 5 is given below, as an example. Those concepts which will be described in the current section are shown in yellow boxes with arrows to show how they are connected to each other. For example two main types of optimal alignments will be described in this section of the chapter: local and global. Those concepts which were described in previous sections of the chapter are shown in grey boxes, so that the links can easily be seen between the topics of the current section and what has already been presented. For example, creating alignments requires methods for scoring gaps and for scoring substitutions, both of which have already been described in the chapter. In this way the major concepts and their inter-relationships are gradually built up throughout the chapter.



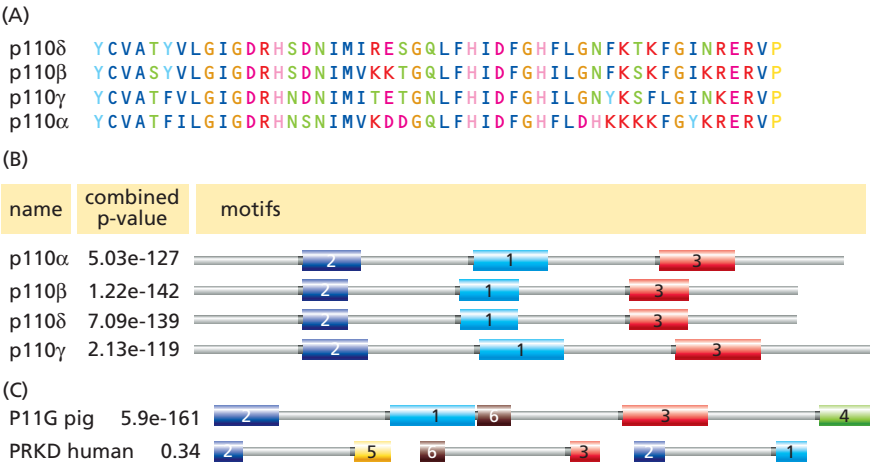


Mind Maps

Each chapter has a mind map, which is a specialized pedagogical feature, enabling the student to visualize and remember the steps that are necessary for specific applications. The mind map for Chapter 4 is given above, as an example. In this example, four main areas of the topic ‘producing and analyzing sequence alignments’ have been identified: measuring matches, database searching, aligning sequences, and families. Each of these areas, colored for clarity, is developed to identify the key concepts involved, creating a visual aid to help the reader see at a glance the range of the material covered in discussing this area. Occasionally there are important connections between distinct areas of the mind map, as here in linking BLAST and PHI-BLAST, with the latter method being derived directly from the former, but having a quite different function, and thus being in a different area of the mind map.

Illustrations

Each chapter is illustrated with four-color figures. Considerable care has been put into ensuring simplicity as well as consistency of representation across the book. Figure 4.16 is given below, as an example.



Further Reading

It is not possible to summarize all current knowledge in the confines of this book, let alone anticipate future developments in this rapidly developing subject. Therefore at the end of each chapter there are references to research literature and specialist monographs to help readers continue to develop their knowledge and skills. We have grouped the books and articles according to topic, such that the sections within the Further Reading correspond to the sections in the chapter itself: we hope this will help the reader target their attention more quickly onto the appropriate extension material.

List of Symbols

Bioinformatics makes use of numerous symbols, many of which will be unfamiliar to those who do not already know the subject well. To help the reader navigate the symbols used in this book, a comprehensive list is given at the back which quotes each symbol, its definition, and where its most significant occurrences in the book are located.

Glossary

All technical terms are highlighted in bold where they first appear in the text and are then listed and explained in the Glossary. Further, each term in the Glossary also appears in the Index, so the reader can quickly gain access to the relevant pages where the term is covered in more detail. The book has been designed to cross-reference in as thorough and helpful a way as possible.

Garland Science Website

Garland Science has made available a number of supplementary resources on its website, which are freely available and do not require a password. For more details, go to www.garlandscience.com/gs_textbooks.asp and follow the link to *Understanding Bioinformatics*.

Artwork

All the figures in *Understanding Bioinformatics* are available to download from the Garland Science website. The artwork files are saved in zip format, with a single zip file for each chapter. Individual figures can then be extracted as jpg files.

Additional Material

The Garland Science website has some additional material relating to the topics in this book. For each of the seven parts a pdf is available, which provides a set of useful weblinks relevant to those chapters. These include weblinks to relevant and important databases and to file format definitions, as well as to free programs and to servers which permit data analysis on-line. In addition to these, the sets of data which were used to illustrate the methods of analysis are also provided. These will allow the reader to reanalyze the same data, reproducing the results shown here and trying out other techniques.

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CONTENTS IN BRIEF

PART 1 Background Basics

Chapter 1:	The Nucleic Acid World	3
Chapter 2:	Protein Structure	25
Chapter 3:	Dealing With Databases	45

PART 2 Sequence Alignments

Chapter 4:	Producing and Analyzing Sequence Alignments	Applications Chapter	71
Chapter 5:	Pairwise Sequence Alignment and Database Searching	Theory Chapter	115
Chapter 6:	Patterns, Profiles, and Multiple Alignments	Theory Chapter	165

PART 3 Evolutionary Processes

Chapter 7:	Recovering Evolutionary History	Applications Chapter	223
Chapter 8:	Building Phylogenetic Trees	Theory Chapter	267

PART 4 Genome Characteristics

Chapter 9:	Revealing Genome Features	Applications Chapter	317
Chapter 10:	Gene Detection and Genome Annotation	Theory Chapter	357

PART 5 Secondary Structures

Chapter 11:	Obtaining Secondary Structure from Sequence	Applications Chapter	411
Chapter 12:	Predicting Secondary Structures	Theory Chapter	461

PART 6 Tertiary Structures

Chapter 13:	Modeling Protein Structure	Applications Chapter	521
Chapter 14:	Analyzing Structure–Function Relationships	Applications Chapter	567

PART 7 Cells and Organisms

Chapter 15:	Proteome and Gene Expression Analysis	599
Chapter 16:	Clustering Methods and Statistics	625
Chapter 17:	Systems Biology	667

APPENDICES Background Theory

Appendix A:	Probability, Information, and Bayesian Analysis	695
Appendix B:	Molecular Energy Functions	700
Appendix C:	Function Optimization	709

CONTENTS

Preface	v	Amino acids are covalently linked together in the protein chain by peptide bonds	29
A Note to the Reader	vii	Secondary structure of proteins is made up of α -helices and β -strands	33
List of Reviewers	xii	Several different types of β -sheet are found in protein structures	35
Contents in Brief	xiii	Turns, hairpins and loops connect helices and strands	36
Part 1 Background Basics			
Chapter 1 The Nucleic Acid World			
1.1 The Structure of DNA and RNA	5	2.2 Implication for Bioinformatics	37
DNA is a linear polymer of only four different bases	5	Certain amino acids prefer a particular structural unit	37
Two complementary DNA strands interact by base pairing to form a double helix	7	Evolution has aided sequence analysis	38
RNA molecules are mostly single stranded but can also have base-pair structures	9	Visualization and computer manipulation of protein structures	38
1.2 DNA, RNA, and Protein: The Central Dogma	10	2.3 Proteins Fold to Form Compact Structures	40
DNA is the information store, but RNA is the messenger	11	The tertiary structure of a protein is defined by the path of the polypeptide chain	41
Messenger RNA is translated into protein according to the genetic code	12	The stable folded state of a protein represents a state of low energy	41
Translation involves transfer RNAs and RNA-containing ribosomes	13	Many proteins are formed of multiple subunits	42
1.3 Gene Structure and Control	14	Summary	43
RNA polymerase binds to specific sequences that position it and identify where to begin transcription	15	Further Reading	44
The signals initiating transcription in eukaryotes are generally more complex than those in bacteria	17	Chapter 3 Dealing with Databases	
Eukaryotic mRNA transcripts undergo several modifications prior to their use in translation	18	3.1 The Structure of Databases	46
The control of translation	19	Flat-file databases store data as text files	48
1.4 The Tree of Life and Evolution	20	Relational databases are widely used for storing biological information	49
A brief survey of the basic characteristics of the major forms of life	21	XML has the flexibility to define bespoke data classifications	50
Nucleic acid sequences can change as a result of mutation	22	Many other database structures are used for biological data	51
Summary	23	Databases can be accessed locally or online and often link to each other	52
Further Reading	24	3.2 Types of Database	52
Chapter 2 Protein Structure		There's more to databases than just data	53
2.1 Primary and Secondary Structure	25	Primary and derived data	53
Protein structure can be considered on several different levels	26	How we define and connect things is very important: Ontologies	54
Amino acids are the building blocks of proteins	27	3.3 Looking for Databases	55
The differing chemical and physical properties of amino acids are due to their side chains	28	Sequence databases	55
		Microarray databases	58

Protein interaction databases	58	4.4 Inserting Gaps	85
Structural databases	59	Gaps inserted in a sequence to maximize similarity require a scoring penalty	85
3.4 Data Quality	61	Dynamic programming algorithms can determine the optimal introduction of gaps	86
Nonredundancy is especially important for some applications of sequence databases	62	4.5 Types of Alignment	87
Automated methods can be used to check for data consistency	63	Different kinds of alignments are useful in different circumstances	87
Initial analysis and annotation is usually automated	64	Multiple sequence alignments enable the simultaneous comparison of a set of similar sequences	90
Human intervention is often required to produce the highest quality annotation	65	Multiple alignments can be constructed by several different techniques	90
The importance of updating databases and entry identifier and version numbers	65	Multiple alignments can improve the accuracy of alignment for sequences of low similarity	91
Summary	66	ClustalW can make global multiple alignments of both DNA and protein sequences	92
Further Reading	67	Multiple alignments can be made by combining a series of local alignments	92
		Alignment can be improved by incorporating additional information	93
Part 2 Sequence Alignments		4.6 Searching Databases	93
APPLICATIONS CHAPTER		Fast yet accurate search algorithms have been developed	94
Chapter 4 Producing and Analyzing Sequence Alignments		FASTA is a fast database-search method based on matching short identical segments	95
4.1 Principles of Sequence Alignment	72	BLAST is based on finding very similar short segments	95
Alignment is the task of locating equivalent regions of two or more sequences to maximize their similarity	73	Different versions of BLAST and FASTA are used for different problems	95
Alignment can reveal homology between sequences	74	PSI-BLAST enables profile-based database searches	96
It is easier to detect homology when comparing protein sequences than when comparing nucleic acid sequences	75	SSEARCH is a rigorous alignment method	97
4.2 Scoring Alignments	76	4.7 Searching with Nucleic Acid or Protein Sequences	97
The quality of an alignment is measured by giving it a quantitative score	76	DNA or RNA sequences can be used either directly or after translation	97
The simplest way of quantifying similarity between two sequences is percentage identity	76	The quality of a database match has to be tested to ensure that it could not have arisen by chance	97
The dot-plot gives a visual assessment of similarity based on identity	77	Choosing an appropriate <i>E</i> -value threshold helps to limit a database search	98
Genuine matches do not have to be identical	79	Low-complexity regions can complicate homology searches	100
There is a minimum percentage identity that can be accepted as significant	81	Different databases can be used to solve particular problems	102
There are many different ways of scoring an alignment	81	4.8 Protein Sequence Motifs or Patterns	103
4.3 Substitution Matrices	81	Creation of pattern databases requires expert knowledge	104
Substitution matrices are used to assign individual scores to aligned sequence positions	81	The BLOCKS database contains automatically compiled short blocks of conserved multiply aligned protein sequences	105
The PAM substitution matrices use substitution frequencies derived from sets of closely related protein sequences	82	4.9 Searching Using Motifs and Patterns	107
The BLOSUM substitution matrices use mutation data from highly conserved local regions of sequence	84	The PROSITE database can be searched for protein motifs and patterns	107
The choice of substitution matrix depends on the problem to be solved	84		

The pattern-based program PHI-BLAST searches for both homology and matching motifs	108	The BLAST algorithm makes use of finite-state automata	147
Patterns can be generated from multiple sequences using PRATT	108	Comparing a nucleotide sequence directly with a protein sequence requires special modifications to the BLAST and FASTA algorithms	150
The PRINTS database consists of fingerprints representing sets of conserved motifs that describe a protein family	109	5.4 Alignment Score Significance	153
The Pfam database defines profiles of protein families	109	The statistics of gapped local alignments can be approximated by the same theory	156
4.10 Patterns and Protein Function	109	5.5 Aligning Complete Genome Sequences	156
Searches can be made for particular functional sites in proteins	109	Indexing and scanning whole genome sequences efficiently is crucial for the sequence alignment of higher organisms	157
Sequence comparison is not the only way of analyzing protein sequences	110	The complex evolutionary relationships between the genomes of even closely related organisms require novel alignment algorithms	159
Summary	111	Summary	159
Further Reading	112	Further Reading	161
THEORY CHAPTER		THEORY CHAPTER	
Chapter 5 Pairwise Sequence Alignment and Database Searching		Chapter 6 Patterns, Profiles, and Multiple Alignments	
5.1 Substitution Matrices and Scoring	117	6.1 Profiles and Sequence Logos	167
Alignment scores attempt to measure the likelihood of a common evolutionary ancestor	117	Position-specific scoring matrices are an extension of substitution scoring matrices	168
The PAM (MDM) substitution scoring matrices were designed to trace the evolutionary origins of proteins	119	Methods for overcoming a lack of data in deriving the values for a PSSM	171
The BLOSUM matrices were designed to find conserved regions of proteins	122	PSI-BLAST is a sequence database searching program	176
Scoring matrices for nucleotide sequence alignment can be derived in similar ways	125	Representing a profile as a logo	177
The substitution scoring matrix used must be appropriate to the specific alignment problem	126	6.2 Profile Hidden Markov Models	179
Gaps are scored in a much more heuristic way than substitutions	126	The basic structure of HMMs used in sequence alignment to profiles	180
5.2 Dynamic Programming Algorithms	127	Estimating HMM parameters using aligned sequences	185
Optimal global alignments are produced using efficient variations of the Needleman–Wunsch algorithm	129	Scoring a sequence against a profile HMM: The most probable path and the sum over all paths	187
Local and suboptimal alignments can be produced by making small modifications to the dynamic programming algorithm	135	Estimating HMM parameters using unaligned sequences	190
Time can be saved with a loss of rigor by not calculating the whole matrix	139	6.3 Aligning Profiles	193
5.3 Indexing Techniques and Algorithmic Approximations	141	Comparing two PSSMs by alignment	193
Suffix trees locate the positions of repeats and unique sequences	141	Aligning profile HMMs	195
Hashing is an indexing technique that lists the starting positions of all k-tuples	143	6.4 Multiple Sequence Alignments by Gradual Sequence Addition	196
The FASTA algorithm uses hashing and chaining for fast database searching	144	The order in which sequences are added is chosen based on the estimated likelihood of incorporating errors in the alignment	198
		Many different scoring schemes have been used in constructing multiple alignments	200

The multiple alignment is built using the guide tree and profile methods and may be further refined	204	Phylogenetic analyses of a small dataset of 16S RNA sequence data	255
6.5 Other Ways of Obtaining Multiple Alignments	207	Building a gene tree for a family of enzymes can help to identify how enzymatic functions evolved	259
The multiple sequence alignment program DIALIGN aligns ungapped blocks	207	Summary	264
The SAGA method of multiple alignment uses a genetic algorithm	209	Further Reading	265
6.6 Sequence Pattern Discovery	211	THEORY CHAPTER	
Discovering patterns in a multiple alignment: eMOTIF and AACC	213	Chapter 8 Building Phylogenetic Trees	
Probabilistic searching for common patterns in sequences: Gibbs and MEME	215	8.1 Evolutionary Models and the Calculation of Evolutionary Distance	268
Searching for more general sequence patterns	217	A simple but inaccurate measure of evolutionary distance is the <i>p</i> -distance	268
Summary	218	The Poisson distance correction takes account of multiple mutations at the same site	270
Further Reading	219	The Gamma distance correction takes account of mutation rate variation at different sequence positions	270
		The Jukes–Cantor model reproduces some basic features of the evolution of nucleotide sequences	271
		More complex models distinguish between the relative frequencies of different types of mutation	272
		There is a nucleotide bias in DNA sequences	275
		Models of protein-sequence evolution are closely related to the substitution matrices used for sequence alignment	276
		8.2 Generating Single Phylogenetic Trees	276
		Clustering methods produce a phylogenetic tree based on evolutionary distances	276
		The UPGMA method assumes a constant molecular clock and produces an ultrametric tree	278
		The Fitch–Margoliash method produces an unrooted additive tree	279
		The neighbor-joining method is related to the concept of minimum evolution	282
		Stepwise addition and star-decomposition methods are usually used to generate starting trees for further exploration, not the final tree	285
		8.3 Generating Multiple Tree Topologies	286
		The branch-and-bound method greatly improves the efficiency of exploring tree topology	288
		Optimization of tree topology can be achieved by making a series of small changes to an existing tree	288
		Finding the root gives a phylogenetic tree a direction in time	291
		8.4 Evaluating Tree Topologies	293
		Functions based on evolutionary distances can be used to evaluate trees	293
		Unweighted parsimony methods look for the trees with the smallest number of mutations	297

Part 3 Evolutionary Processes

APPLICATIONS CHAPTER

Chapter 7 Recovering Evolutionary History

7.1 The Structure and Interpretation of Phylogenetic Trees	225
Phylogenetic trees reconstruct evolutionary relationships	225
Tree topology can be described in several ways	230
Consensus and condensed trees report the results of comparing tree topologies	232
7.2 Molecular Evolution and its Consequences	235
Most related sequences have many positions that have mutated several times	236
The rate of accepted mutation is usually not the same for all types of base substitution	236
Different codon positions have different mutation rates	238
Only orthologous genes should be used to construct species phylogenetic trees	239
Major changes affecting large regions of the genome are surprisingly common	247
7.3 Phylogenetic Tree Reconstruction	248
Small ribosomal subunit rRNA sequences are well suited to reconstructing the evolution of species	249
The choice of the method for tree reconstruction depends to some extent on the size and quality of the dataset	249
A model of evolution must be chosen to use with the method	251
All phylogenetic analyses must start with an accurate multiple alignment	255

Mutations can be weighted in different ways in the parsimony method	300	Prokaryotic promoter regions contain relatively well-defined motifs	339
Trees can be evaluated using the maximum likelihood method	302	Eukaryotic promoter regions are typically more complex than prokaryotic promoters	340
The quartet-puzzling method also involves maximum likelihood in the standard implementation	305	A variety of promoter-prediction methods are available online	340
Bayesian methods can also be used to reconstruct phylogenetic trees	306	Promoter prediction results are not very clear-cut	341
8.5 Assessing the Reliability of Tree Features and Comparing Trees	307	9.6 Confirming Predictions	342
The long-branch attraction problem can arise even with perfect data and methodology	308	There are various methods for calculating the accuracy of gene-prediction programs	342
Tree topology can be tested by examining the interior branches	309	Translating predicted exons can confirm the correctness of the prediction	343
Tests have been proposed for comparing two or more alternative trees	310	Constructing the protein and identifying homologs	343
Summary	311	9.7 Genome Annotation	346
Further Reading	312	Genome annotation is the final step in genome analysis	347
		Gene ontology provides a standard vocabulary for gene annotation	348
		9.8 Large Genome Comparisons	353
		Summary	354
		Further Reading	355
Part 4 Genome Characteristics		THEORY CHAPTER	
APPLICATIONS CHAPTER		Chapter 10 Gene Detection and Genome Annotation	
Chapter 9 Revealing Genome Features		10.1 Detection of Functional RNA Molecules Using Decision Trees	361
9.1 Preliminary Examination of Genome Sequence	318	Detection of tRNA genes using the tRNAscan algorithm	361
Whole genome sequences can be split up to simplify gene searches	319	Detection of tRNA genes in eukaryotic genomes	362
Structural RNA genes and repeat sequences can be excluded from further analysis	319	10.2 Features Useful for Gene Detection in Prokaryotes	364
Homology can be used to identify genes in both prokaryotic and eukaryotic genomes	322	10.3 Algorithms for Gene Detection in Prokaryotes	368
9.2 Gene Prediction in Prokaryotic Genomes	322	GeneMark uses inhomogeneous Markov chains and dicodon statistics	368
9.3 Gene Prediction in Eukaryotic Genomes	323	GLIMMER uses interpolated Markov models of coding potential	371
Programs for predicting exons and introns use a variety of approaches	323	ORPHEUS uses homology, codon statistics, and ribosome-binding sites	372
Gene predictions must preserve the correct reading frame	324	GeneMark.hmm uses explicit state duration hidden Markov models	373
Some programs search for exons using only the query sequence and a model for exons	327	EcoParse is an HMM gene model	376
Some programs search for genes using only the query sequence and a gene model	332	10.4 Features Used in Eukaryotic Gene Detection	377
Genes can be predicted using a gene model and sequence similarity	334	Differences between prokaryotic and eukaryotic genes	377
Genomes of related organisms can be used to improve gene prediction	336	Introns, exons, and splice sites	379
9.4 Splice Site Detection	337	Promoter sequences and binding sites for transcription factors	381
Splice sites can be detected independently by specialized programs	338		
9.5 Prediction of Promoter Regions	338		

10.5 Predicting Eukaryotic Gene Signals	381	that incorporate additional information about protein structure	414
Detection of core promoter binding signals is a key element of some eukaryotic gene-prediction methods	381	Machine-learning approaches to secondary structure prediction mainly make use of neural networks and HMM methods	415
A set of models has been designed to locate the site of core promoter sequence signals	383		
Predicting promoter regions from general sequence properties can reduce the numbers of false-positive results	387		
Predicting eukaryotic transcription and translation start sites	389		
Translation and transcription stop signals complete the gene definition	389		
10.6 Predicting Exon/Intron Structure	389	11.2 Training and Test Databases	416
Exons can be identified using general sequence properties	390	There are several ways to define protein secondary structures	417
Splice-site prediction	392		
Splice sites can be predicted by sequence patterns combined with base statistics	393	11.3 Assessing the Accuracy of Prediction Programs	417
GenScan uses a combination of weight matrices and decision trees to locate splice sites	394	Q ₃ measures the accuracy of individual residue assignments	417
GeneSplicer predicts splice sites using first-order Markov chains	394	Secondary structure predictions should not be expected to reach 100% residue accuracy	418
NetPlantGene uses neural networks with intron and exon predictions to predict splice sites	395	The Sov value measures the prediction accuracy for whole elements	419
Other splicing features may yet be exploited for splice-site prediction	396	CAFASP/CASP: Unbiased and readily available protein prediction assessments	419
Specific methods exist to identify initial and terminal exons	396		
Exons can be defined by searching databases for homologous regions	397	11.4 Statistical and Knowledge-Based Methods	421
10.7 Complete Eukaryotic Gene Models	397	The GOR method uses an information theory approach	422
		The program Zpred includes multiple alignment of homologous sequences and residue conservation information	425
10.8 Beyond the Prediction of Individual Genes	399	There is an overall increase in prediction accuracy using multiple sequence information	426
Functional annotation	400	The nearest-neighbor method: The use of multiple nonhomologous sequences	428
Comparison of related genomes can help resolve uncertain predictions	403	PREDATOR is a combined statistical and knowledge-based program that includes the nearest-neighbor approach	428
Evaluation and reevaluation of gene-detection methods	405		
Summary	405	11.5 Neural Network Methods of Secondary Structure Prediction	430
Further Reading	406	Assessing the reliability of neural net predictions	432
		Several examples of Web-based neural network secondary structure prediction programs	432
		PROF: Protein forecasting	434
		PSIPRED	434
		Jnet: Using several alternative representations of the sequence alignment	434
		11.6 Some Secondary Structures Require Specialized Prediction Methods	435
		Transmembrane proteins	436
		Quantifying the preference for a membrane environment	437
		11.7 Prediction of Transmembrane Protein Structure	438
		Multi-helix membrane proteins	439
		A selection of prediction programs to predict transmembrane helices	441

Part 5 Secondary Structures

APPLICATIONS CHAPTER

Chapter 11 Obtaining Secondary Structure from Sequence

11.1 Types of Prediction Methods	413
Statistical methods are based on rules that give the probability that a residue will form part of a particular secondary structure	414
Nearest-neighbor methods are statistical methods	

Statistical methods	443	12.4 Neural Networks Have Been Employed	
Knowledge-based prediction	443	Successfully for Secondary Structure Prediction	492
Evolutionary information from protein families improves the prediction	444	Layered feed-forward neural networks can transform a sequence into a structural prediction	494
Neural nets in transmembrane prediction	445	Inclusion of information on homologous sequences improves neural network accuracy	502
Predicting transmembrane helices with hidden Markov models	446	More complex neural nets have been applied to predict secondary and other structural features	503
Comparing the results: What to choose	447		
What happens if a non-transmembrane protein is submitted to transmembrane prediction programs	448	12.5 Hidden Markov Models Have Been Applied to Structure Prediction	504
Prediction of transmembrane structure containing β -strands	448	HMM methods have been found especially effective for transmembrane proteins	506
		Nonmembrane protein secondary structures can also be successfully predicted with HMMs	509
11.8 Coiled-coil Structures	451	12.6 General Data Classification Techniques Can Predict Structural Features	510
The COILS prediction program	452	Support vector machines have been successfully used for protein structure prediction	511
PAIRCOIL and MULTICOIL are an extension of the COILS algorithm	453	Discriminants, SOMs, and other methods have also been used	512
Zippering the Leucine zipper: A specialized coiled coil	453		
11.9 RNA Secondary Structure Prediction	455	Summary	514
Summary	458	Further Reading	515
Further Reading	459		
THEORY CHAPTER			
Chapter 12 Predicting Secondary Structures			
12.1 Defining Secondary Structure and Prediction		Part 6 Tertiary Structures	
Accuracy	463	APPLICATIONS CHAPTER	
The definitions used for automatic protein secondary structure assignment do not give identical results	464	Chapter 13 Modeling Protein Structure	
There are several different measures of the accuracy of secondary structure prediction	469	13.1 Potential Energy Functions and Force Fields	524
		The conformation of a protein can be visualized in terms of a potential energy surface	525
12.2 Secondary Structure Prediction Based on Residue Propensities	472	Conformational energies can be described by simple mathematical functions	525
Each structural state has an amino acid preference which can be assigned as a residue propensity	473	Similar force fields can be used to represent conformational energies in the presence of averaged environments	526
The simplest prediction methods are based on the average residue propensity over a sequence window	476	Potential energy functions can be used to assess a modeled structure	527
Residue propensities are modulated by nearby sequence	479	Energy minimization can be used to refine a modeled structure and identify local energy minima	527
Predictions can be significantly improved by including information from homologous sequences	484	Molecular dynamics and simulated annealing are used to find global energy minima	528
		13.2 Obtaining a Structure by Threading	529
12.3 The Nearest-Neighbor Methods are Based on Sequence Segment Similarity	485	The prediction of protein folds in the absence of known structural homologs	531
Short segments of similar sequence are found to have similar structure	487	Libraries or databases of nonredundant protein folds are used in threading	531
Several sequence similarity measures have been used to identify nearest-neighbor segments	488	Two distinct types of scoring schemes have been used in threading methods	531
A weighted average of the nearest-neighbor segment structures is used to make the prediction	490	Dynamic programming methods can identify optimal alignments of target sequences and structural folds	533
A nearest-neighbor method has been developed to predict regions with a high potential to misfold	491		

Several methods are available to assess the confidence to be put on the fold prediction	534	MolIDE is a downloadable semi-automatic modeling package	560
The C2-like domain from the Dictyostelia: A practical example of threading	535	Automated modeling on the Web illustrated with p110 α kinase	561
13.3 Principles of Homology Modeling	537	Modeling a functionally related but sequentially dissimilar protein: mTOR	563
Closely related target and template sequences give better models	539	Generating a multidomain three-dimensional structure from sequence	564
Significant sequence identity depends on the length of the sequence	540	Summary	564
Homology modeling has been automated to deal with the numbers of sequences that can now be modeled	541	Further Reading	565
Model building is based on a number of assumptions	541		
13.4 Steps in Homology Modeling	542	APPLICATIONS CHAPTER	
Structural homologs to the target protein are found in the PDB	543	Chapter 14 Analyzing Structure-Function Relationships	
Accurate alignment of target and template sequences is essential for successful modeling	543	14.1 Functional Conservation	568
The structurally conserved regions of a protein are modeled first	544	Functional regions are usually structurally conserved	569
The modeled core is checked for misfits before proceeding to the next stage	545	Similar biochemical function can be found in proteins with different folds	570
Sequence realignment and remodeling may improve the structure	545	Fold libraries identify structurally similar proteins regardless of function	571
Insertions and deletions are usually modeled as loops	545	14.2 Structure Comparison Methods	574
Nonidentical amino acid side chains are modeled mainly by using rotamer libraries	547	Finding domains in proteins aids structure comparison	574
Energy minimization is used to relieve structural errors	548	Structural comparisons can reveal conserved functional elements not discernible from a sequence comparison	576
Molecular dynamics can be used to explore possible conformations for mobile loops	548	The CE method builds up a structural alignment from pairs of aligned protein segments	576
Models need to be checked for accuracy	549	The Vector Alignment Search Tool (VAST) aligns secondary structural elements	577
How far can homology models be trusted?	551	DALI identifies structure superposition without maintaining segment order	578
13.5 Automated Homology Modeling	552	FATCAT introduces rotations between rigid segments	579
The program MODELLER models by satisfying protein structure constraints	553	14.3 Finding Binding Sites	580
COMPOSER uses fragment-based modeling to automatically generate a model	553	Highly conserved, strongly charged, or hydrophobic surface areas may indicate interaction sites	582
Automated methods available on the Web for comparative modeling	554	Searching for protein-protein interactions using surface properties	584
Assessment of structure prediction	554	Surface calculations highlight clefts or holes in a protein that may serve as binding sites	585
13.6 Homology Modeling of PI3 Kinase p110α	557	Looking at residue conservation can identify binding sites	586
Swiss-Pdb Viewer can be used for manual or semi-manual modeling	557	14.4 Docking Methods and Programs	587
Alignment, core modeling, and side-chain modeling are carried out all in one	558	Simple docking procedures can be used when the structure of a homologous protein bound to a ligand analog is known	588
The loops are modeled from a database of possible structures	559	Specialized docking programs will automatically dock a ligand to a structure	588
Energy minimization and quality inspection can be carried out within Swiss-Pdb Viewer	559		

Scoring functions are used to identify the most likely docked ligand	590	The changes in a set of protein spots can be tracked over a number of different samples	618
The DOCK program is a semirigid-body method that analyzes shape and chemical complementarity of ligand and binding site	590	Databases and online tools are available to aid the interpretation of 2D gel data	620
Fragment docking identifies potential substrates by predicting types of atoms and functional groups in the binding area	591	Protein microarrays allow the simultaneous detection of the presence or activity of large numbers of different proteins	621
GOLD is a flexible docking program, which utilizes a genetic algorithm	591	Mass spectrometry can be used to identify the proteins separated and purified by 2D gel electrophoresis or other means	621
The water molecules in binding sites should also be considered	592	Protein-identification programs for mass spectrometry are freely available on the Web	622
Summary	593	Mass spectrometry can be used to measure protein concentration	623
Further Reading	594	Summary	623
		Further Reading	624
 Part 7 Cells and Organisms			
Chapter 15 Proteome and Gene Expression Analysis			
15.1 Analysis of Large-scale Gene Expression	601	Chapter 16 Clustering Methods and Statistics	
The expression of large numbers of different genes can be measured simultaneously by DNA microarrays	602	16.1 Expression Data Require Preparation Prior to Analysis	626
Gene expression microarrays are mainly used to detect differences in gene expression in different conditions	602	Data normalization is designed to remove systematic experimental errors	627
Serial analysis of gene expression (SAGE) is also used to study global patterns of gene expression	604	Expression levels are often analyzed as ratios and are usually transformed by taking logarithms	628
Digital differential display uses bioinformatics and statistics to detect differential gene expression in different tissues	605	Sometimes further normalization is useful after the data transformation	630
Facilitating the integration of data from different places and experiments	606	Principal component analysis is a method for combining the properties of an object	631
The simplest method of analyzing gene expression microarray data is hierarchical cluster analysis	606		
Techniques based on self-organizing maps can be used for analyzing microarray data	608	16.2 Cluster Analysis Requires Distances to be Defined Between all Data Points	633
Self-organizing tree algorithms (SOTAs) cluster from the top down by successive subdivision of clusters	610	Euclidean distance is the measure used in everyday life	634
Clustered gene expression data can be used as a tool for further research	610	The Pearson correlation coefficient measures distance in terms of the shape of the expression response	635
		The Mahalanobis distance takes account of the variation and correlation of expression responses	636
15.2 Analysis of Large-scale Protein Expression	612	16.3 Clustering Methods Identify Similar and Distinct Expression Patterns	637
Two-dimensional gel electrophoresis is a method for separating the individual proteins in a cell	613	Hierarchical clustering produces a related set of alternative partitions of the data	639
Measuring the expression levels shown in 2D gels	614	<i>k</i> -means clustering groups data into several clusters but does not determine a relationship between clusters	641
Differences in protein expression levels between different samples can be detected by 2D gels	615	Self-organizing maps (SOMs) use neural network methods to cluster data into a predetermined number of clusters	644
Clustering methods are used to identify protein spots with similar expression patterns	615	Evolutionary clustering algorithms use selection, recombination, and mutation to find the best possible solution to a problem	646
Principal component analysis (PCA) is an alternative to clustering for analyzing microarray and 2D gel data	618		

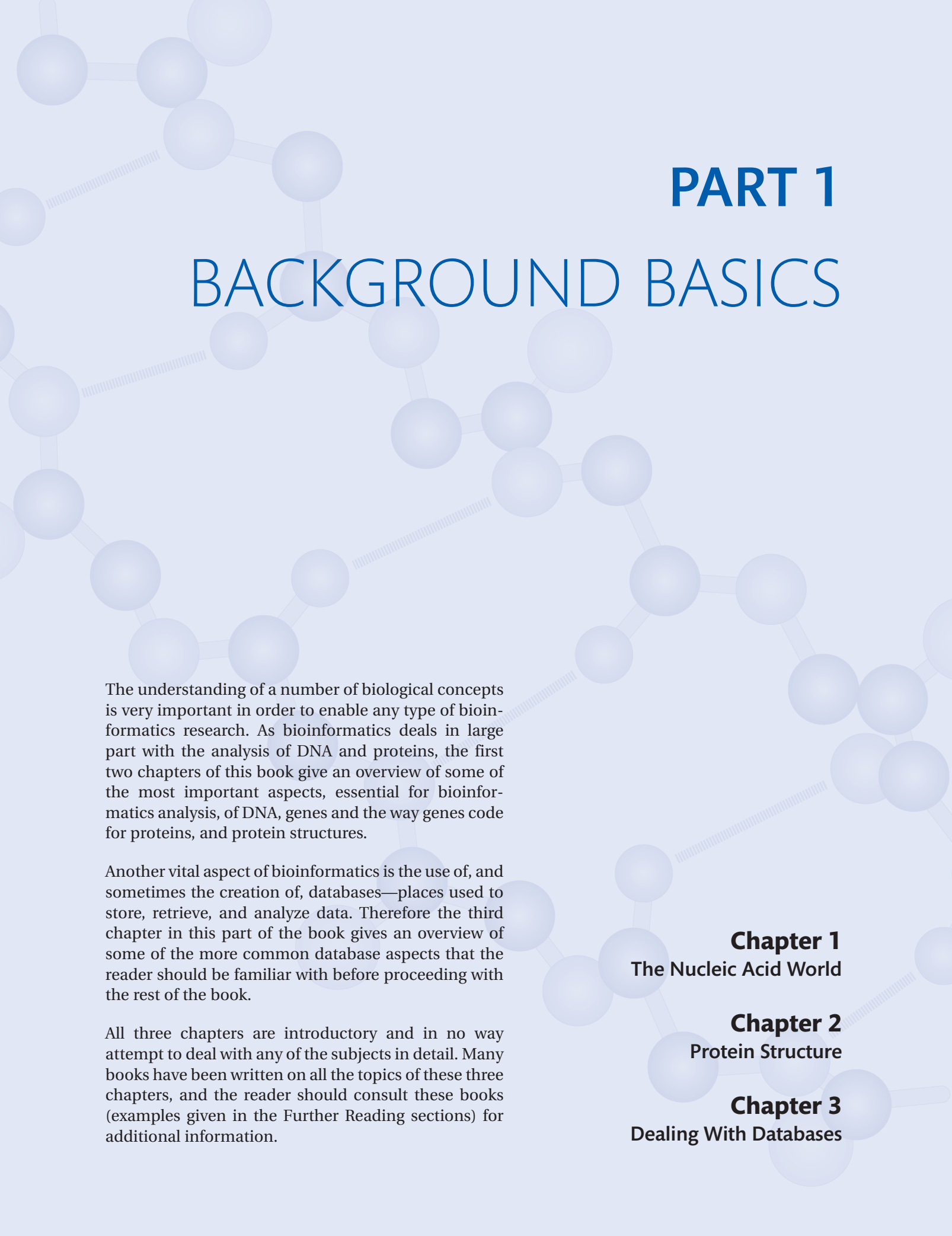
The self-organizing tree algorithm (SOTA) determines the number of clusters required	648	17.4 Storing and Running System Models	689
Biclustering identifies a subset of similar expression level patterns occurring in a subset of the samples	649	Specialized programs make simulating systems easier	691
The validity of clusters is determined by independent methods	650	Standardized system descriptions aid their storage and reuse	692
16.4 Statistical Analysis can Quantify the Significance of Observed Differential Expression	651	Summary	692
<i>t</i> -tests can be used to estimate the significance of the difference between two expression levels	654	Further Reading	693
Nonparametric tests are used to avoid making assumptions about the data sampling	656	APPENDICES Background Theory	
Multiple testing of differential expression requires special techniques to control error rates	657	Appendix A: Probability, Information, and Bayesian Analysis	
16.5 Gene and Protein Expression Data Can be Used to Classify Samples	659	Probability Theory, Entropy, and Information	695
Many alternative methods have been proposed that can classify samples	660	Mutually exclusive events	695
Support vector machines are another form of supervised learning algorithms that can produce classifiers	661	Occurrence of two events	696
Summary	662	Occurrence of two random variables	696
Further Reading	664	Bayesian Analysis	697
Chapter 17 Systems Biology		Bayes' theorem	697
17.1 What is a System?	669	Inference of parameter values	698
A system is more than the sum of its parts	669	Further Reading	699
A biological system is a living network	670	Appendix B: Molecular Energy Functions	
Databases are useful starting points in constructing a network	671	Force Fields for Calculating Intra- and Intermolecular Interaction Energies	701
To construct a model more information is needed than a network	672	Bonding terms	702
There are three possible approaches to constructing a model	674	Nonbonding terms	704
Kinetic models are not the only way in systems biology	678	Potentials used in Threading	706
17.2 Structure of the Model	679	Potentials of mean force	706
Control circuits are an essential part of any biological system	680	Potential terms relating to solvent effects	707
The interactions in networks can be represented as simple differential equations	680	Further Reading	708
17.3 Robustness of Biological Systems	683	Appendix C: Function Optimization	
Robustness is a distinct feature of complexity in biology	684	Full Search Methods	710
Modularity plays an important part in robustness	685	Dynamic programming and branch-and-bound	710
Redundancy in the system can provide robustness	686	Local Optimization	710
Living systems can switch from one state to another by means of bistable switches	688	The downhill simplex method	711
		The steepest descent method	711
		The conjugate gradient method	714
		Methods using second derivatives	714
		Thermodynamic Simulation and Global Optimization	715
		Monte Carlo and genetic algorithms	716
		Molecular dynamics	718
		Simulated annealing	719
		Summary	719
		Further Reading	719
		List of Symbols	721
		Glossary	734
		Index	751



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A faint, light blue molecular structure graphic is overlaid on the page. It consists of numerous spheres of varying sizes connected by lines of different thicknesses, some of which are dashed. The spheres have a slight gradient and shadow, giving them a three-dimensional appearance. The overall pattern is complex and organic, resembling a network or a molecular model.

PART 1

BACKGROUND BASICS

The understanding of a number of biological concepts is very important in order to enable any type of bioinformatics research. As bioinformatics deals in large part with the analysis of DNA and proteins, the first two chapters of this book give an overview of some of the most important aspects, essential for bioinformatics analysis, of DNA, genes and the way genes code for proteins, and protein structures.

Another vital aspect of bioinformatics is the use of, and sometimes the creation of, databases—places used to store, retrieve, and analyze data. Therefore the third chapter in this part of the book gives an overview of some of the more common database aspects that the reader should be familiar with before proceeding with the rest of the book.

All three chapters are introductory and in no way attempt to deal with any of the subjects in detail. Many books have been written on all the topics of these three chapters, and the reader should consult these books (examples given in the Further Reading sections) for additional information.

Chapter 1
The Nucleic Acid World

Chapter 2
Protein Structure

Chapter 3
Dealing With Databases



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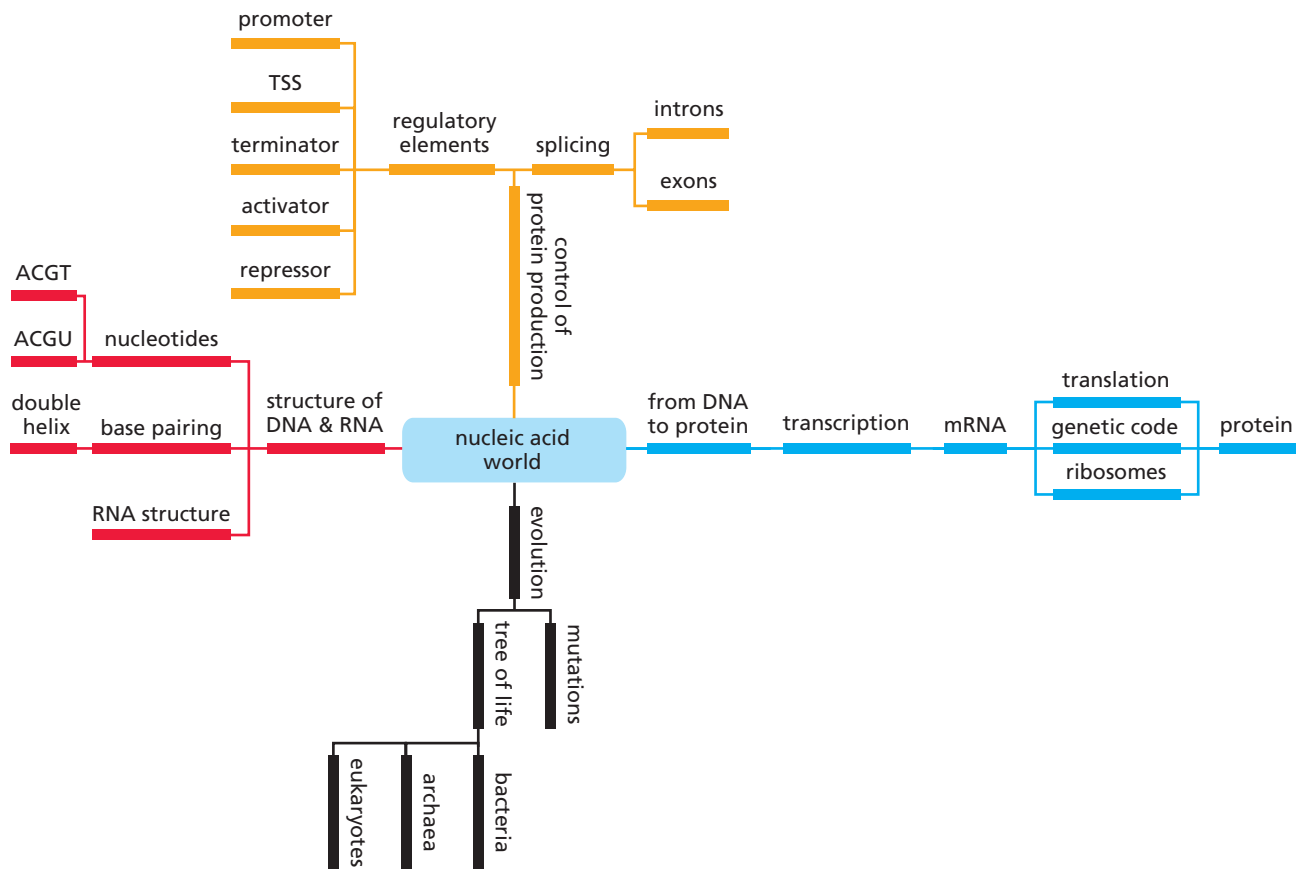
THE NUCLEIC ACID WORLD

When you have read Chapter 1, you should be able to:

- State the chemical structures of nucleic acids.
- Explain base-pairing and the double helix.
- Explain how DNA stores genetic information.
- Summarize the intermediate role of mRNA between DNA and proteins.
- Outline how mRNA is translated into protein by ribosomes.
- Outline how gene control is exercised by binding to short nucleotide sequences.
- Show that eukaryotic mRNA often has segments (introns) removed before translation.
- Discuss how all life probably evolved from a single common ancestor.
- Summarize how evolution occurs by changes to the sequence of genomic DNA.

It is amazing to realize that the full diversity of life on this planet—from the simplest bacterium to the largest mammal—is captured in a linear code inside all living cells. In almost exactly the way the vivid detail of a musical symphony or a movie can be digitally recorded in a binary code, so the four base units of the DNA molecule capture and control all the complexity of life. The crucially important discovery of the link between DNA, proteins, and the diversity of life came during the twentieth century and brought about a revolution in the understanding of genetics. Since that time we have amassed increasing amounts of information on the sequences of DNA, RNA, and proteins and the great variety of these molecules in cells under differing conditions. The growth of this information threatens to outstrip our ability to analyze it. It is one of the key challenges facing biologists today to organize, study, and draw conclusions from all this information: the patterns within the sequences and experimental data, the structure and the function of the various types of molecules, and how everything interacts to produce a correctly functioning organism. Bioinformatics is the name we give to this study and our aim is to use it to obtain greater understanding of living systems.

Information about nucleic acids and proteins is the raw material of bioinformatics, and in the first two chapters of this book we will briefly review these two types of biomolecules and their complementary roles in reproducing and maintaining life. This review is not a comprehensive introduction to cell and molecular biology; for that you should consult one of the excellent introductory textbooks of cell and molecular biology listed in Further Reading (p. 24). Rather, it is intended simply as a reminder of those aspects of nucleic acids and proteins that you will need as a background to the bioinformatics methods described in the rest of the book. More



Mind Map 1.1

A mind map schematic of the topics covered in this chapter and divided, in general, according to the topic sections. This is to help you visualize the topics, understand the structure of the chapter, and memorize the important elements.

information about the biological context of the bioinformatics problems we discuss is also given in the biology boxes and glossary items throughout the book.

This chapter will deal with the nucleic acids—**deoxyribonucleic acid (DNA)** and **ribonucleic acid (RNA)**—and how they encode proteins, while the structure and functions of proteins themselves will be discussed in Chapter 2. In these two chapters we shall also discuss how DNA changes its information-coding and functional properties over time as a result of the processes of **mutation**, giving rise to the enormous diversity of life, and the need for bioinformatics to understand it.

The main role of DNA is information storage. In all living cells, from unicellular bacteria to multicellular plants and animals, DNA is the material in which genetic instructions are stored and is the chemical structure in which these instructions are transmitted from generation to generation; all the information required to make and maintain a new organism is stored in its DNA. Incredibly, the information required to reproduce even very complex organisms is stored on a relatively small number of DNA molecules. This set of molecules is called the organism's **genome**. In humans there are just 46 DNA molecules in most cells, one in each chromosome. Each DNA molecule is copied before cell division, and the copies are distributed such that each daughter cell receives a full set of genetic information. The basic set of 46 DNA molecules together encode everything needed to make a human being. (We will skip over the important influence of the environment and the nature–nurture debate, as they are not relevant to this book.)

Proteins are manufactured using the information encoded in DNA and are the molecules that direct the actual processes on which life depends. Processes essential to life, such as energy metabolism, biosynthesis, and intercellular communication, are all carried out through the agency of proteins. A few key processes such as the

synthesis of proteins also involve molecules of RNA. Ignoring for a moment some of the complexity that can occur, a gene is the information in DNA that directs the manufacture of a specific protein or RNA molecular form. As we shall see, however, the organization of genes within the genome and the structure of individual genes are somewhat different in different groups of organisms, although the basic principles by which genes encode information are the same in all living things.

Organisms are linked together in evolutionary history, all having evolved from one or a very few ancient ancestral life forms. This process of evolution, still in action, involves changes in the genome that are passed to subsequent generations. These changes can alter the protein and RNA molecules encoded, and thus change the organism, making its survival more, or less, likely in the circumstances in which it lives. In this way the forces of evolution are inextricably linked to the genomic DNA molecules.

1.1 The Structure of DNA and RNA

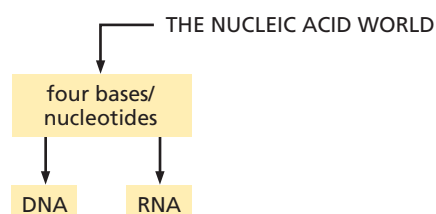
Considering their role as the carrier of genetic information, DNA molecules have a relatively simple chemical structure. They are linear polymers of just four different **nucleotide** building blocks whose differences are restricted to a substructure called the **base** (see Flow Diagram 1.1). For efficient encoding of information, one might have expected there to be numerous different bases, but in fact there are only four. This was one of the reasons why the true role of DNA as the carrier of genetic information was not appreciated until the 1940s, long after the role of the chromosomes in heredity was apparent. But although chemically simple, genomic DNA molecules are immensely long, containing millions of bases each, and it is the order of these bases, the **nucleotide sequence** or **base sequence** of DNA, which encodes the information for making proteins.

The three-dimensional structure of DNA is also relatively simple, involving regular double helices. There are also larger-scale regular structures, but it has been clearly established that the information content of DNA is held at the level of the base sequence itself.

RNA molecules are also linear polymers, but are much smaller than genomic DNA. Most RNA molecules also contain just four different base types. However, several classes of RNA molecules are known, some of which have a small proportion of other bases. RNA molecules tend to have a less-regular three-dimensional structure than DNA.

DNA is a linear polymer of only four different bases

The building blocks of DNA and RNA are nucleotides, which occur in different but chemically related forms. A nucleotide consists of a nitrogen-containing base that is linked to a five-carbon sugar ring at the 1' position on the ring, which also carries a phosphate group at the 5' position (see Figure 1.1A). In DNA the sugar is deoxyribose, in RNA it is ribose, the difference between the two being the absence or presence,



Flow Diagram 1.1

The key concept introduced in this section is that DNA and RNA are composed of subunits called nucleotides, with only four different nucleotide types in a molecule, but a different set of four nucleotides in each of the two types of nucleic acid.

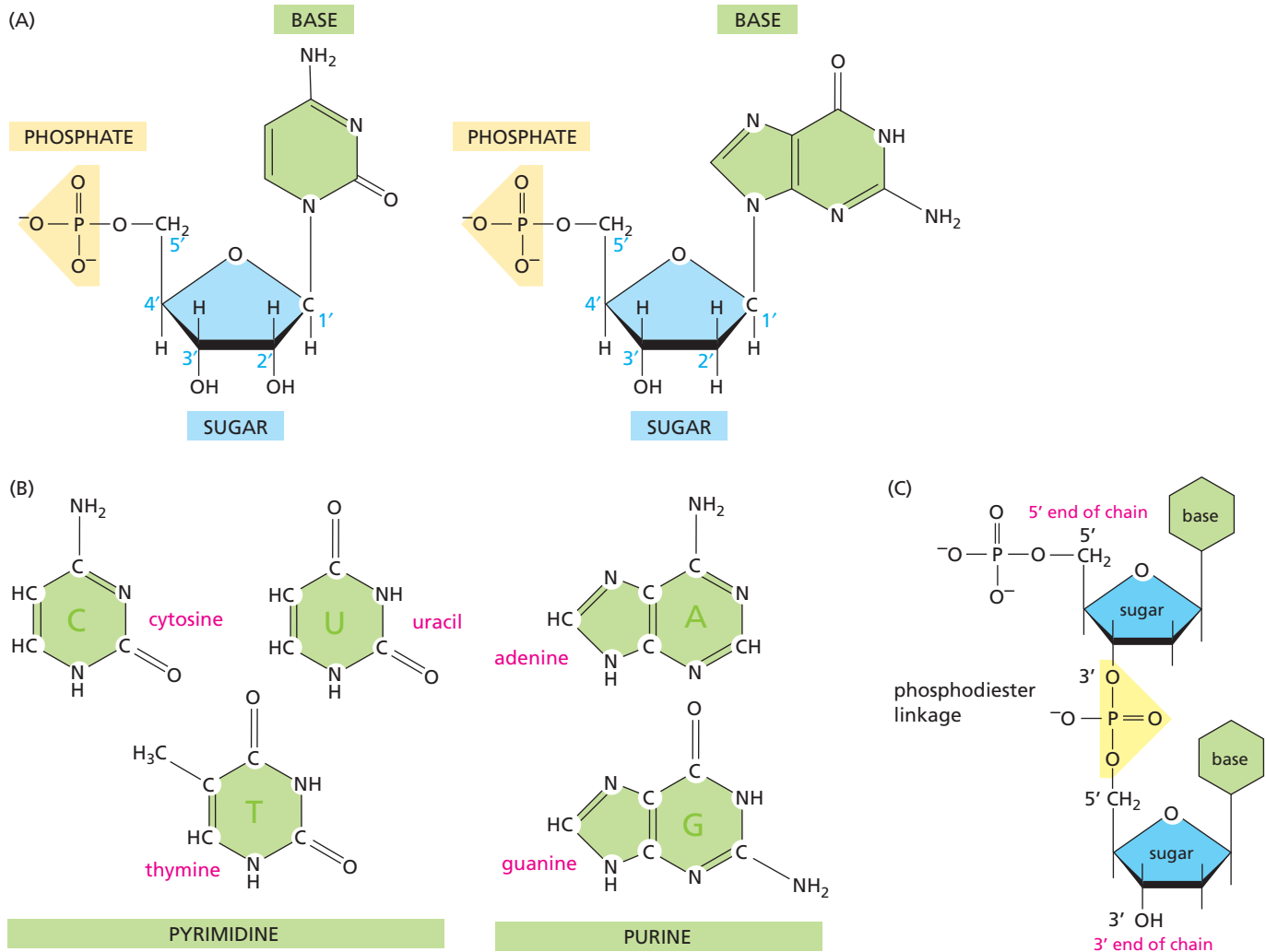
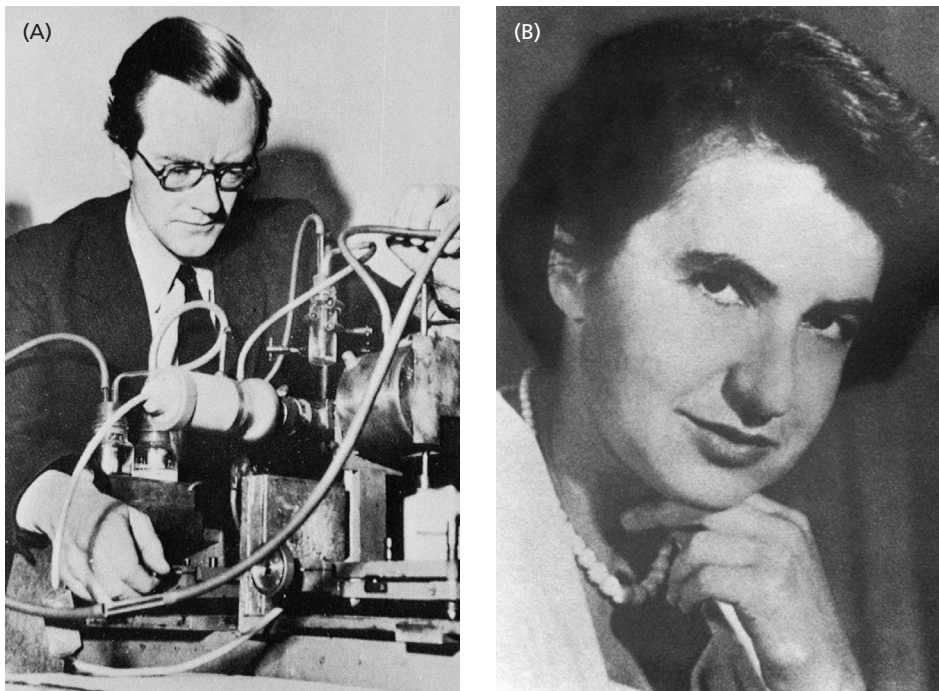


Figure 1.1
The building blocks of DNA and RNA. (A) Left, cytidylate (ribo-CMP); right, deoxyguanylate (dGMP). Each consists of three main parts: a phosphate group, a pentose sugar, and a base. It is the base part that is responsible for the differences in the nucleotides. (B) The bases fall into two groups: the purines and the pyrimidines. The purines consist of a 6- plus a 5-membered nitrogen-containing ring while the pyrimidines have only one 6-membered nitrogen-containing ring. (C) It is the phosphate group that is involved in linking the building blocks together by a phosphodiester linkage in DNA. (From B. Alberts et al., *Molecular Biology of the Cell*, 4th ed. New York: Garland Science, 2002.)

respectively, of a hydroxyl group at the 2' position. These two types of nucleotide are referred to as **deoxyribonucleotides** and **ribonucleotides**, respectively. Apart from this, the only difference between nucleotides is the base, which is a planar ring structure, either a pyrimidine or a purine (see Figure 1.1B). In DNA only four different bases occur: the purines guanine (G) and adenine (A) and the pyrimidines cytosine (C) and thymine (T). In most forms of RNA molecule there are also just four bases, three being the same as in DNA, but thymine is replaced by the pyrimidine uracil (U).

Each nucleic acid chain, or strand, is a linear polymer of nucleotides linked together by phosphodiester linkages through the phosphate on one nucleotide and the hydroxyl group on the 3' carbon on the sugar of another (see Figure 1.1C). This process is carefully controlled in living systems so that a chain is exclusively made with either deoxyribonucleotides (DNA) or ribonucleotides (RNA). The resulting chain has one end with a free phosphate group, which is known as the **5' end**, and one end with a free 3' hydroxyl group, which is known as the **3' end**. The base sequence or nucleotide sequence is defined as the order of the nucleotides along the chain from the 5' to the 3' end. It is normal to use the one-letter code given above to identify the bases, starting at the 5' end, for example AGCTTAG.

There are instances of bases within a nucleic acid chain being modified in a living cell. Although relatively rare, they can be of great interest. In the case of DNA, in vertebrates cytosine bases can be methylated (addition of a $-CH_3$ group). This can

**Figure 1.2**

Two scientists whose work was influential on James Watson and Francis Crick when they elucidated the structure of DNA.

(A) Maurice Wilkins.

(B) Rosalind Franklin. (A and B courtesy of Science Photo Library.)

result in certain genes being rendered inactive, and is involved in the newly discovered phenomenon known as **genomic imprinting**, where the change that may occur in the offspring depends on whether the gene is maternally or paternally inherited. The class of RNA molecules called tRNA (see Section 1.2) have modifications to approximately 10% of their bases, and many different modifications have been seen involving all base types. These changes are related to the function of the tRNA.

Two complementary DNA strands interact by base-pairing to form a double helix

The key discovery for which the Nobel Prize was awarded to James Watson and Francis Crick, who drew on the work of Rosalind Franklin and others (see Figure 1.2), was the elucidation in 1953 of the double-helical structure of the DNA molecule, in which two strands of DNA are wound around each other and are held together by hydrogen bonding between the bases of each strand. Their structure was an early example of model building of large molecules, and was based on knowledge of the chemical structure of the constituent nucleotides and experimental X-ray diffraction data on DNA provided by, among others, Maurice Wilkins, who was also awarded a share in the Nobel Prize. (Current methods of model building as applied to proteins are discussed in Chapter 13.)

All the bases are on the inside of the double helix, with the phosphate-linked sugars forming a backbone on the outside (see Figure 1.3A). Crucial to Watson and Crick's success was their realization that the DNA molecules contained two strands and that the base-pairing follows a certain set of rules, now called **Watson–Crick base-pairing**, in which a specific purine pairs only with a specific pyrimidine: A with T, and C with G. Each strand of a DNA double helix therefore has a base sequence that is **complementary** to the base sequence of its partner strand. The bases interact through hydrogen bonding; two hydrogen bonds are formed in a T–A base pair, and three in a G–C base pair (see Figure 1.3B). This complementarity means that if you know the base sequence of one strand of DNA, you can deduce the base sequence of the other. Note, however, that the two strands are antiparallel, running in opposite directions, so that the complementary sequence to AAG is not TTC but CTT (see Figure 1.3A).

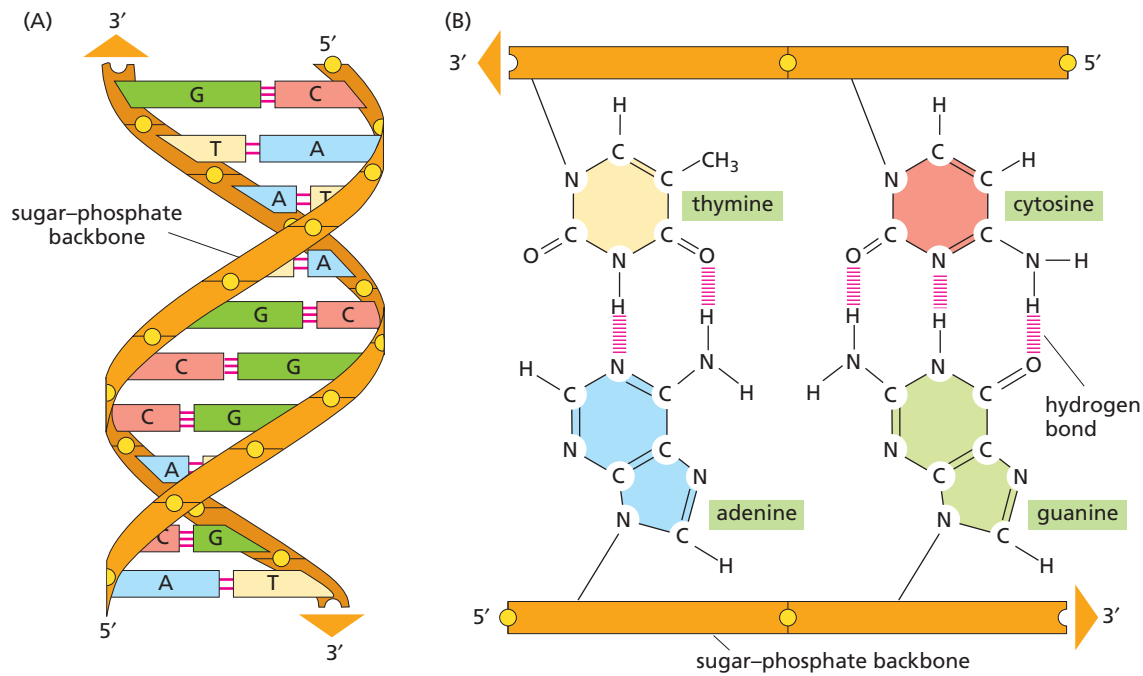
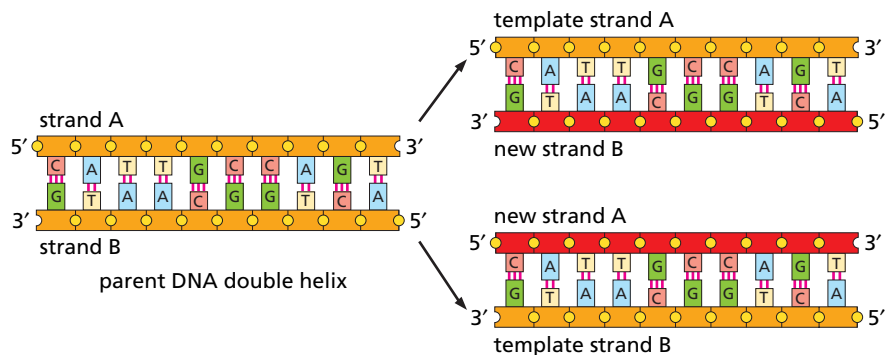


Figure 1.3
The double helical structure of DNA. (A) DNA exists in cells mainly as a two-stranded coiled structure called the double helix. (B) The two strands of the helix are held together by hydrogen bonds (shown as red lines) between the bases; these bonds are referred to as base-pairing. (From B. Alberts et al., *Molecular Biology of the Cell*, 4th ed. New York: Garland Science, 2002.)

Hydrogen bonds are noncovalent bonds, which in biomolecules are much weaker than covalent bonds, often involving a binding energy of only $5\text{--}20\text{ kJ mol}^{-1}$. As a consequence the two strands of the DNA double helix can be relatively easily separated. There are a number of processes in which this strand separation is required, for example when the molecules of DNA are copied as a necessary prelude to cell division, a process called **DNA replication**. The separated strands each serve as a template on which a new complementary strand is synthesized by an enzyme called DNA polymerase, which moves along the template successively matching each base in the template to the correct incoming nucleotide and joining the nucleotide to the growing new DNA strand. Thus each double helix gives rise to two new identical DNA molecules (see Figure 1.4). Genetic information is thus preserved intact through generations of dividing cells. The actual biochemical process of DNA replication is complex, involving a multitude of proteins, and will not concern us here, but at its heart is this simple structural property of complementarity.

One of the truly remarkable features of DNA in living systems is how few errors there are in the complementarity of the two strands. The error rate is approximately 1 base in 10^9 . This is very important, as it is vital to transmit the genome as accurately as possible to subsequent generations. Many alternative base-pairings are possible, although these are less favorable than the Watson-Crick base-pairing. These energies can be used to predict the expected rate of incorrect base-pairing. However, genomic DNA shows a much lower error rate than expected. This is a

Figure 1.4
DNA replication. DNA duplicates itself by first separating the two strands, after which enzymes come and build complementary strands base-paired to each of the old strands. (From B. Alberts et al., *Molecular Biology of the Cell*, 4th ed. New York: Garland Science, 2002.)



result of the controlled way in which the DNA polymerase builds the second strand, including mechanisms for checking and correcting erroneous base-pairings.

DNA strands can pair with a DNA or RNA strand of complementary sequence to make a double-stranded DNA or DNA/RNA hybrid. This property forms the basis of a set of powerful experimental molecular biology techniques. Known generally as nucleic acid **hybridization**, it is exploited in applications such as DNA **microarrays** (described in Chapter 15), *in situ* hybridization to detect the activity of specific genes in cells and tissues, and fluorescence *in situ* hybridization (FISH) for visually locating genes on chromosomes.

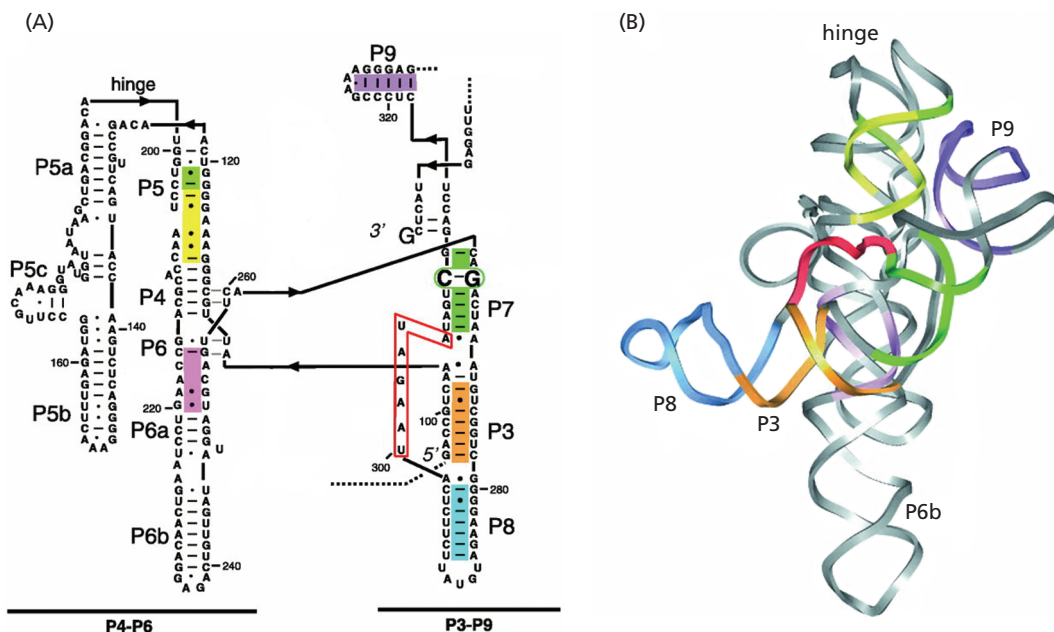
RNA molecules are mostly single stranded but can also have base-pair structures

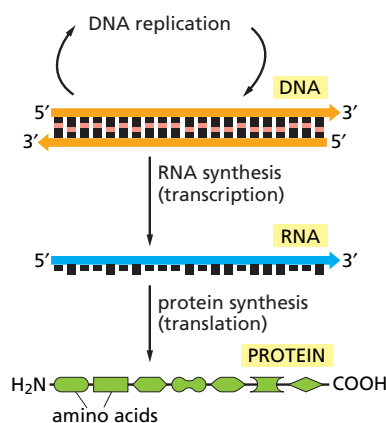
In contrast to DNA, almost all RNA molecules in living systems are single stranded. Because of this, RNA has much more structural flexibility than DNA, and some RNAs can even act as enzymes, catalyzing a particular chemical reaction. The large number of hydrogen bonds that can form if the RNA molecule can double back on itself and create base-pairing makes such interactions highly energetically favorable. Often, short stretches of an RNA strand are almost or exactly complementary to other stretches nearby. Two such stretches can interact with each other to form a double-helix structure similar to that of DNA, with loops of unpaired nucleotides at the ends of the helices. The interactions stabilizing such helices are often not standard Watson–Crick pairing. These structures are often referred to as **RNA secondary structure** by analogy with protein secondary structure described in Section 2.1.

It is likely that all RNA molecules in a cell have some regions of stable three-dimensional structure due to limited base-pairing, the unpaired regions being very flexible and not taking up any specific structure. In some cases RNA folds up even further, packing helices against each other to produce a molecule with a rigid structure. The three-dimensional structures that result have a functional role for many RNA molecules. An example of such structures is shown in Figure 1.5, and is fairly typical in that a significant fraction of the sequence is involved in base-pairing interactions. The prediction of ribosomal secondary structures and three-dimensional structure is not covered in this book, but introductory references are given in Further Reading. Even more complex interactions are possible, one example of

Figure 1.5

Example three-dimensional structure of RNA. The structure shown is the Tetrahymena ribozyme. (A) Nucleotide sequence showing base-pairing and loops. (B) Three-dimensional structure as determined by x-ray crystallography. Entry 1GRZ of MSD database. (From B.L. Golden et al., A preorganized active site in the crystal structure of Tetrahymena ribozyme, *Science* 282:259–264, 1998. Reprinted with permission from AAAS.)



**Figure 1.6**

Central dogma. This is the basic scheme of the transcription of DNA to RNA (mRNA) which is then translated to proteins. Many enzymes are involved in these processes. (From B. Alberts et al., *Molecular Biology of the Cell*, 4th ed. New York: Garland Science, 2002.)

which is illustrated in Figure 1.5A. To the right of the P4 and P6 labels are four bases that are involved in two separate base-pairings, in each case forming a base triplet. Such interactions have been observed on several occasions, as have interactions involving four bases.

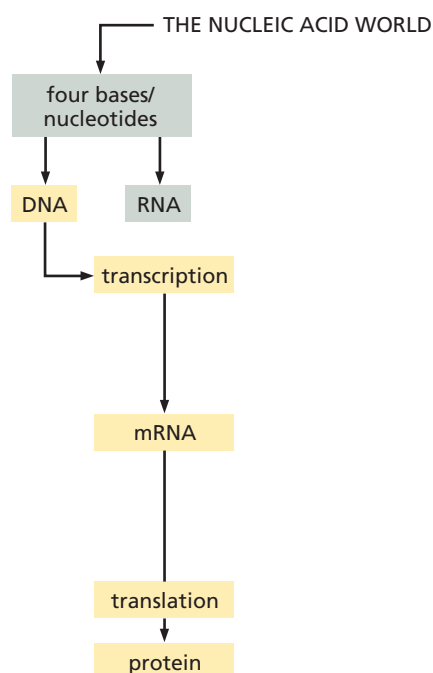
1.2 DNA, RNA, and Protein: The Central Dogma

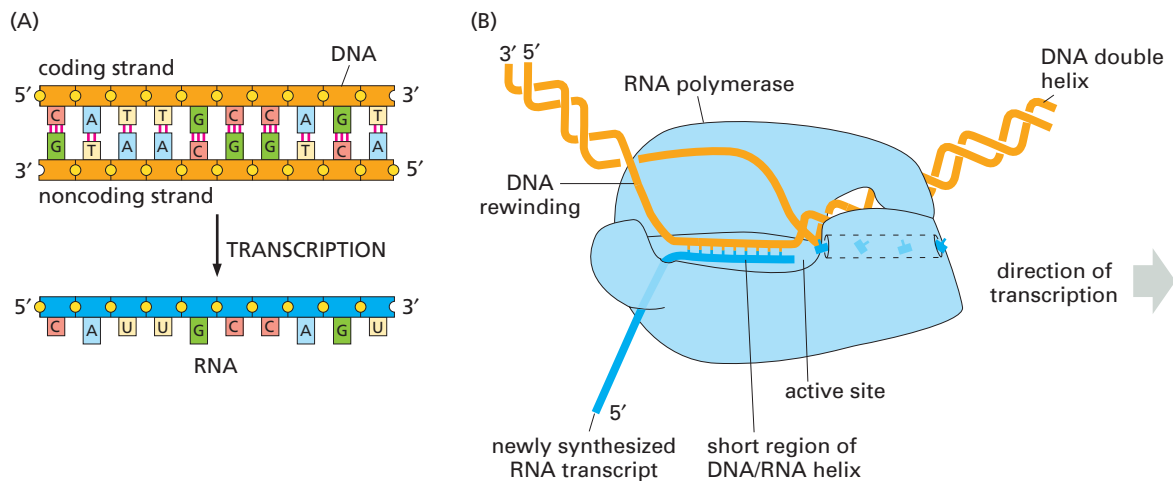
There is a key relationship between DNA, RNA, and the synthesis of proteins, which is often referred to as the central dogma of molecular biology. According to this concept, there is essentially a single direction of flow of genetic information from the DNA, which acts as the information store, through RNA molecules from which the information is translated into proteins (see Figure 1.6 and Flow Diagram 1.2). This basic scheme holds for all known forms of life, although there is some variation in the details of the processes involved. The proteins are the main working components of organisms, playing the major role in almost all the key processes of life. However, not all the genetic information in the DNA encodes proteins. Molecules such as RNA can also be the end product, and other regions of genomes have as yet no known function or product. The genomic DNA encodes all molecules that are necessary for life, whether they are the proteins (such as enzymes) involved in nearly all biological activities or RNA important for translation and transcription.

For example, all the information needed to build and maintain a human being is contained in just 23 pairs of DNA molecules, comprising the chromosomes of the human genome. These molecules are amongst the largest and longest known, the smallest having 47 million bases and the largest 247 million bases, with the entire human genome being composed of approximately 3 billion bases. Even bacterial genomes, which are much smaller than this, tend to have several million bases. The DNA of each chromosome encodes hundreds to thousands of proteins, depending on the chromosome, each protein being specified by a distinct segment of DNA. In simple terms, this segment is the gene for that protein. In practice, a **gene** is considered also to include surrounding regions of noncoding DNA that act as control regions, as will be described in Section 1.3. These are involved in determining whether the gene is active—in which case the protein is produced—or is inactive.

Flow Diagram 1.2

The key concept introduced in this section is that DNA is transcribed into mRNA which is then translated to make protein molecules. The shaded boxes show the concepts introduced in the previous section, the yellow-colored boxes refer to concepts in this section. This color coding is used throughout the book.





The sequence of the protein-coding region of a gene carries information about the protein sequence in a coded form called the genetic code. This DNA sequence is decoded in a two-stage process (see Figure 1.6), the stages being called transcription and translation. Both stages involve RNA molecules and will now be described.

DNA is the information store, but RNA is the messenger

The information encoded in DNA is expressed through the synthesis of other molecules; it directs the formation of RNA and proteins with specific sequences. As is described in detail in Chapter 2, proteins are linear polymers composed of another set of chemical building blocks, the **amino acids**. There are some 20 different amino acids in proteins, and their varied chemistry (see Figure 2.3) makes proteins much more chemically complex and biochemically versatile molecules than nucleic acids.

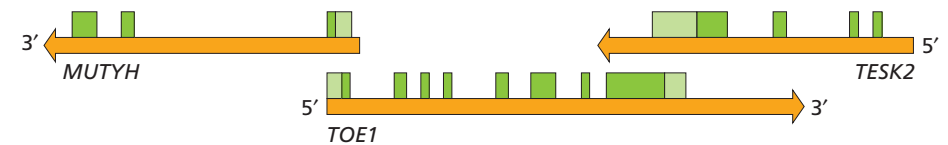
The sequence of bases in the DNA of a gene specifies the sequence of amino acids in a protein chain. The conversion does not occur directly, however. After a signal to switch on a gene is received, a single-stranded RNA copy of the gene is first made in a process called **transcription**. Transcription is essentially similar to the process of DNA replication, except that only one of the DNA strands acts as a template in this case, and the product is RNA not DNA (see Figure 1.7). RNA synthesis is catalyzed by enzymes called RNA polymerases, which, like DNA polymerases, move along the template, matching incoming ribonucleotides to the bases in the template strand and joining them together to make an RNA chain. Only the relevant region of DNA is transcribed into RNA, therefore the RNA is a much smaller molecule than the DNA it comes from. So while the DNA carries information about many proteins, the RNA carries information from just one part of the DNA, usually information for a single protein. RNA transcribed from a protein-coding gene is called **messenger RNA (mRNA)** and it is this molecule that directs the synthesis of the protein chain, in the process called translation, which will be described in more detail below. When a gene is being transcribed into RNA, which is in turn directing protein synthesis, the gene is said to be **expressed**. Expression of many genes in a cell or a set of cells can be measured using DNA or RNA expression microarrays (see Chapter 15).

Only one of the DNA strands in any given gene is transcribed into RNA. As the RNA must have a **coding** sequence that can be correctly translated into protein, the DNA strand that acts as the physical template for RNA synthesis does not carry the coding sequence itself, but its complement. It is therefore known as the **noncoding strand** or **anticoding** or **antisense strand**. The sequence of the other, non-template DNA strand is identical to that of the messenger RNA (with T replacing U), and this strand is called the coding or **sense strand**. This is the DNA sequence that is written out to represent a gene, and from which the protein sequence can be

Figure 1.7

Transcription. (A) One strand of the DNA is involved in the synthesis of an RNA strand complementary to the strand of the DNA. (B) The enzyme RNA polymerase is involved in the transcription process. It reads the DNA and recruits the correct building blocks of RNA to string them together based on the DNA code. (From B. Alberts et al., *Molecular Biology of the Cell*, 4th ed. New York: Garland Science, 2002.)

Figure 1.8
Overlapping genes. A schematic showing the overlap of three human genes. The dark green boxes show protein-coding regions of the DNA (exons) while the light green boxes show regions that are untranslated. (Adapted from V. Veeramachaneni et al., Mammalian overlapping genes: the comparative perspective, *Genome Res.* 14:280–286, 2004.)



deduced according to the rules of the genetic code. Note that the polymerase transcribes the anticoding strand in the direction from 3' to 5', so that the mRNA strand is produced from the 5' to the 3' end.

Although only one segment of the DNA strand is transcribed for any given gene, it is also possible for genes to overlap so that one or both strands at the same location encode parts of different proteins. This most commonly occurs in **viruses** as a means of packing as much information as possible into a very small genome. However, overlapping genes can also occur in mammals; recently 774 pairs of overlapping genes were identified in the human genome (see Figure 1.8).

The genomic DNA sequence contains more information than just the protein sequences. The transcriptional apparatus has to locate the sites where gene transcription should begin, and when to transcribe a given gene. At any one time, a cell is only expressing a few thousand of the genes in its genome. To accomplish this regulated gene expression, the DNA contains control sequences in addition to coding regions. We shall return to these regulatory regions later, after first discussing the details of the coding of protein sequences.

Table 1.1
Standard genetic code. The corresponding amino acid is given next to each codon. The three-letter amino acid code defined in Table 2.1 is used.

Messenger RNA is translated into protein according to the genetic code

The **genetic code** refers to the rules governing the correspondence of the base sequence in DNA or RNA to the amino acid sequence of a protein. The essential

First letter of the codon	5' end	Second letter of the codon								3' end
		U		C		A		G		
	U	UUU	Phe	UCU	Ser	UAU	Tyr	UGU	Cys	U
		UUC	Phe	UCC	Ser	UAC	Tyr	UGC	Cys	C
		UUA	Leu	UCA	Ser	UAA	Stop	UGA	Stop	A
		UUG	Leu	UCG	Ser	UAG	Stop	UGG	Trp	G
	C	CUU	Leu	CCU	Pro	CAU	His	CGU	Arg	U
		CUC	Leu	CCC	Pro	CAC	His	CGC	Arg	C
		CUA	Leu	CCA	Pro	CAA	Gln	CGA	Arg	A
		CUG	Leu	CCG	Pro	CAG	Gln	CGG	Arg	G
A	AUU	Ile	ACU	Thr	AAU	Asn	AGU	Ser	U	
	AUC	Ile	ACC	Thr	AAC	Asn	AGC	Ser	C	
	AUA	Ile	ACA	Thr	AAA	Lys	AGA	Arg	A	
	AUG	Met	ACG	Thr	AAG	Lys	AGG	Arg	G	
G	GUU	Val	GCU	Ala	GAU	Asp	GGU	Gly	U	
	GUC	Val	GCC	Ala	GAC	Asp	GGC	Gly	C	
	GUA	Val	GCA	Ala	GAA	Glu	GGA	Gly	A	
	GUG	Val	GCG	Ala	GAG	Glu	GGG	Gly	G	
									Third letter of the codon	

problem is how a code of four different bases in nucleic acids can specify proteins made up of 20 different types of amino acids. The solution is that each amino acid is encoded by a set of three consecutive bases. The three-base sets in RNA are called **codons**, and genetic code tables conventionally give the genetic code in terms of RNA codons. The standard genetic code is given in Table 1.1. From this table you can see that the genetic code is **degenerate**, in that most amino acids can be specified by more than one codon. The degeneracy of the genetic code means that you can deduce the protein sequence from a DNA or RNA sequence, but you cannot unambiguously deduce a nucleic acid sequence from a protein sequence.

There are three codons that do not encode an amino acid but instead signal the end of a protein-coding sequence. These are generally called the stop codons. The signal to start translating is more complex than a single codon, but in most cases translation starts at an AUG codon, which codes for the amino acid methionine. This initial methionine residue is often subsequently removed from the newly synthesized protein. In general, all life uses the same genetic code, but there are some exceptions and care should be taken to use the appropriate code when deducing amino acid sequences from DNA sequences. The code tables can be accessed through many of the sequence databases and through the National Center for Biotechnology Information (NCBI) Taxonomy section.

The translation of bases to amino acids occurs in nonoverlapping sets of three bases. There are thus three possible ways to translate any nucleotide sequence, depending on which base is chosen as the start. These three **reading frames** give three different protein sequences (see Figure 1.9). In the actual translation process the detailed control signals ensure that only the appropriate reading frame is translated into protein. When trying to predict protein-coding sequences in DNA sequences, information about the control signals is often lacking so that one needs to try six possible reading frames, three for each DNA strand. Usually, only one of these reading frames will produce a functional protein. Proteins tend (with notable exceptions) to be at least 100 amino acids in length. The incorrect reading frames often have a stop codon, which results in a much shorter translated sequence. When analyzing bacterial genome sequences, for example, to predict protein-coding sequences, reading frames are identified that give a reasonable length of uninterrupted protein code, flanked by appropriate start and stop signals, called **open reading frames** or **ORFs**. Gene prediction is discussed in Chapters 9 and 10.

Translation involves transfer RNAs and RNA-containing ribosomes

RNAs have a variety of roles in cells but are mainly involved in the transfer of information from DNA and the use of this information to manufacture proteins. There are three main classes of RNA in all cells—messenger RNA (mRNA), **ribosomal RNA (rRNA)**, and **transfer RNA (tRNA)**—as well as numerous smaller RNAs with a variety of roles, some of which we will encounter in this book. The role of mRNA has been described above. rRNAs and tRNAs are involved in the process of mRNA translation and protein synthesis.

The mRNA produced by transcription is translated into protein by ribosomes, large multimolecular complexes formed of rRNA and proteins, in a process called **translation**. Ribosomes consist of two main subunit complexes, one of which binds the mRNA (see Figure 1.10A). Amino acids do not recognize the codons in mRNA directly and their addition in the correct order to a new protein chain is mediated by the tRNA molecules, which transfer the amino acid to the growing protein chain when bound to the ribosome. These small tRNA molecules have a three-base **anticodon** at one end that recognizes a codon in mRNA, and at the other end a site to which the corresponding amino acid becomes attached by a specific enzyme. This system is the physical basis for the genetic code. There are different tRNAs corresponding to every

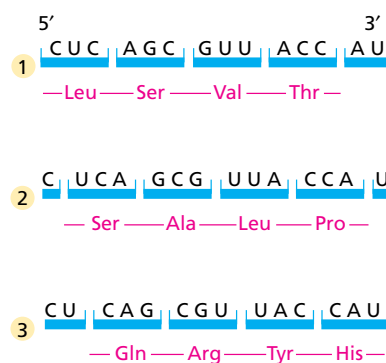
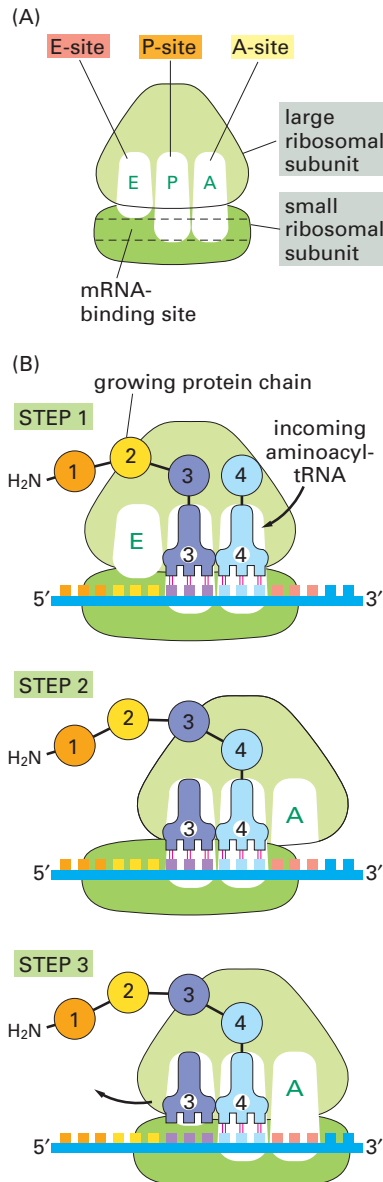


Figure 1.9

The three reading frames of a strand of mRNA. Each reading frame starts one nucleotide further giving rise to a different protein sequence. (From B. Alberts et al., *Molecular Biology of the Cell*, 4th ed. New York: Garland Science, 2002.)

Figure 1.10

Translation. (A) Schematic structure of the ribosome showing the binding site for the mRNA and the three tRNA binding sites. (B) A simplified view of the three steps of translation during which residue 4 is added to the C-terminal end of the growing protein chain. These steps are repeated for every residue in the protein. (From B. Alberts et al., *Molecular Biology of the Cell*, 4th ed. New York: Garland Science, 2002.)



amino acid, but some tRNA anticodons can bind to several similar codons. One common mechanism for this flexibility, which is called **wobble base-pairing**, involves the occurrence of modified bases in the anticodon. The human tRNA set has 48 different anticodons, but some species have less than 40. tRNA is a good example of an RNA molecule with a specific three-dimensional structure that is crucial for its function, and an example structure is shown in Figure 1.11. The identification of tRNA genes is described in Chapter 10, and the generalized sequence and secondary structure of tRNA molecules are shown in Figure 10.2.

A simplified outline of the process of translation is given in Figure 1.10B and we shall not go into details of it here. The decoding of mRNA starts at its 5' end, which is the end of the mRNA that is first synthesized at transcription. This is the key justification for the DNA sequence of a gene being conventionally written as the sequence of the sense or coding strand starting at its 5' end. The site of binding of the ribosome gradually moves along the mRNA toward the 3' end. The tRNA molecules bind to the ribosome and are the physical link between the mRNA and the growing protein chain. The enzymatic activity in the ribosome that joins amino acids together to form a protein chain is due to the rRNA, not to the ribosomal proteins.

1.3 Gene Structure and Control

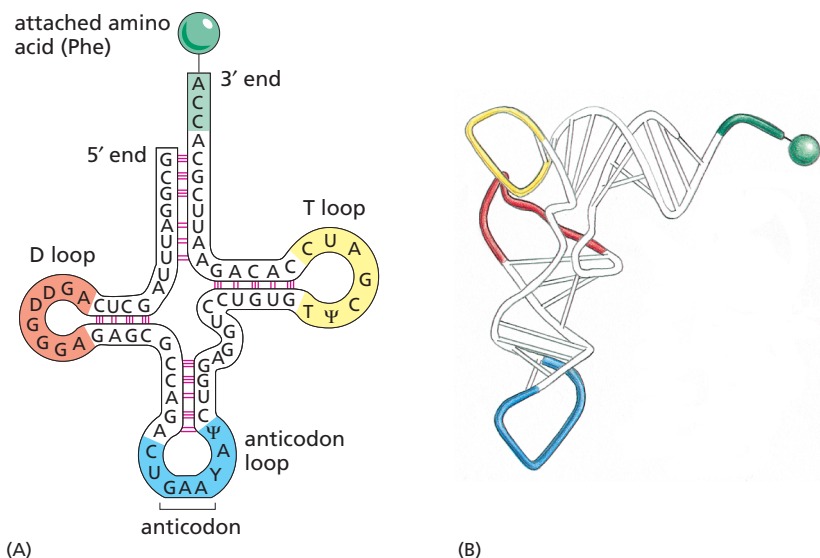
The description in the previous section of the details of the central dogma focused almost exclusively on the way the protein sequence information is stored in the genome and its conversion from nucleotides to amino acids. Little attention was paid to the ways in which these processes are controlled. Additionally, there are further complications, especially in **eukaryotes**, whose genes often have a more complicated structure including noncoding regions called introns between protein-coding regions. Expressing such genes involves an extra step in converting the DNA information to proteins, called RNA splicing, in which the mRNA produced initially is modified to remove the introns.

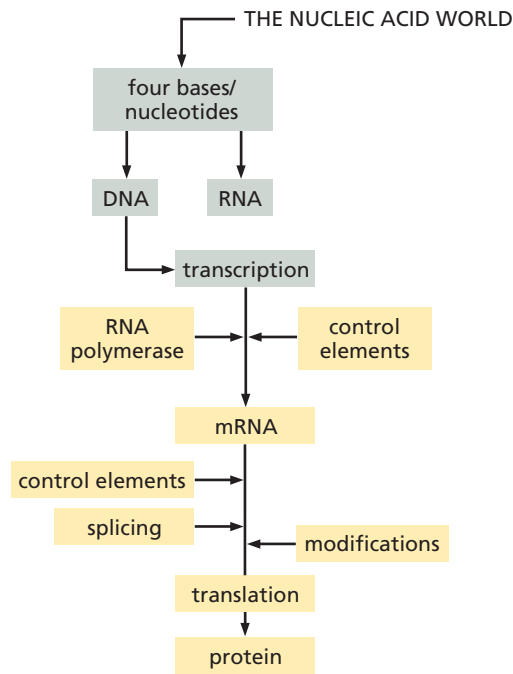
Figure 1.11

The structure of a tRNA molecule.

(A) A schematic of the Phe-tRNA molecule showing the arrangement of the base-pairing and loops. (B) The actual three-dimensional structure of the same

tRNA molecule colored to show the equivalent regions as in the schematic. (From B. Alberts et al., *Molecular Biology of the Cell*, 4th ed. New York: Garland Science, 2002.)



**Flow Diagram 1.3**

The key concept introduced in this section is that the processes of transcription and translation are subject to several distinct and sophisticated control mechanisms.

The regulation of many processes that interpret the information contained in a DNA sequence relies on the presence of short signal sequences in the DNA (see Flow Diagram 1.3). There are many different signal sequences, the general term for which is a **regulatory element**. For example, the molecules involved in transcription and translation require signals to identify where they should start and stop. In addition there are signals that are involved in the control of whether transcription occurs or not. The majority of these regulatory sequences are binding sites for specialized regulatory proteins that interact with the DNA to regulate processes such as transcription, RNA splicing, and translation. In order to interpret a DNA sequence correctly, it is important to have some understanding of the nature of these signals and their roles, although the precise mechanisms by which gene regulatory proteins act are not relevant to this book and are not discussed further. In this section we will survey the main aspects of the control of genes and how control structures occur in gene structure.

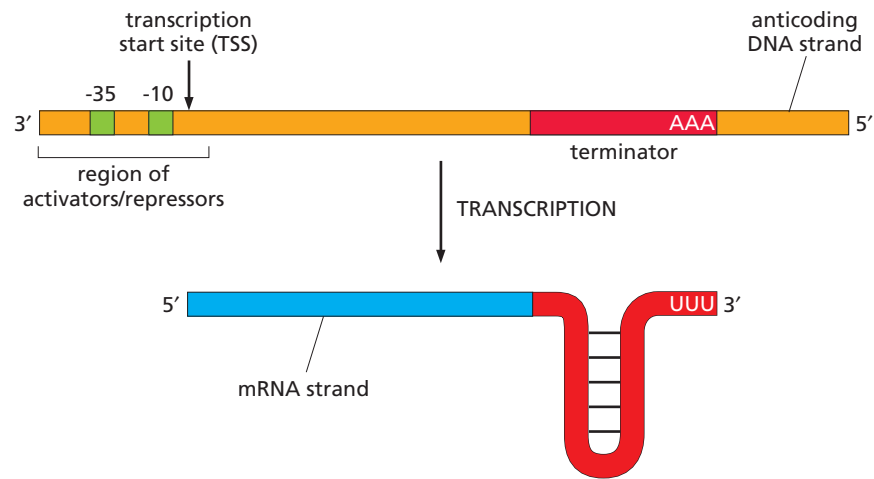
There are two distinct classes of organism—**prokaryotes** and eukaryotes—whose translation and transcription processes need to be considered separately. For a description of the general characteristics of prokaryotic and eukaryotic organisms see Section 1.4. We shall briefly review the general types of noncoding regulatory sequences found in prokaryotic and eukaryotic genes and introduce their roles. As prokaryotic control regions are, in general, less complicated than those of eukaryotic genes, we shall use them first to describe the basic signals that direct RNA polymerase to bind and start transcription in the appropriate place.

RNA polymerase binds to specific sequences that position it and identify where to begin transcription

The control regions in DNA at which RNA polymerase binds to initiate transcription are called **promoters**. RNA polymerase binds more tightly to these regions than to the rest of the DNA and this triggers the start of transcription. Additional proteins binding with the polymerase are also required in order to activate transcription. Bacterial promoters typically occur immediately before the position of the **transcription start site (TSS)** and contain two characteristic short sequences, or motifs, that are the same or very similar in the promoters for different genes.

Figure 1.12

The start and stop signals for prokaryotic transcription. The signals to start transcription are short nucleotide sequences that bind transcription enzymes. The signal to stop transcription is a short nucleotide sequence that forms a loop structure preventing the transcription apparatus from continuing.



One of these motifs is centered at approximately 10 bases before the start, conventionally written as -10 , and the other at approximately 35 bases before, written as -35 (see Figure 1.12). These two sequences are essential for the tight binding of the RNA polymerase, and they position it at the appropriate location and in the correct orientation to start transcription on the right strand and in the right direction. Sequences located before the start point of transcription are commonly called **upstream sequences**. Sequences located after the start point of transcription are called **downstream sequences**. This terminology is easy to remember if one thinks of the direction of transcription as the flow of a river.

One of the problems in finding promoters in DNA sequences is that the sequence outside the particular conserved motifs varies considerably, and even the motifs vary somewhat from gene to gene. Motifs like these are often described in terms of their **consensus sequence**, which is made up of the bases that are most frequently found at each position. In *Escherichia coli*, for example, the consensus sequence for the -10 motif is TATAAT, and that of the -35 motif is TTGACA. Furthermore, the separation between these two motifs can be between 15 and 19 bases. Note that by convention these are the sequences found on the coding DNA strand, whereas in reality the polymerase binds to double-stranded DNA, which also contains the complementary anticoding strand sequence. RNA polymerase binds to a variety of sequences but binds more strongly the closer the promoter is to a consensus sequence. The tighter the binding, the more frequently the region will be transcribed. Such variation in sequence is a common feature of many control sites, and makes it harder to locate them.

The termination of transcription is also controlled by sequence signals. In bacteria the **terminator signal** is distinct from the promoter in that it is active when transcribed to form the end of the mRNA strand. In the mRNA the terminator sequence produces two short stretches of complementary sequence that can base-pair to form an RNA double helix, usually involving at least four to five CG base pairs; this is followed, usually within five bases, by at least three consecutive U nucleotides (see Figure 1.12). The prokaryotic terminator sequence is more variable than the promoter sequence, and is usually not included in genome annotations, as described in Chapter 9.

In addition to the promoter sequences, many bacterial genes have additional controls, including binding sites for proteins other than the RNA polymerase. Some of these proteins improve the efficiency of correct binding of the RNA polymerase and are called **activators**, while others called **repressors** block the promoter sites and thus inhibit the expression of the gene. These additional repressor and activator proteins have a profound influence on whether, and when, transcription actually

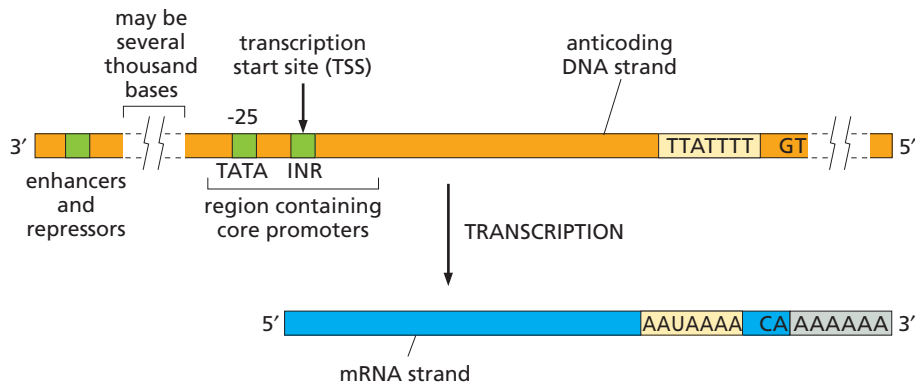


Figure 1.13
The start and stop signals for eukaryotic transcription. All the signals are short sequences that bind enzymes involved in this complex process.

occurs, and so are of crucial biological importance. In this book, however, we shall only be concerned with locating genes, not in trying to dissect their higher-level control, so this aspect of gene control will not be discussed in any detail.

The signals initiating transcription in eukaryotes are generally more complex than those in bacteria

In bacteria all genes are transcribed by a single type of RNA polymerase, whereas in eukaryotes there are three different RNA polymerases, each of which transcribes a different type of gene. All the mRNA that codes for proteins is produced by RNA polymerase II, and in this and later chapters we shall focus mainly on this class of genes. The other RNA polymerases are concerned with transcribing genes for tRNAs, rRNAs, and other types of RNA and these have different types of promoters from genes transcribed by RNA polymerase II.

There are differences in the mechanisms for initiating transcription in eukaryotes compared with bacteria, but the principles are the same. Perhaps the most important difference is the much greater degree of variation in the set of promoters present to initiate transcription. There is a set of **core promoter** DNA sequence signals located in the region of the transcription start site (see Figure 1.13) that are initially bound by a complex of proteins known as **general transcription initiation factors**, which in turn recruit the RNA polymerase to the DNA in the correct position to start transcription. The most important core promoter sequence in genes transcribed by RNA polymerase II is about 25 nucleotides upstream from the start of transcription; this is called the **TATA box** and is characterized by a TATA sequence motif. Details of the sequence variation of this promoter are given in Figure 10.12A, but it should also be noted that this signal is not present in all eukaryotic genes. The transcription factor that binds this motif is called the TATA-binding protein (TBP). Many other protein components are involved in initiating and regulating RNA polymerase activity, but any given gene will only require a small subset for activation, and thus only have a subset of the promoter signals. There appears to be no promoter ubiquitous in eukaryotic genes. In further contrast to the situation in prokaryotes, some of the eukaryotic protein-binding sites that modify RNA polymerase activity can be more than a thousand bases away from the promoter. It is thought that the intervening DNA loops round so that all the gene regulatory proteins come together in a complex that regulates the activity of the RNA polymerase and determines whether or not it starts transcription.

Although termination signals have been identified for both RNA polymerases I and III, no specific signal has been identified for RNA polymerase II. However, it may yet remain to be identified because following transcription the mRNA transcript contains a AAUAAA sequence signal that results in cleavage of the 3' end of the transcript at a site some 10–30 bases after the signal and addition of a series of adenosine nucleotides to form a new 3' end. This occurs very quickly and removes

information about any signal which may be present to terminate transcription. In eukaryotes, this initial mRNA transcript is further modified before translation, as will be described below.

Eukaryotic mRNA transcripts undergo several modifications prior to their use in translation

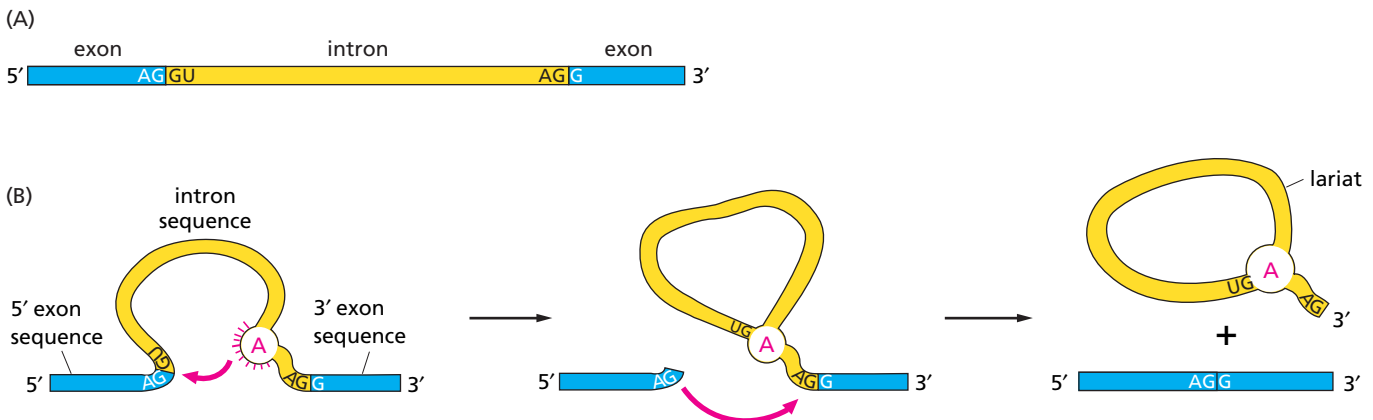
The major difference between eukaryotes and prokaryotes in terms of their transcription and translation processes is that the eukaryotic mRNA transcripts are substantially modified before translation. Two of these modifications have no effect on the final protein product. The first modification, which occurs whilst transcription is still in progress, involves the addition of a modified guanosine nucleotide (7-methylguanosine) to the 5' end of the transcript, a process called **RNA capping**. The last modification, which also occurs while transcription continues, is that which produces the 3' end of the transcript as mentioned previously; this modification consists of two separate steps. The first step is the cleavage of the mRNA transcript after a CA sequence. The second step, called **polyadenylation**, results in approximately 200 adenosine nucleotides being added to the 3' end.

The other mRNA modification that occurs in eukaryotes has a significant effect on the final protein products. The major structural difference between the protein-coding genes of prokaryotes and those of eukaryotes is that the protein-coding DNA of most plant and animal genes is interrupted by stretches of noncoding DNA called **introns**; the blocks of protein-coding sequence between the introns are known as **exons** (see Figure 1.14). Introns have lengths which vary from 10 to over 100,000 nucleotides, whereas exons tend to an average length of 100–200 nucleotides and rarely exceed 1000 nucleotides. Most bacterial protein-coding genes, on the other hand, have an uninterrupted coding sequence. Introns are found in the genes of most eukaryotes but are less frequent in some genomes, for example that of the yeast *Saccharomyces cerevisiae*. They occur very rarely in prokaryotes.

The existence of introns necessitates an extra step between transcription and translation in eukaryotes, which is known as **RNA splicing**. The complete gene is initially transcribed into RNA; the introns are then excised and the exons joined, or spliced, together to provide a functional mRNA that will give the correct protein sequence when translated (Figure 1.14). In most protein-coding genes, RNA splicing is carried out by a complex called a spliceosome, which consists of small nuclear RNA molecules (snRNAs) and proteins. This complex contains the enzymatic activity that cleaves and rejoins the RNA transcript. The excised intron forms a circular structure called a lariat with a branching point usually at an adenine base (Figure 1.14). The lariat RNA is subsequently degraded. There are particular sequence motifs present at the sites at which the RNA transcript is spliced, as well as the position which will become the lariat branch point. However, these

Figure 1.14

The simple schematic of the splicing of an intron. (A) A linear schematic that shows a segment of pre-mRNA with an intron in yellow. The donor splice site has the conserved dinucleotide GU at the start of the intron. The acceptor splice site has the conserved dinucleotide AG at the end of the intron. (B) The intron is spliced out in a two-stage process: firstly creating a loop structure, the lariat, involving the adenine base (colored red), followed by joining the two exons together releasing the intron. In this case the intron occurred within the codon AGG, which is formed when the exons are spliced together. (B, from B. Alberts et al., *Molecular Biology of the Cell*, 4th ed. New York: Garland Science, 2002.)



sequence motifs show considerable variability, with the exception of the first and last two bases of each intron. In most cases these are GU and AG as shown in Figure 1.14, but a few instances of another signal, namely AU and AC, have been found in some complex eukaryotes. These are known as AT-AC or U12 introns, after the DNA sequences or one of the components of the spliceosome, respectively. In some even rarer cases the intron RNA itself has splicing activity.

It should be noted that although protein-coding sequences require a whole number of codons to encode the protein sequence, and thus are a multiple of three bases in length, individual exons do not have this requirement. Codons can be derived from the spliced ends of two consecutive exons, as shown in Figure 1.14. This can lead to further complications in gene prediction, as it adds to the difficulty of correctly identifying the exons and from them the protein sequence. For this and other reasons the possible existence of introns in eukaryotic genes significantly complicates the process of gene prediction in eukaryotic compared with prokaryotic genomes, as is discussed in more detail in Chapters 9 and 10.

Although usually all the exons are retained and joined together during RNA splicing, there are cases where this does not happen and **alternative splicing** occurs, excluding some exons or parts of exons, and thus producing different versions of a protein from the same gene. Alternative splicing is quite common in the genes of humans and other mammals, and is thought to be one means by which the relatively small number of genes present in the genome can specify a much greater number of proteins.

The control of translation

There are various sequence motifs in the mRNA transcripts that indicate the beginning and end of a protein-coding sequence and the site at which the mRNA initially binds to a ribosome. Most protein-coding sequences start with a methionine codon, typically AUG, and invariably end with one of the stop codons of the genetic code (see Table 1.1). In bacterial DNA there is also a distinct short sequence at the 5' end of the mRNA known as the **Shine-Dalgarno sequence** that indicates the ribosome-binding site. This has a typical consensus sequence of AGGAGGU and occurs a few bases upstream of the AUG translation start codon.

Eukaryotes do not use a Shine-Dalgarno sequence to position the ribosome, but instead have components that bind specifically to the 7-methylguanosine nucleotide at the 5' end of all eukaryotic mRNA transcripts. The ribosome binds to this and then starts to search for an AUG codon. Occasionally, the first AUG is missed and another downstream codon is used instead. Termination of translation occurs on encountering a stop codon as in prokaryotes.

There is one feature of gene organization in bacteria that is rarely found in eukaryotes, and which relates to the way the ribosome binds near the translation start site. Functionally related protein-coding sequences are often clustered together into **operons** (see Figure 1.15). Each operon is transcribed as a single mRNA transcript and the proteins are then separately translated from this one long molecule. This has the advantage that only one control region is required to activate the simultaneous

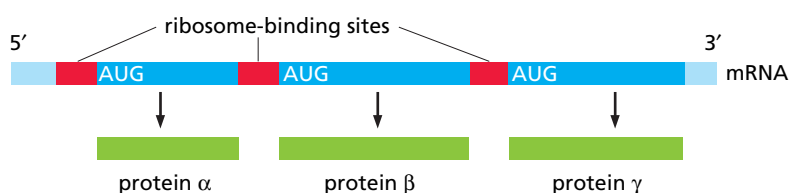
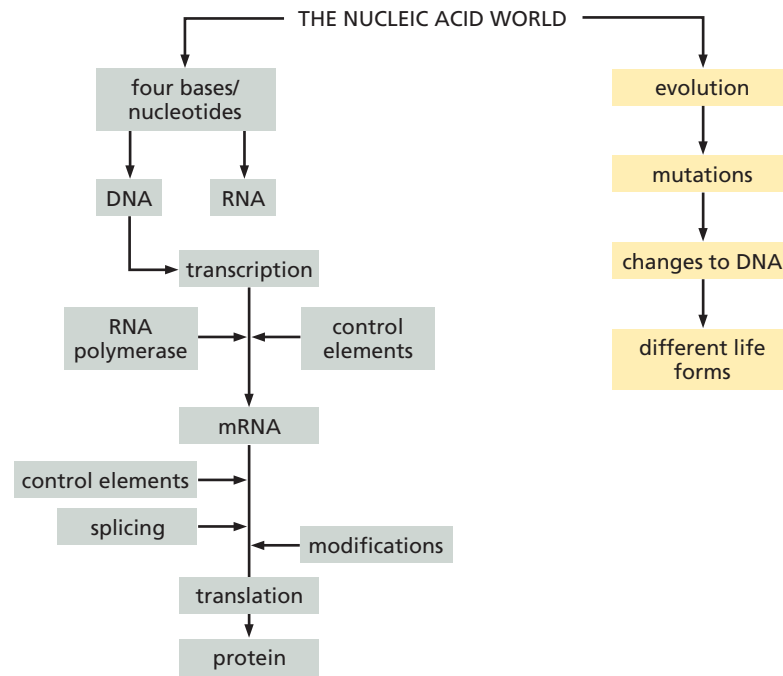


Figure 1.15

A schematic of operon structure. This single mRNA molecule contains three separate protein-coding regions that produce proteins α , β , and γ on translation. The mRNA contains three separate ribosome-binding sites (red).

Flow Diagram 1.4

The key concept introduced in this section is that the process of evolution is based on mutations of the DNA that result in different life forms, which may then be subjected to different evolutionary selective pressures.



expression of, say, all the enzymes required for a particular metabolic pathway. Not all bacterial genes are contained in operons; many are transcribed individually and have their own control regions.

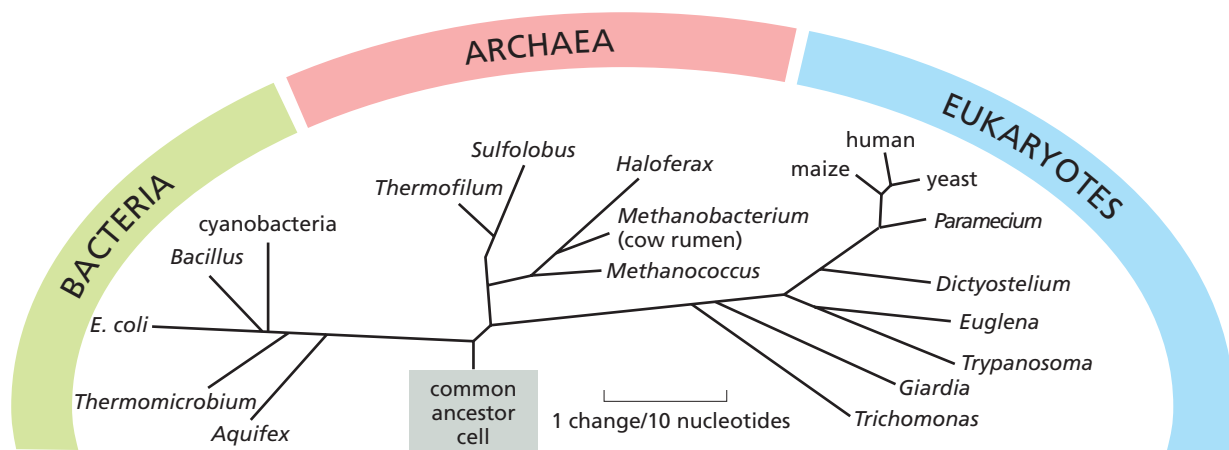
1.4 The Tree of Life and Evolution

The integrity of the genetic information in DNA is carefully protected, and without this protection life would soon disintegrate. Nevertheless, errors inevitably arise in the genome, and these are extremely important as they also provide the genetic variation on which natural selection and evolution can act. Over very long periods of time some of those changes that do not prove to cause their carriers a disadvantage are likely to spread and eventually occur in all genomes of the species (see Flow Diagram 1.4). In this way species can evolve, and ultimately they can evolve into entirely new species. It is generally thought that all existing life has evolved from a single common, very distant ancestor. The evolutionary relationship of known species to each other is commonly described as the tree of life (see Figure 1.16). In this section we will briefly describe the most basic divisions of the tree of life, and some of the modifications which are seen in genomes and their consequences.

Figure 1.16

Tree of life. Evolution branches out like a real tree where the roots are the origin and the branches are the different groups of life form.





A brief survey of the basic characteristics of the major forms of life

In this context, all living organisms can be divided into two vast groups: the prokaryotes, which are further divided into the unicellular bacteria and archaea, and the eukaryotes, which include all other living organisms (see Figure 1.17). Another class of objects that contain nucleic acid instructions for their reproduction is the viruses. These have very small genomes that encode the proteins that make up the virus structure, but viruses can only replicate inside a living cell of another organism, as they hijack the cell's biochemical machinery for replicating DNA and synthesizing proteins. Viruses may have either DNA or RNA genomes. Although viral genes follow the basic rules by which DNA encodes RNAs and proteins, it is worth noting that some viral genomes have unusual features not commonly present in cellular genomes, such as overlapping genes, which need careful interpretation.

The prokaryotes are a vast group of unicellular microorganisms. Their cells are simple in structure, lacking a nucleus and other intracellular organelles such as mitochondria and chloroplasts. Taxonomically, the prokaryotes comprise two **superkingdoms**, or **domains**, called the Bacteria and the Archaea, which in evolutionary terms are as distinct from each other as both are from the rest of the living world. Their DNA is usually in the form of a single circular chromosome (although linear chromosomes are known in prokaryotes), containing a single circular DNA molecule, and is not enclosed in a nucleus. In favorable growing conditions, prokaryotes reproduce rapidly by simple cell division, replicating their DNA at each division so that each new cell receives a complete set of genetic instructions. Many bacteria also contain extrachromosomal DNAs in the form of **plasmids**, small circular DNAs that carry additional genes and which can often be transmitted from bacterium to bacterium. Genes for drug resistance, the ability to utilize unusual compounds, or the ability to cause disease are often carried on plasmids. The best-studied bacterium, and the one that for many years provided virtually all our knowledge about the processes of transcription and translation, is the gut bacterium *Escherichia coli*, abbreviated to *E. coli*.

All other living organisms are eukaryotes and belong to the domain **Eukarya**. All animals, plants, fungi, algae, and protozoa are eukaryotes. The eukaryotes include both multicellular and unicellular organisms. Unicellular eukaryotes widely used as model organisms for genetic and genomic studies are the yeasts and unicellular algae such as *Chlamydomonas*. Eukaryotic cells are larger and more complicated than those of prokaryotes. The DNA is contained inside a nucleus, and is highly packaged with histones and other proteins into a number of linear chromosomes, ranging from two to hundreds depending on the organism. Humans have 46 chromosomes in their

Figure 1.17

The tree of life based on analysis of molecular sequences. Note that multicellular organisms are all in a small region of the tree. (From B. Alberts et al., *Molecular Biology of the Cell*, 4th ed. New York: Garland Science, 2002.)

body cells (made up of two sets of 23 chromosomes inherited from each parent), the fruit fly *Drosophila* has 8, petunias have 14, while the king crab has 208. There appears to be no particular reason why the DNA is divided up into such different numbers of chromosomes in different organisms; the actual numbers of genes in the genomes of multicellular organisms are much more similar and vary between 20,000 and 30,000 for those organisms whose genomes have been sequenced to date.

Eukaryotic cells are highly compartmentalized, with different functions being carried out in specialized organelles. Two of these are of particular interest here, as they contain their own small genomes. **Mitochondria** contain the components for the process of energy generation by aerobic respiration, and **chloroplasts** contain the molecular components for photosynthesis in plant and algal cells. These two organelles are believed to be the relics of prokaryotic organisms engulfed by the ancestors of eukaryotic cells and still retain small DNA genomes of their own—mitochondrial DNA (mitDNA) and chloroplast DNA—and the protein machinery to transcribe and translate them. These genomes encode some of the proteins specific to mitochondria and chloroplasts, but most of their proteins are now encoded by genes in the eukaryotic cell nucleus.

Nucleic acid sequences can change as a result of mutation

There are a number of occasions, such as DNA replication, when the genomic DNA is actively involved in processes that leave it vulnerable to damage. Sometimes this damage will be on a large scale, such as the duplication of whole genes, but often it involves just a single base being incorrectly replicated. The general term used to describe such damage is mutation. Depending on which part of the DNA sequence is affected, mutations can have a drastic effect on the information encoded, leading to changes in the sequence of encoded proteins, or the loss of control signals. Genes can be rendered inactive or proteins dysfunctional, although mutations can also have beneficial effects (see Box 1.1). In organisms that use sexual reproduction, unless the DNA affected is in a germ cell, the DNA will not be used to generate the genomes of future generations, and so will only affect the organism in which the damage occurred. In such cases, the organism might suffer, especially if the mutation causes uncontrolled cell growth and division, as happens in tumors. The alternative is that the mutation is transmitted through to the next generation, in which case it has a chance to eventually become part of the normal genome of the species.

Box 1.1 There is more to genomes than protein and RNA genes

Not all the DNA sequence in a genome contains a meaningful message or a known function. Regions without a message are sometimes referred to as junk DNA, although this term should not be taken too literally as we have much still to learn, and these regions may yet come to be seen as functional in some way. Mammalian genomes contain large amounts of this type of DNA, both in the form of introns and between genes. Simpler eukaryotic organisms have less, and bacteria have very little.

Much of the so-called junk DNA is in the form of highly repeated DNA sequences, which form significant percentages of the genomes of many higher organisms. Many of these **DNA repeats** are due to DNA elements known as transposons, which can copy themselves and

insert themselves into other parts of the genome. Transposons are present in both bacteria and eukaryotes, but in mammalian genomes they have multiplied to a much greater extent and appear to have largely lost their function, existing now simply as apparently functionless sequence repeats.

On a final note, changes in DNA sequences that occur during evolution occasionally destroy some of the control sequences needed for a gene to be expressed. When this happens, the resultant inactive gene is called a **pseudogene**. Soon after the initial inactivation, pseudogenes can be hard to distinguish from active genes, but over time they accumulate many more mutations than active genes, and so diverge to (ultimately) random sequence.

The fate of the mutation, to be lost or to be retained, depends on the process of natural selection that is the cornerstone of the theory of evolution.

The existence of similar DNA and protein sequences in different organisms is a consequence of the process of evolution that has generated the multitude of living organisms from an ancient ancestor held in common. The sequence similarities reveal details of the ways that mutations arise and of the balance of forces which will result in only a small group of mutations being preserved through evolutionary selection. Therefore some details of the mechanism of mutation are of relevance in a number of areas of bioinformatics, including sequence alignment (see especially Sections 4.3 and 5.1) and phylogenetic analysis (see especially Sections 7.2 and 8.1). In the phylogenetic analysis described in Chapters 7 and 8 an attempt is made to reconstruct the evolutionary history of a set of sequences. This requires a detailed model of evolution, which requires a comprehensive understanding of the kinds of mutations that occur, their effects, and the process of natural selection by which they are either accepted or lost.

Summary

We have tried in this chapter to give a brief introduction to the nucleic acids and their role in living systems. We have focused exclusively on the role of nucleic acids in genomes, although a look at any introductory textbook on molecular biology will show that, in addition, single nucleotides play a part in many other processes. We have described how the sequence of DNA can encode proteins, and how simple sequence signals are used to control the interpretation of the genomic DNA. The material is sufficient to allow a novice to appreciate the techniques discussed in this book, particularly those for gene detection. You should be aware, however, that there are probably exceptions to every general statement we have made in this chapter! (See Box 1.2.) For example, under certain circumstances, the codon UGA can code for the unusual amino acid selenocysteine instead of being understood as a stop signal. Many organisms have their own peculiarities, and one should preferably seek out an expert for advice.

Box 1.2 Things are usually not that simple!

This chapter has given a brief introduction to the key role of nucleic acids in the storage, interpretation, and transmission of genetic information. The many genome sequencing projects are producing results at a phenomenal rate, and the techniques of bioinformatics, such as are described in Chapters 9 and 10, are required to identify and characterize the functional components of the genomes. When working on such projects, it must be remembered that the descriptions given in this and the other chapters are general, and wherever possible care must be taken to discover the specific details applicable to the organism of interest.

A further warning is required in that even some of the fundamental concepts described in this chapter are

much less well defined than might be supposed. Two brief illustrations of this will be given, dealing with the definition of a gene and of the human genome. Processes such as alternative splicing are making it ever harder to agree on a definition of a gene. The human genome is another concept whose definition is proving harder to agree on than expected. Recent studies have shown that there is much greater variation in the genome in humans than was expected. As well as many small point mutations that were anticipated, it was found that there were a surprising number of large-scale differences between humans. As a result it is no longer clear how to define a fully representative human genome. There are reviews in Further Reading that explain these points in more detail.

Further Reading

General References

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Box 1.2

Check E (2005) Patchwork people. *Nature* 437, 1084–1086.

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PROTEIN STRUCTURE

When you have read Chapter 2, you should be able to:

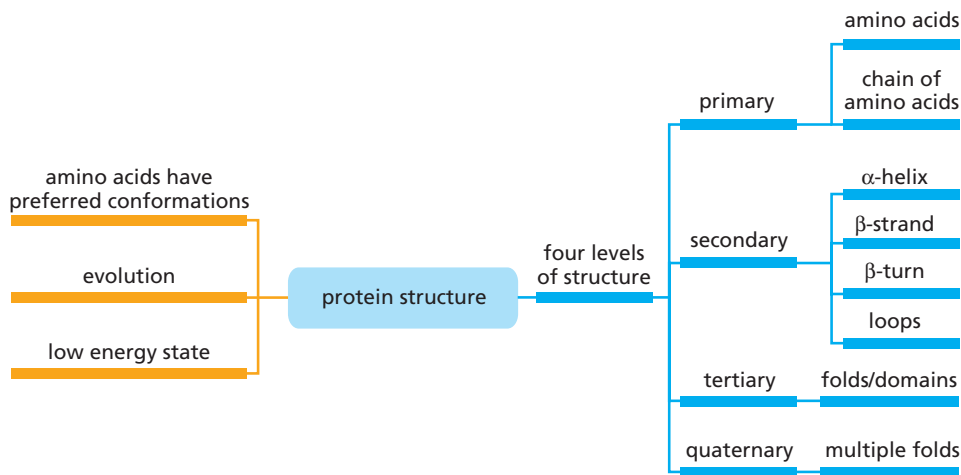
- Discuss how proteins are important for function.
- Describe how amino acids are the building blocks of proteins.
- Show how amino acids are joined together to form a protein chain.
- Explain that secondary structures are repeating structures found in most proteins.
- Describe how the whole protein chain folds up into a fold.
- Discuss how the structure of the fold is important for its function.
- Outline how proteins often exist as a complex of more than one fold.

If there is one class of molecules which could be said to live life it would be the proteins. They are responsible for catalyzing almost all the chemical reactions in the cell (RNA has a more limited but important role, as we saw in Chapter 1), they regulate all gene activity, and they provide much of the cellular structure. There is speculation that life may have started with nucleic acid chemistry only, but it is the extraordinary functional versatility of proteins that has enabled life to reach its current complex state. Proteins can function as enzymes catalyzing a wide variety of reactions necessary for life, and they can be important for the structure of living systems, such as those proteins involved in the cytoskeleton. The size of a protein can vary from relatively small to quite large macromolecules.

As we saw in Chapter 1, the DNA sequence of a gene can be analyzed to give the **amino acid sequence** of the protein product. In that aspect alone, the ready availability of DNA sequences of genes and whole genomes from the 1980s onward revolutionized biology, as it opened up this vital shortcut to determining the amino acid sequence of virtually any protein. Bioinformatics uses this sequence information to find related proteins and thus gather together knowledge that can help deduce the likely properties of unknown proteins, plus their structures and functions. Knowing the relationship between a protein's structure and its function provides a greater understanding of how the protein works, and thus often enables the researcher to propose experiments to explore how modifying the structure will affect the function. As the vast majority of currently marketed pharmaceuticals act by interacting with proteins, structure–function studies are vital to the design of new drugs, and bioinformatics has an important role in speeding up this process and enabling computer modeling of these interactions, as we shall see in Chapter 14.

Mind Map 2.1

A mind map visualization of aspects of protein structure.



There are many excellent books that describe protein structures, form, and classification in detail (see Further Reading). In this chapter we will introduce proteins and outline some concepts that are important for bioinformatics, providing a minimum amount of information required to understand the techniques and algorithms described in later chapters. Additional information is given where needed in those chapters.

2.1 Primary and Secondary Structure

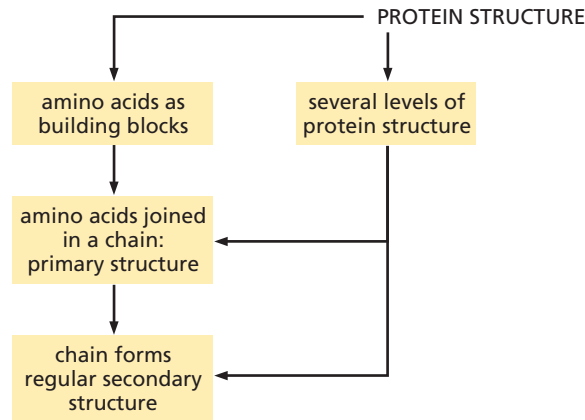
A protein folds into a three-dimensional structure, which is determined by its protein sequence. The fold of the protein consists of repeating structural units called secondary structures, that will be discussed in this section (see Flow Diagram 2.1). The fold of the protein is very important for the way the protein will function, and whether it will function correctly. Therefore the study of the ways in which proteins fold and understanding how they fold is an important area of bioinformatics, as well as predicting the fold of a protein from its sequence.

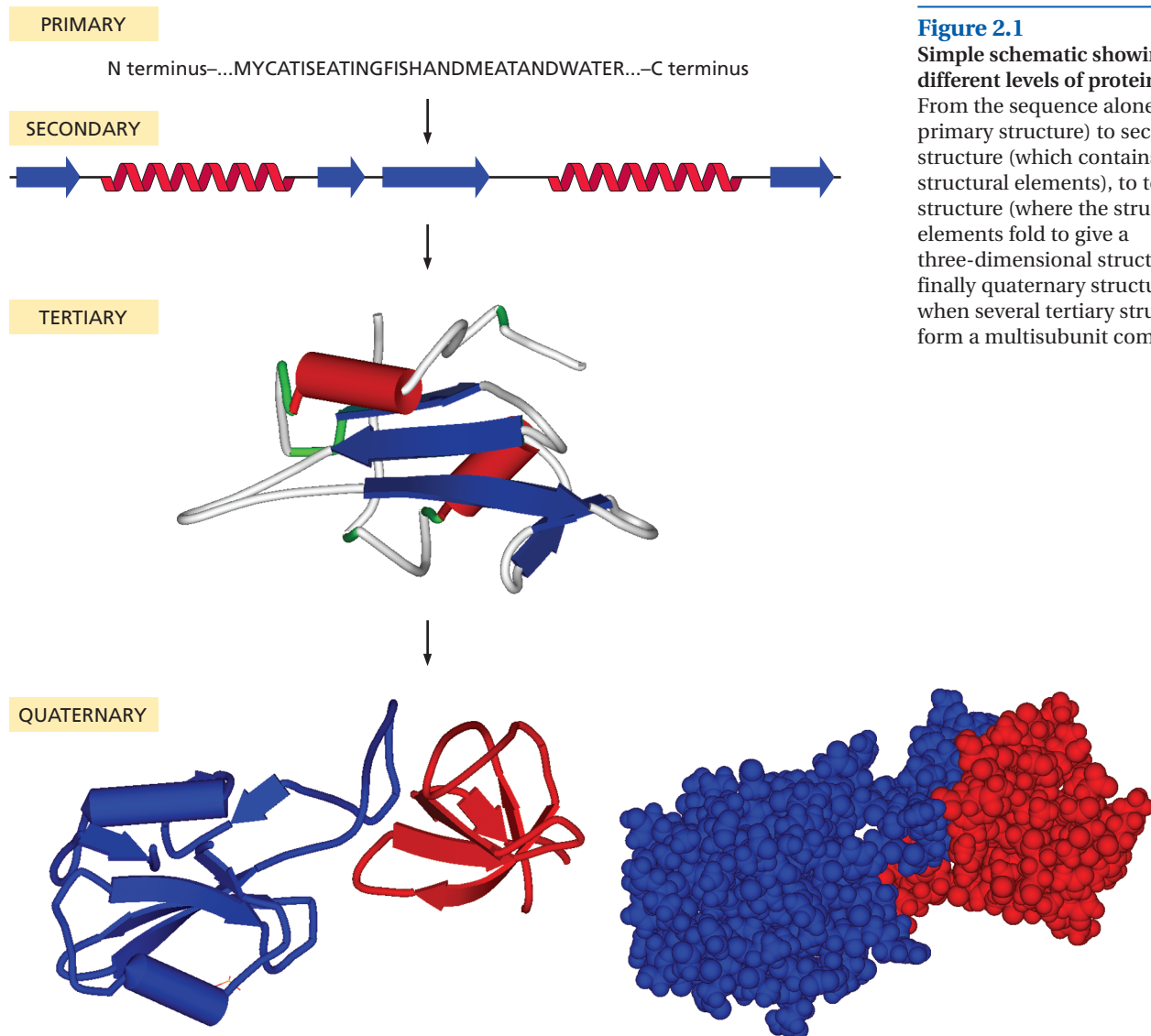
Protein structure can be considered on several different levels

The analysis of protein structure by experimental techniques such as X-ray crystallography and nuclear magnetic resonance (NMR) has shown that proteins adopt distinct structural elements. In general there are four levels of protein structure to consider (see Figure 2.1). The **primary structure** is the protein sequence, the types

Flow Diagram 2.1

The key concepts introduced in this section are that proteins are linear polymers of amino acids whose order is often called the primary structure or sequence, and that proteins can fold into a limited number of regular three-dimensional forms called secondary structures.



**Figure 2.1**

Simple schematic showing the different levels of protein structure. From the sequence alone (the primary structure) to secondary structure (which contains local structural elements), to tertiary structure (where the structural elements fold to give a three-dimensional structure), to finally quaternary structure found when several tertiary structures form a multisubunit complex.

and order of the amino acids in the protein chain; the **secondary structure** is the first level of protein folding, in which parts of the chain fold to form generic structures that are found in all proteins. The **tertiary structure** is formed by the further folding and packing together of these elements to give the final three-dimensional **conformation** unique to the protein. Many functional proteins are formed of more than one protein chain, in which case the individual chains are called protein subunits. The subunit composition and arrangement in such multisubunit proteins is called the **quaternary conformation**. The structure adopted by a protein chain, and thus its function, is determined entirely by its amino acid sequence, but the rules that govern how a protein chain of a given sequence folds up are not yet understood and it is impossible to predict the folded structure of a protein *de novo* from its amino acid sequence alone. Helping to solve this problem is one of the challenges facing bioinformatics.

Amino acids are the building blocks of proteins

Proteins are made up of 20 types of naturally occurring amino acids (see Table 2.1), with a few other amino acids occurring infrequently. These 20 amino acids consist solely of the elements carbon (C), nitrogen (N), oxygen (O), and hydrogen (H), with the exception of cysteine and methionine, which also contain sulfur (S). The structure

Table 2.1
The 20 amino acids colored according to their properties. The amino acid name, the three-letter code, and the one-letter code are given. Note that the one-letter code for amino acids can have the same letter as found in the one-letter codes that denote nucleotides, but they mean different things. So for example the letters C, G, A, and T in a DNA sequence would stand for cytosine, guanine, adenine, and thymine, respectively. In protein sequences these same letters would refer to cysteine, glycine, alanine, and threonine, respectively.

- Nonpolar
- Polar
- Acidic
- Basic

Amino acid	Three-letter code	One-letter code	Comment
Glycine	Gly	G	Only -H as side chain
Alanine	Ala	A	
Valine	Val	V	
Leucine	Leu	L	
Isoleucine	Ile	I	
Proline	Pro	P	Side chain to N bond
Phenylalanine	Phe	F	
Methionine	Met	M	
Tryptophan	Trp	W	
Cysteine	Cys	C	Forms disulfide bonds
Asparagine	Asn	N	Amide N polar
Glutamine	Gln	Q	Amide N polar
Serine	Ser	S	-OH group polar
Threonine	Thr	T	-OH group polar
Tyrosine	Tyr	Y	-OH group polar
Aspartic acid	Asp	D	
Glutamic acid	Glu	E	
Histidine	His	H	
Lysine	Lys	K	
Arginine	Arg	R	

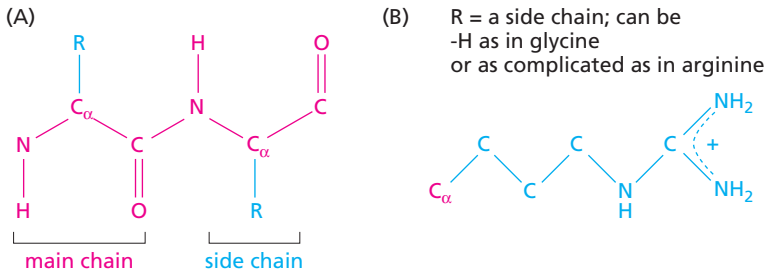
of an amino acid can be divided into a common **main chain** part (see Figure 2.2) and a side chain that differs in chemical structure among the different amino acids. The side chain is attached to the main chain carbon atom known as the α -carbon (C_α).

The 20 amino acids are given in Table 2.1 along with the abbreviations that will be used in this book. When the amino acid sequence of a protein is written out, it is usually given in the one-letter code. We will first look at the side-chain properties that distinguish one amino acid from another, and then explore in some detail the main chain component, whose properties are responsible for most of the general features of protein structure.

The differing chemical and physical properties of amino acids are due to their side chains

The functional properties of proteins are almost entirely due to the side chains of the amino acids. Each type of amino acid has specific chemical physical properties

Figure 2.2
Diagram of an amino acid. (A) shows the chemical structure of two amino acids, where R represents the side chains, which can be different as shown in (B). The amino acid consists of a central C_α atom with a main chain N and C at either side of it. The C is bonded to an O with a double bond. The main chain is colored red and the side chain blue.



that are conferred on it by the structure and chemical properties of its side chain. They can, however, be classified into overlapping groups that share some common physical and chemical properties, such as size and electrical charge. The smallest amino acid is glycine, which has only a hydrogen atom as its side chain. This endows it with particular properties such as great flexibility. The other extreme of side-chain flexibility is represented by proline, an amino acid that has a side chain bonded to the main-chain nitrogen atom, resulting in a rigid structure. Some amino acids have uncharged side chains and these are generally **hydrophobic** (not liking water, therefore tend to be buried within the protein surrounded by other hydrophobic amino acids) while others are positively or negatively charged. The charged or polar amino acids are **hydrophilic**; they like to be surrounded by water molecules with which they can form interactions. Buried inside the protein they will often have another oppositely charged hydrophilic residue to form an interaction with, for example, the long side chain of the amino acid arginine, with its $-\text{NH}_3^+$ groups is (relatively) large and basic (positively charged). The amino acid side chains are shown in Figure 2.3.

As there are 20 distinct amino acids that occur in proteins, there can be 20^n different polypeptide chains of length n . For example, a polypeptide chain 250 amino acids in length will be one of more than 10^{325} alternative different sequences. Clearly, the sequences that do occur are only a tiny fraction of those possible. Often only a few sequence modifications are needed to destabilize the three-dimensional conformation of a protein, and so it is probable that the majority of these alternative sequences will not adopt a stable conformation.

Amino acids are covalently linked together in the protein chain by peptide bonds

The primary structure of a protein is the sequence of amino acids in the linear protein chain, which consists of covalently linked amino acids. This linear chain is often called a polypeptide chain. The amino acids are linked by peptide bonds, which are formed by a **condensation reaction** (the loss of a water molecule) between the backbone carboxyl group of one amino acid and the amino group of another (see Figure 2.4). When linked together in this way, the individual amino acids are conventionally called amino acid residues. There are two key properties of peptide bonds that are important in determining protein structure, and will be discussed in this chapter. First, the bond is essentially flat, or planar; that is to say that the carbon, nitrogen, and carbonyl oxygen atoms involved in the bond all essentially lie in the same plane. This limits rotation around the bond. Second, both the $-\text{NH}$ and $-\text{C}=\text{O}$ groups are capable of noncovalent hydrogen-bonding interactions; this type of interaction is key to the folding up of the protein chain into a three-dimensional structure.

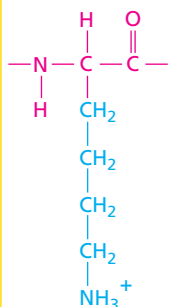
The peptide bonds and connecting C_α atoms are referred to as the protein backbone or main chain. A polypeptide will have a reactive amino group at one end and a reactive carboxyl group at the other end. These are referred to as the **amino terminus** (or **N terminus**) and **carboxy terminus** (or **C terminus**), respectively. When a protein is synthesized by translation of an mRNA (see Chapter 1), the amino terminus is synthesized first. Protein sequences are thus conventionally written starting with the N terminus on the left. The relation of DNA coding-strand sequence to mRNA sequence to protein sequence is shown in Figure 2.5

In principle, the atoms at either end of a bond are free to rotate in any direction but due to atomic constraints this is not the case. **Torsion angles** are used to define the conformation around bonds that can rotate and are defined for four atoms (see Figure 2.6A). Torsion angles are usually described in the range of -180° to $+180^\circ$ and are the number of degrees that atoms 1 and 2 have to move to eclipse atoms 3 and 4. The groups on either side of the peptide bond can rotate to some extent around the

BASIC SIDE CHAINS

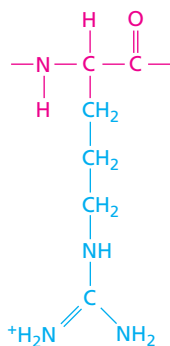
lysine

(Lys, or K)



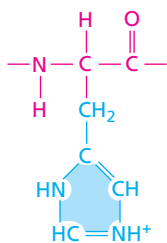
arginine

(Arg, or R)



histidine

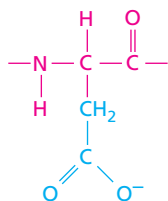
(His, or H)



ACIDIC SIDE CHAINS

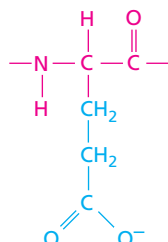
aspartic acid

(Asp, or D)



glutamic acid

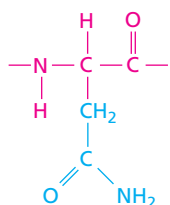
(Glu, or E)



UNCHARGED POLAR SIDE CHAINS

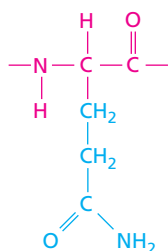
asparagine

(Asn, or N)



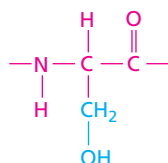
glutamine

(Gln, or Q)



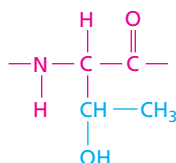
serine

(Ser, or S)



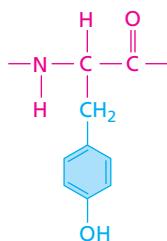
threonine

(Thr, or T)



tyrosine

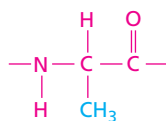
(Tyr, or Y)



NONPOLAR SIDE CHAINS

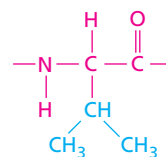
alanine

(Ala, or A)



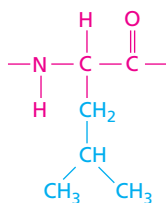
valine

(Val, or V)



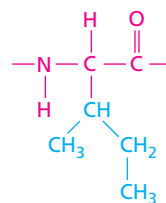
leucine

(Leu, or L)



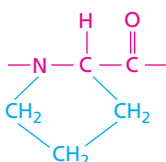
isoleucine

(Ile, or I)



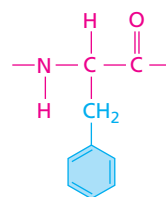
proline

(Pro, or P)



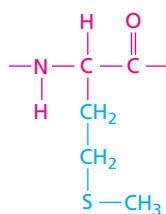
phenylalanine

(Phe, or F)



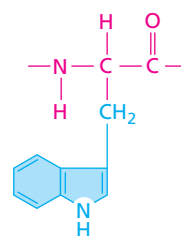
methionine

(Met, or M)



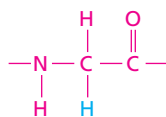
tryptophan

(Trp, or W)



glycine

(Gly, or G)



cysteine

(Cys, or C)

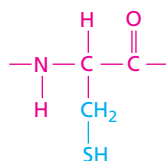
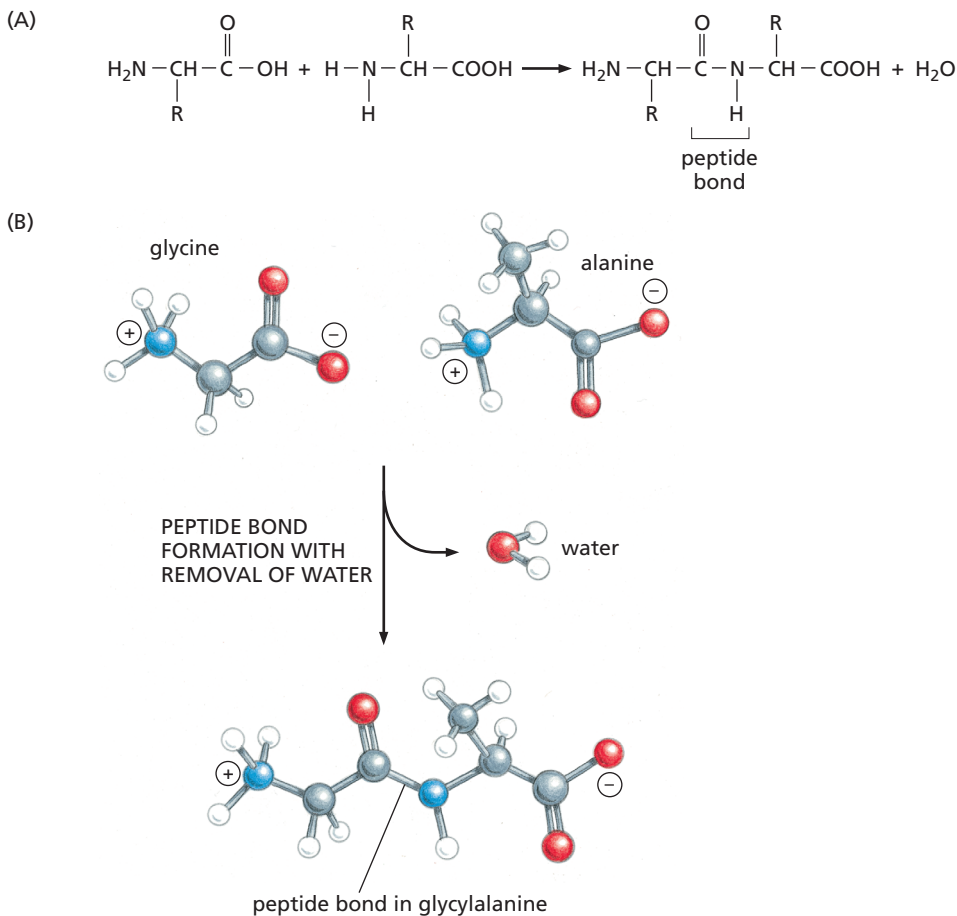


Figure 2.3

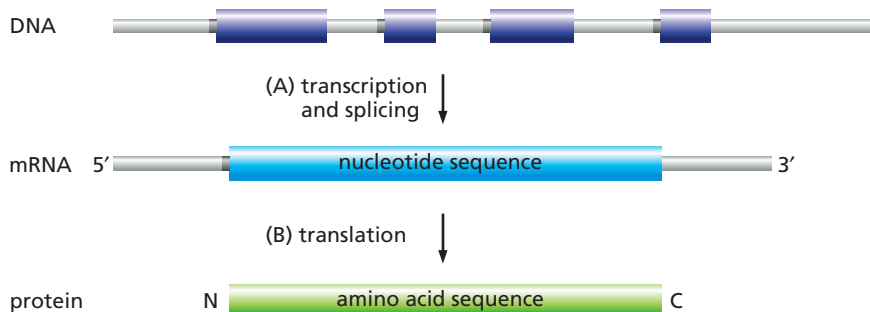
The side chains of the 20 commonly occurring amino acids. They are arranged according to their physicochemical properties: acidic, basic, uncharged polar, and nonpolar (hydrophobic). Some have

additional properties, such as aromaticity (for example Phe). (Adapted from B. Alberts et al., *Molecular Biology of the Cell*, 4th ed. New York: Garland Science, 2002.)

bond. But the peptide bond itself is planar due to the phenomenon known as electronic resonance (the movement of delocalized electrons within the bond) that gives it a partial double-bond character, and leads to a slightly shorter C-N bond than might otherwise be expected. The torsion angle within the peptide bond (usually called ω) is rarely more than 10° from planarity (see Figure 2.6B). The polypeptide

**Figure 2.4**

Peptide bonds. (A) gives the chemical formulae of the peptide bond that is formed between amino acids to make a polypeptide chain. (B) illustrates the above in a diagrammatic form. (B, from B. Alberts et al., *Molecular Biology of the Cell*, 4th ed. New York: Garland Science, 2002.)

**Figure 2.5**

Transcription and translation. The relation of DNA coding-strand sequence to mRNA sequence to protein sequence. The exons (purple boxes) of the DNA are transcribed into mRNA which, using other molecules (see Chapter 1), directs the protein sequence.

Box 2.1 Interactions between protein atoms

Interactions between protein atoms can be either covalent or noncovalent. The covalent bond, which occurs when an electron pair is shared between two atoms, is a stronger bond than the noncovalent interaction. Noncovalent interactions can take place by different means, such as hydrogen bonding or van der Waals or electrostatic interactions.

Atoms can be thought of as a nucleus surrounded by a cloud of constantly moving electrons. Sometimes there are more electrons at one end of the atom than at the other; this forms a dipole. A van der Waals interaction occurs when atoms that have oppositely oriented dipoles are near each other (see Figure B2.1). These interactions are weak.

Less weak are the hydrogen bonds, where there is a weak sharing of electrons between a hydrogen atom

(donor) attached to a nitrogen or an oxygen and an acceptor atom, which in proteins is usually a nitrogen or an oxygen, although other atoms can act as acceptors as well.

Electrostatic interactions occur between groups that are of opposite charge.

Figure B2.1
Two atoms with opposing dipoles.

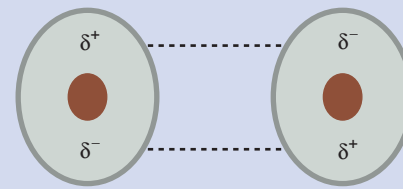
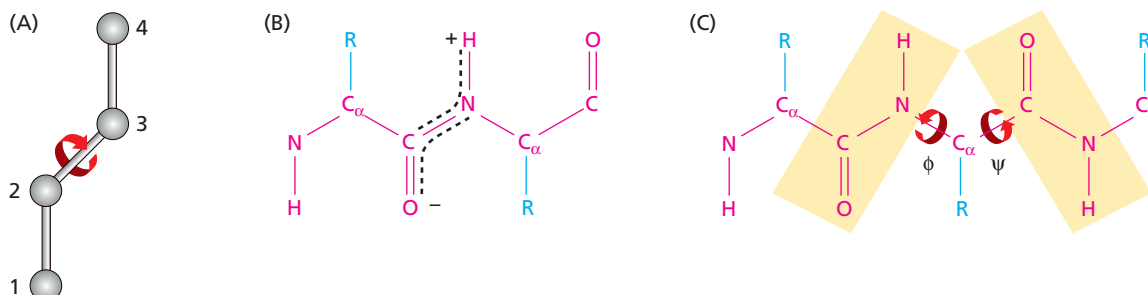


Figure 2.6

Torsion angles. (A) Angles between atoms can be calculated and are referred to as dihedral or torsion angles. (B) The angle of the peptide bond (O-C-N-H atoms) has a partial double-bond character and is therefore more rigid. (C) The angles of the bonds flanking the peptide bond can rotate more freely and are called phi (ϕ) and psi (ψ); these are the main source of conformational flexibility of the polypeptide chain. Preferences for a limited number of values for these angles arise from steric hindrance and hydrogen bonding.

backbone can therefore be thought of as a chain of flat peptide units connected by C_α atoms. Each C_α atom makes two single bonds, within this backbone, one to the N and one to the carbonyl C, giving rise to two torsion angles per residue. These angles are known as ϕ (phi) and ψ (psi) (see Figure 2.6C) and are the main source of flexibility of the polypeptide chain, allowing it to fold up. Although, in principle, ϕ and ψ could take any value between -180° and $+180^\circ$, they usually fall between certain limits, and residues with a given side chain have a strong preference for a particular value giving rise to repeating regular structural units in a protein. These preferences arise in two ways. First, **steric hindrance** prevents certain combinations of angles from occurring. Second, the hydrogen-bonding ability of the peptide groups can lead to selective stabilization for other angle choices. If several consecutive residues have the same (ϕ, ψ) values, the resulting structure will be regular and repetitive. Some regular structures will be stabilized by hydrogen bonding, and we will describe the most commonly occurring structures below.

Additionally, the peptide bond is usually in the atomic conformation called the **trans conformation**, as this is thermodynamically more favorable than the alternative **cis conformation** (see Figure 2.7). This preference is because of clashes between two adjacent side chains (steric hindrance) in the cis conformation. Because of the particular nature of its side chain, proline is found in the cis conformation more frequently than other amino acids.



Box 2.2 Covalent bonds

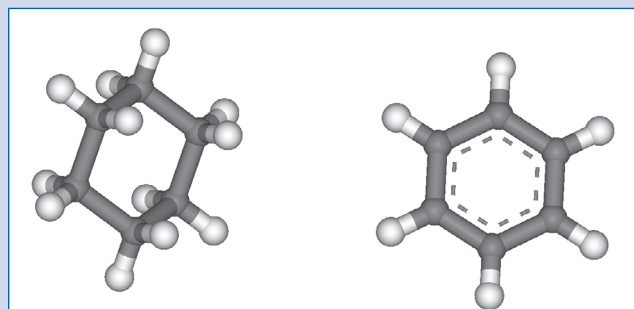
Covalent bonds are formed between two atoms by sharing electrons. The number of bonds an atom can make depends on how many electrons are available for this type of sharing. Most atoms form single bonds sharing one electron each between them. But there are some atoms that can form multiple bonds, such as double bonds. Covalent bonds can be classified according to their bond order. Single bonds have a bond order of 1; double bonds a bond order of 2. Non-integral bond orders are also possible, explained by the theory of resonance. In proteins many bonds with orders intermediate between 1 and 2 occur (usually shown as a dashed line beside a solid line), such as in peptide bonds and in carbon rings where the resonance makes them planar (Figure B2.2).

The arrangement of covalent bonds in a molecule describes the **molecular configuration**, but this does not necessarily define a single three-dimensional structure. Single covalent bonds allow for free rotation about the torsion angle of the bond, subject to any steric hindrance.

Multiple bonds create constraints to the movement around the bond, resulting in some cases in distinct conformations such as the trans and cis forms of peptide bonds.

Figure B2.2

A six-membered ring with single bonds and resonant bonds. The ring with single bonds has two hydrogens per carbon atom, while the ring with resonant bonds has only one hydrogen per carbon atom and is planar.



Perhaps the most important conformational property of a protein residue is its (ϕ, ψ) value. To reflect this, Ramachandran proposed plotting these values in a two-dimensional plot for all the residues of protein structures, and calculating the regions of (ϕ, ψ) values where most of the amino acids were found (these are referred to as favorable regions) and those (ϕ, ψ) value areas of the plot where it is rare to find an amino acid (these regions are referred to as unfavorable). Such plots are called the Ramachandran plots and are very useful tools to estimate how plausible a predicted protein structure is, as described in Chapter 13. The regions that were highly populated correspond to the regular structures (α -helix, β -strand) that are formed by the amino acid sequence in a protein structure (see Figure 2.8).

Secondary structure of proteins is made up of α -helices and β -strands

In nature in general, regular structures are often helical, and a helix is found as a secondary structure in proteins. Helices arise as a result of energetically favorable hydrogen bonding between atoms of the backbone of the protein chain. In stretches of the chain with repetitive (ϕ, ψ) values of approximately $(-60^\circ, -60^\circ)$, an

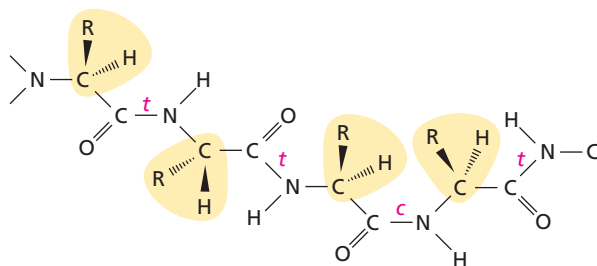
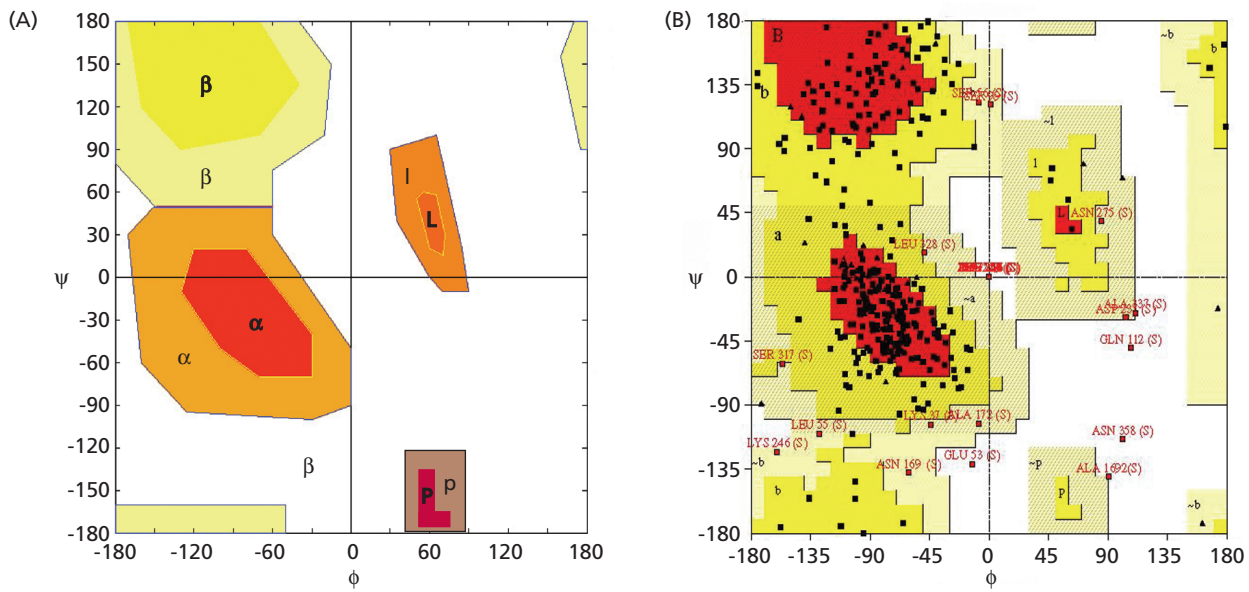


Figure 2.7

Trans versus cis conformation of a peptide bond. The figure illustrates the reasons for a preference for the trans (t). A cis (c) conformation results in a considerable increase in steric hindrance (atom clashes) between the two side-chain atoms.

**Figure 2.8**

Ramachandran plot. (A) An ideal Ramachandran plot with the angle ϕ plotted on the x-axis and the angle ψ plotted on the y-axis. The yellow areas indicate the preferred conformation for β -sheet residues, the regular α -helix, the left-handed helix, and P (p) the epsilon structure. The darker colors show more favorable conformations. (B) The actual Ramachandran plot as calculated for a model of a kinase enzyme. Some residues fall within the disallowed areas (non-colored) and are identified by the residue type and sequence number.

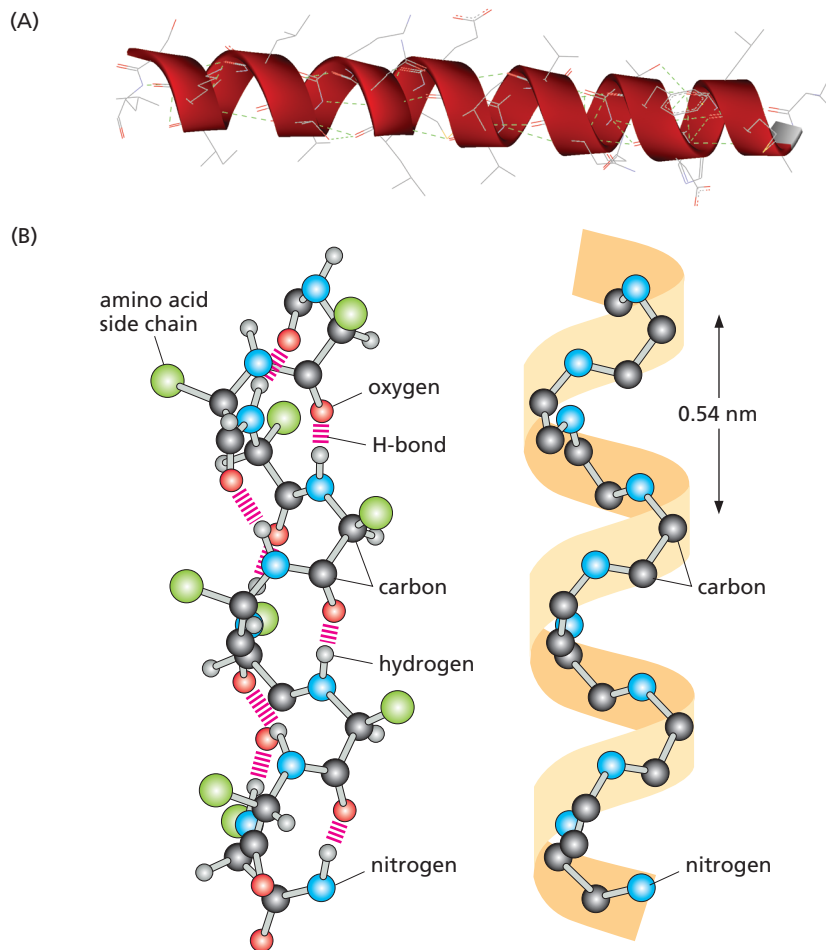
especially stable right-handed helix is formed called the **α -helix** (see Figure 2.9). Left-handed helices can exist in proteins, but side-chain interactions favor the right-handed helix, and this is overwhelmingly predominant.

The proteins we will deal with in this book are composed of one or more globular domains, with most α -helices limited to a single domain, which restricts them usually to 2 nm in length. This is useful to bear in mind when predicting secondary structure from protein sequences, as described in Chapters 11 and 12. However, there are some instances of globular proteins with long α -helices extending across domains, of which an extreme example is hemagglutinin of the influenza virus, which contains an α -helix over 100 μm long. Many α -helices in globular proteins are only one or two turns long.

In an α -helix, the planes of the peptide bonds are nearly parallel with the helix axis and the dipoles within the helix are aligned such that all C=O groups point in the **C-terminal** direction and all N-H groups point in the **N-terminal** direction. The side chains point outward from the helix axis and are generally oriented toward the N terminus. The main chain -C=O and -NH groups in an α -helix are hydrogen bonded to a peptide bond four residues away [O(*i*) to N(*i*+4)]. Within the helix, all the backbone groups that can be are involved in hydrogen bonding. This gives a very regular and stable arrangement (see Figure 2.9).

It should be noted that the structures found in globular proteins are not perfectly regular, so it is frequently difficult to define the precise ends of the helices, and in some cases the hydrogen-bonding patterns are intermediate between these idealized forms. Therefore prediction of these structures using bioinformatics programs is made more difficult. A detailed discussion of this issue is given in Section 12.1.

In the 1960s another common regular repeating structure was found in globular proteins that consisted of extended strands aligned with each other to permit favorable hydrogen bonding. A single extended chain of this type is called a **β -strand**, and a set of β -strands hydrogen bonded together side by side forms a β -sheet (Figure 2.10). A β -strand has two sides that can participate in hydrogen bonding. The -NH groups occupy all of one side, and the -C=O groups the other; one side is proton accepting and the other proton donating. This dictates how β -strands are added to the β -sheet. Often, β -sheets are displayed by drawing out the pattern of

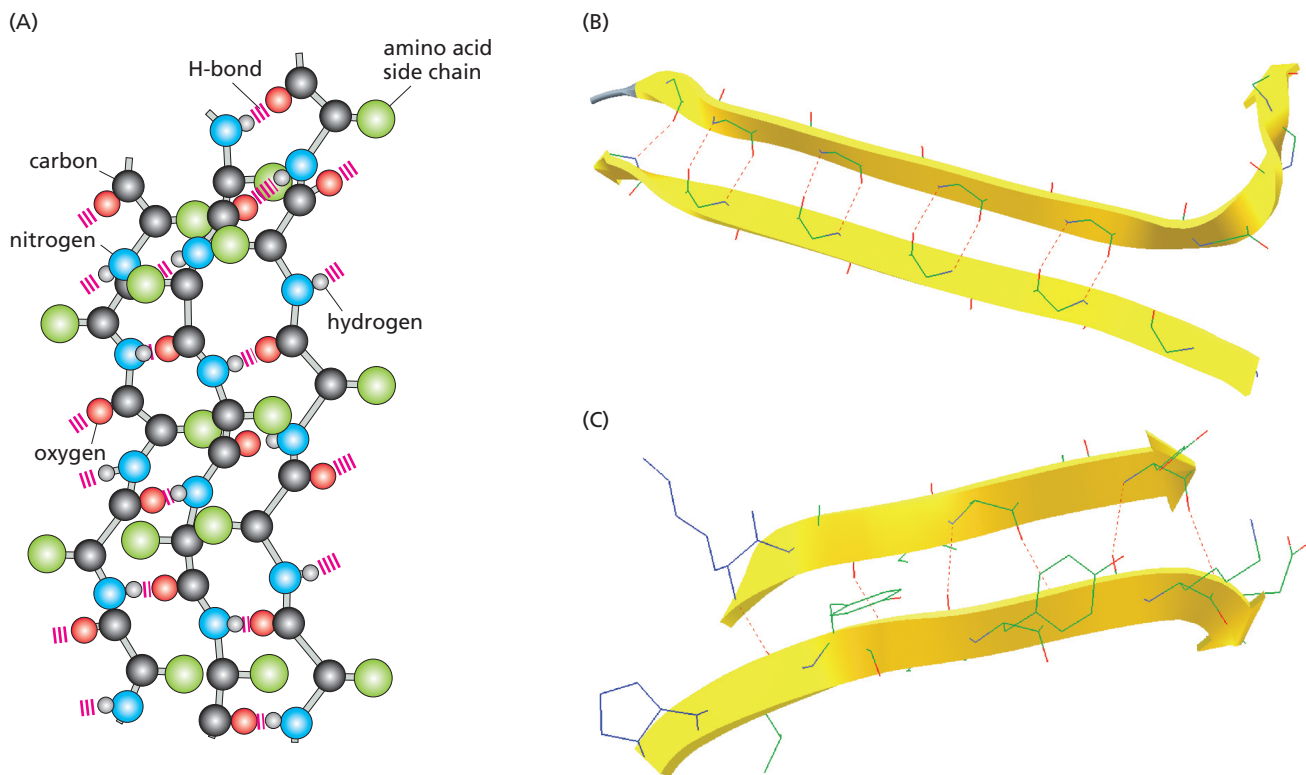
**Figure 2.9****Hydrogen bonding in the α -helix.**

In the α -helix all hydrogen (H-) bonds involve the same element of secondary structure. (A) A representation of the α -helix. (B) The helical structure repeats itself every 0.54 nm (5.4 Å) along the helix axis, therefore we say that the α -helix has a pitch of 0.54 nm. α -Helices have 3.6 amino acid residues per helical turn. The separation of residues along the helix axis is $0.54/3.6$ or 0.15 nm (1.5 Å); i.e., the α -helix has a rise per residue of 0.15 nm (1.5 Å).

hydrogen bonding involved. The backbone in a β -strand is not perfectly flat, but undulates. It is possible for the backbone to be flat, with the chain fully extended, but this is not the energetically favored conformation. One key difference between the β -strand and α -helices is that there are no short-range hydrogen-bonding interactions within an individual β -strand. The strands are stabilized by hydrogen bonding between different strands.

Several different types of β -sheet are found in protein structures

As noted above, β -strands must face each other with correct orientation for hydrogen bonding to occur. Even when this criterion is met, there are still two alternative orientations: the adjoining strands can run parallel (see Figure 2.10C) to each other, or in opposite directions (antiparallel) as is seen in Figure 2.10B. The difference between them is most easily seen by locating the N and C termini of the strands. β -Strands are often drawn in schematics of protein structure as arrows pointing in the N-terminal to C-terminal direction. In their ideal form, β -sheets consist of an all-parallel or all-antiparallel arrangement of β -strands. These ideal β -sheets rarely occur in real proteins, and often there is a mixture of parallel and antiparallel β -strands in the same sheet. The order of the β -strands in the β -sheet is often apparently random when the β -strands are numbered from the N terminus onward. β -Sheets are usually not flat, being rather twisted instead, and a continuous β -sheet can even form a complete cylinder. The protein pores, the porins, that allow substances to pass through the outer membrane of some bacteria are formed of such cylinders. Like α -helices, β -strands are usually restricted to a single globular

**Figure 2.10**

β -Sheets are composed of sets of β -strands hydrogen bonded together. A β -sheet may be formed from the folding of a continuous stretch of the protein chain, or can be formed by β -strands from different parts of the chain coming together. As with other regular protein structures, the hydrogen bonds are formed mainly between backbone groups. (A) A β -strand has two sides that can participate in hydrogen bonding. The -NH groups occupy all of one side, and the -C=O groups the other; one side is proton-accepting and the other proton-donating. This property dictates how β -strands are added to the β -sheet. Often β -sheets are displayed by drawing out the pattern of hydrogen bonding involved. In parallel β -sheets the strands all run in one direction, whereas in antiparallel sheets they all run

in opposite directions. (B) An antiparallel β -sheet with the hydrogen bonds in red dots and (C) a parallel β -sheet. Note the different but specific hydrogen-bonding patterns in the antiparallel and parallel β -sheets. From an examination of many β -sheets seen in globular proteins, it appears that the conformation of the β -strands is less fixed in terms of the (ϕ, ψ) angles than is the case with α -helices. The (ϕ, ψ) angles for idealized (i.e., perfectly straight) β -strands is $(-120^\circ, +120^\circ)$, but almost all β -strands are significantly curved and thus have some deviation from these values. As mentioned above in the case of helices, real structures in globular proteins are considerably distorted from the ideal, and deciding which residues are actually in a β -strand is nontrivial (see Section 12.1).

domain, and so rarely exceed 2 nm in length. However, β -sheets have been found that extend over more than one domain, comprising many β -strands.

Turns, hairpins, and loops connect helices and strands

Usually somewhere in the range of 50–80% of residues in globular proteins can be classified into one of the regular structures described above. The third type of secondary structure element is the **β -turn**. These are short regions where the protein chain takes a 180° change in direction, doubling back on itself (see Figure 2.11). Turns are found, for example, between two adjacent β -strands in β -sheets formed from a continuous stretch of protein chain. The remainder of the protein structure has much less order, and can be viewed as simply the connecting pieces (**loops**) that allow the α -helices and β -strands to pack, creating the protein structure. Whilst this view may not be too unrealistic from the viewpoint of structure alone, it is not at all the case when considering function. Most of the surface of proteins (and hence the accessible regions) is not regular structure. These surface parts often hold substrate-binding

Box 2.3 Chameleon sequences

Some short sequences of amino acids can adopt either an α -helical or a β -strand conformation, depending on the protein in which they occur. These sequences are known as chameleon sequences. Recently it has been found that this ability of a protein sequence played a part in bovine spongiform encephalopathy. The prion protein associated with this disease can adopt different stable conformations. The conformation found in the

diseased state is a mixture of helices and sheets, while in the normal state the prion protein consists of a bundle of helices. Consequently, predictions based on local sequence information will not be able to resolve such conflicts as found in the prion protein. For a further discussion of the consequences for secondary structure prediction, see Figure 12.12 and Section 12.3.

residues and catalytic residues, and are therefore extremely important. For example, the ligand-binding site of the phosphoinositol kinases (which will be used later in this book) is in part formed by a relatively large loop on the surface. This loop is so mobile that its structure could not be seen in a crystallographic analysis.

Any chain between two regular structures is referred to as a loop. In many cases a loop will contain a turn (or even several). In general there are no classifications for loops, but there is an important exception. In antibody recognition, immunoglobulins employ loops at the edge of a β -sheet to recognize the antigen. There are vast numbers of different immunoglobulin structures, all with the same overall chain fold, but it is the difference at these loops that results in different affinities. With many structures known, it has been observed that the loops take up one of a limited number of structures (called canonical forms), so that in this particular case the loops have been classified. This type of classification is important when trying to predict both the structure and function of the protein.

2.2 Implication for Bioinformatics

In part, bioinformatics concerns itself with the analysis of protein sequence to predict the secondary structure, the tertiary structure, and the function of the protein, as well as its relationship to other proteins. Different secondary structures tend to have subtle differences in chemical environments, resulting in amino acid preferences. In addition, amino acid preferences are seen at particular locations in proteins due to the functional role they play, for example as catalytic residues or stabilizing the overall protein structure. These aspects of proteins, which influence structure prediction and analysis, are described in more detail below.

Certain amino acids prefer a particular structural unit

Due to the various properties of the amino acid side chains, certain residues or certain types are found more often in one or the other of the structural units. Some residues have been classified, for example, as α -helix breakers (poor formers) or formers. Proline, for example, is a poor helix former due to the fact that its backbone nitrogen atom is already bonded to its own side chain and cannot form hydrogen bonds within the helix, and its rigid structure causes some steric hindrance. Other poor α -helix formers are Gly and Tyr, while Ala, Glu, Leu, and Met are good α -helix formers, whereas Val, Ile, Tyr, Cys, Trp, Phe, and Thr are more frequently found in β -strands than in α -helices. In addition, Gly, Asn, Pro, Ser, and Asp are more likely to be found in turn segments than in any other structure. These types of preferences have been the driving forces in secondary structure prediction programs (especially in the early development of the field) and are discussed further in Chapters 11 and 12.

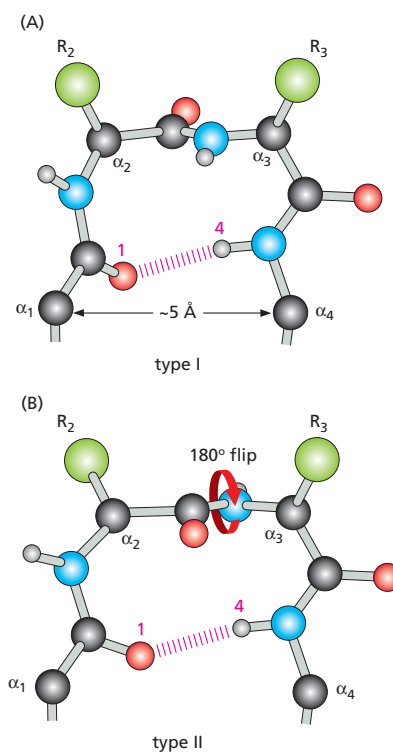
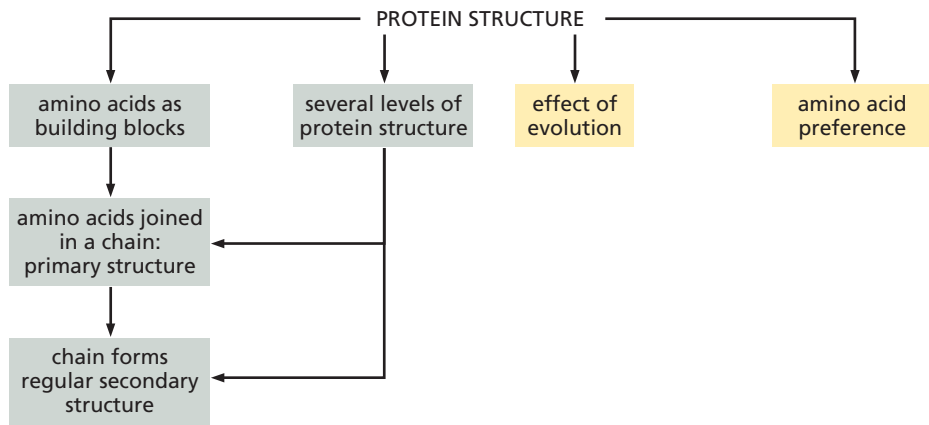


Figure 2.11

Turns. The two common types of tight turn, called β -turns type I and type II, respectively. The side chain of R3 is usually hydrogen (glycine).



Flow Diagram 2.2

The key concept introduced in this section is that, for a variety of reasons related to protein structure and function, each position in a protein sequence has an associated amino acid preference that is a reflection of the evolutionary selection pressure on the organism.

Evolution has aided sequence analysis

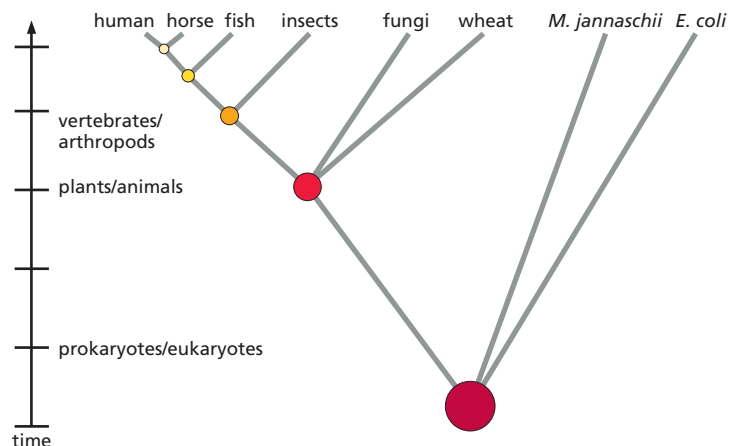
Protein sequence similarity is a powerful tool for characterizing protein function and structure since an enormous amount of information is conserved throughout the evolutionary process (Chapters 7 and 8). Proteins that have a common ancestor are referred to as being **homologous** (see Figure 2.12). Sequence alignment and database search techniques (Chapters 4 and 5) can identify homologous proteins. Homologous proteins usually have a similar three-dimensional structure with related active sites and binding domains. Therefore homologous proteins will also often have related functions, although this is not always the case. Most amino acids that change during evolution are found in regions that are not structurally or functionally important, such as many of the loops (or variable) regions. If the homologous protein is also functionally related then the amino acids involved in function are often conserved during evolution, which helps in identifying the function of a new protein (see Flow Diagram 2.2).

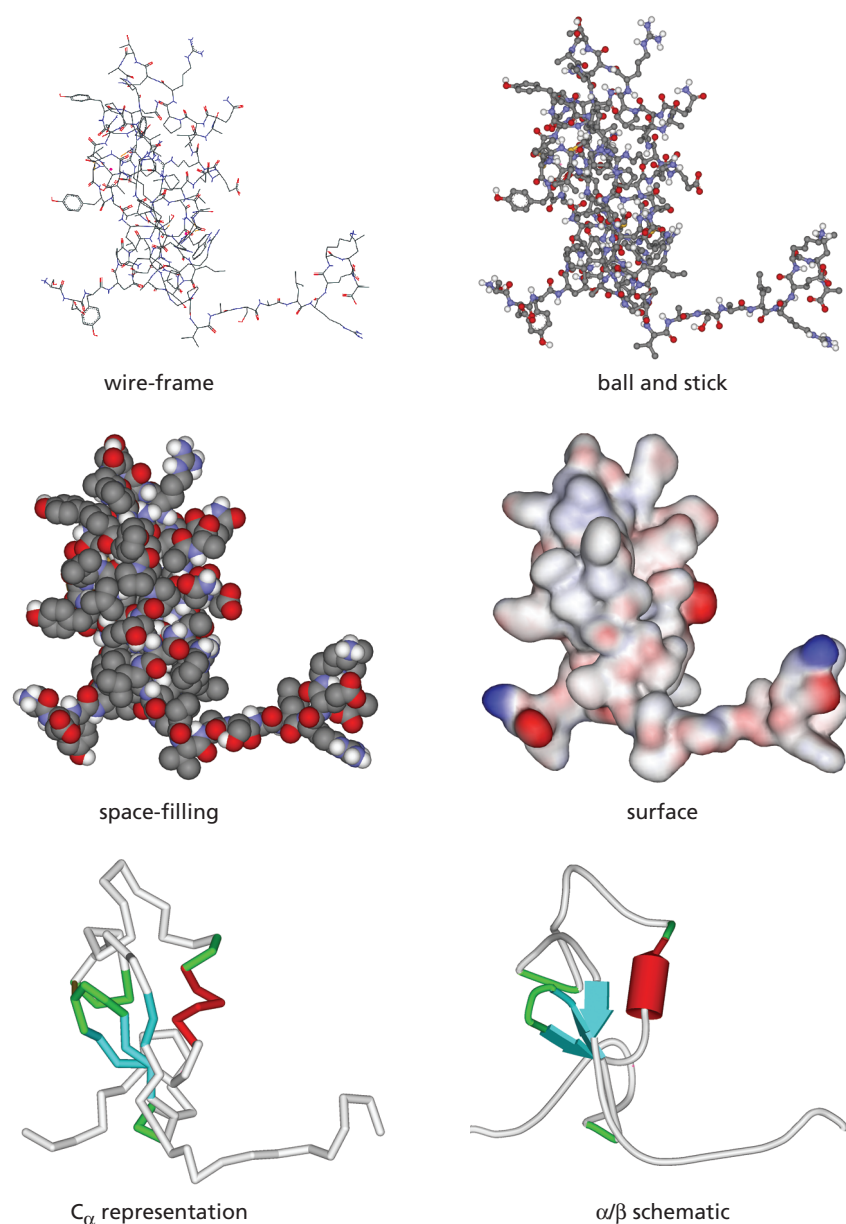
Visualization and computer manipulation of protein structures

There are a number of programs available that read the coordinate file and convert it to a visible three-dimensional representation of the protein. The protein can be rotated, specific regions highlighted, and some measurements can be calculated. Some of these programs are very powerful and can be of great use in analyzing the structural properties and molecular function, as well as allowing for the manual modification of the molecule. Some of the programs are free or low cost, such as Chimera, Yasara, and DeepView. Others are extremely powerful programs that allow

Figure 2.12

Tree of life. This tree shows how life may have evolved from a common ancestor giving the diversity of life forms that inhabit Earth.

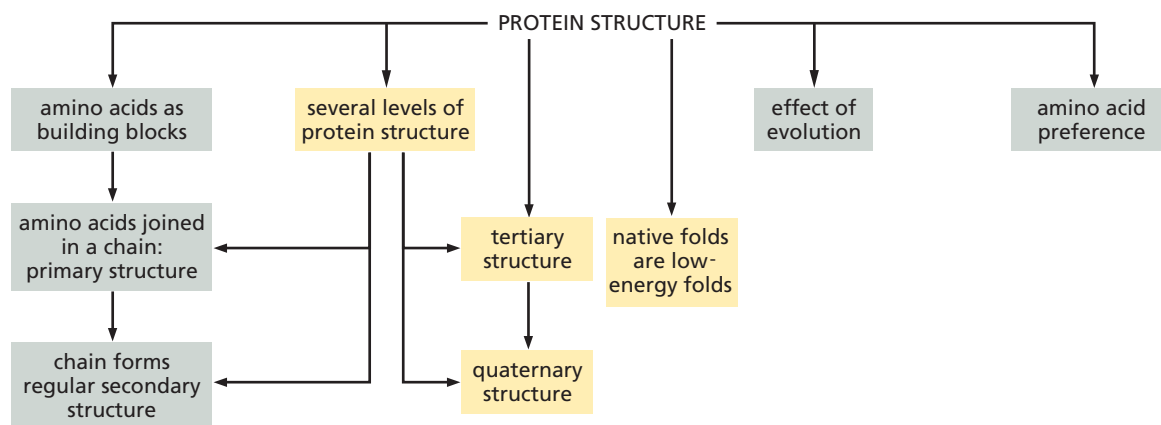


**Figure 2.13**

Molecular representations. The different representations that can be used to illustrate molecules, from very simple ones that only use the C_α or backbone atoms to space-filling models of all atoms in the structure.

the user to carry out computationally intensive modifications to the molecule, but are expensive. It would be helpful for the reader to gain some experience of using these programs before reading the secondary and tertiary structure prediction chapters (see Chapters 11 to 14).

There are many styles for viewing molecular structures, including those with atomic-level detail such as space-filling models, ball and stick models, and wire-frame models (also called stick models or skeletal models), as well as surface models. However, it is often desirable to have a simplified model of the protein, such as **backbone** or **C_α models** and schematic (cartoon) models. Such models can be represented on a computer screen and can be represented in different styles and colors. Molecular models are usually based upon an atomic coordinate file, which in general give the (*x,y,z*) coordinates of each atom. Figure 2.13 shows some of the more common structural representations.

**Flow Diagram 2.3**

The key concepts introduced in this section are that most proteins exist in a well-defined three-dimensional structure that is thermodynamically favorable, and that proteins often occur as complexes of several protein chains.

2.3 Proteins Fold to Form Compact Structures

Most **protein folds** seem to be dominated by the appearance of regular structures such as α -helices. It is now known that these are the first elements to form during protein folding, which confirms that they are the most stable components of the protein fold. Furthermore, the overall structure of a protein is so complex that these smaller, more ordered elements are useful as an aid to understanding and classifying the fold. Most proteins are composed of some combination of α -helices and β -strands. There are also proteins that are all α -helix and some that are all β -sheet.

Protein chains themselves rarely have any biological function. Only when the chain has folded up into a three-dimensional structure (however small) does the protein have functional activity. Some proteins are enzymes that bind other molecules (ligands) and catalyze their biochemical reactions (see Figure 2.14), others act by

Box 2.4 Supersecondary structure

Supersecondary structures are fold-elements composed of specific combinations of α -helices and/or β -strands that are often repeated throughout protein structures.

The figure illustrates schematically the most commonly occurring supersecondary structures: (A) a $\beta\alpha\beta$ repeat; (B) a β -meander; and (C) a special β -unit called the Greek Key structure, which occurs when an antiparallel sheet doubles back on itself. Part D in the figure is a cartoon representation of a Greek Key form gamma β -crystallin [Protein Data Bank (PDB) code 1AMM].

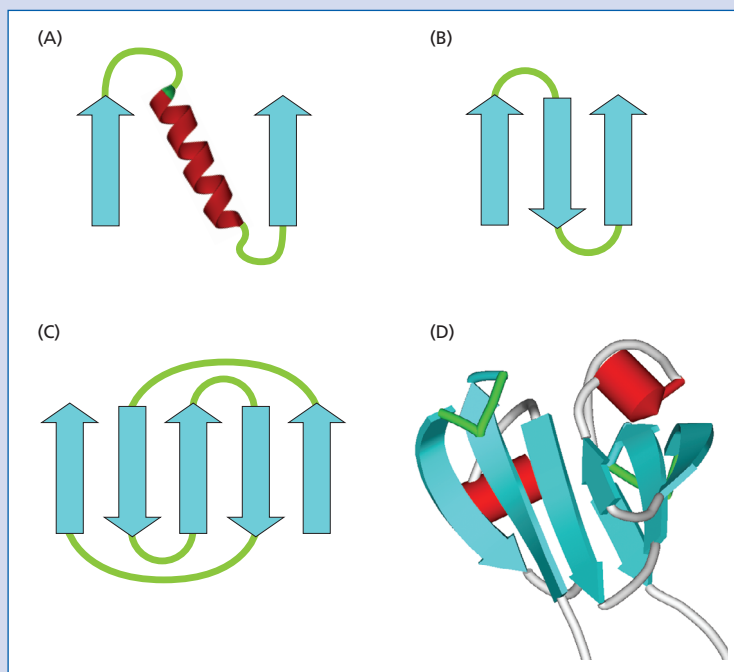


Figure B2.3
Some examples of supersecondary structure.

binding other proteins and influencing their activity, and yet others bind to DNA and regulate gene expression. Some proteins have a purely structural function, making up the fabric of the cell. Large numbers of proteins are released, or secreted, from cells and act as chemical messengers, influencing the behavior of other cells by acting on yet another large functional class of proteins, known as receptors, on cell surfaces.

The proteins that will concern us most in this book are globular proteins, proteins that are roughly spherical when folded up. The other main class of proteins are fibrous proteins, such as the keratin of wool and hair, and the silk protein. As globular proteins, or proteins composed of multiple globular units, are the proteins that carry out most cellular functions, they are the most studied. Some globular proteins are composed of a single, folded, globular mass; in others, the protein chain folds up into several such discrete structural units, each of which is termed a protein **domain**. Many proteins are composed of a multiplicity of different domains, each with a particular biochemical or binding function. The possibility of multidomain structure poses challenges when comparing protein sequences or interpreting the amino acid sequence of an unknown protein, as we shall see in Chapter 4.

The tertiary structure of a protein is defined by the path of the polypeptide chain

In the tertiary structure of a protein, various combinations of secondary structure pack together to form a compactly folded mass. For simplicity we will discuss tertiary structure here as if a protein were folding up into just one globular domain, but many proteins are composed of multiple domains joined by stretches of polypeptide chain. In a multidomain protein it is thought that each domain folds independently of the others. As will soon become clear in later chapters, bioinformatics questions are often concerned with comparing the sequences and structures of different domains rather than whole proteins. A domain can be anything from 50 to around 350 amino acids in length. The core of each domain is mainly composed of tightly packed α -helices, β -sheets, or a mixture of both. The three-dimensional structure of a protein is known as its conformation. More specifically, the spatial path of any given folded polypeptide chain is known as its fold.

As proteins exist in an aqueous environment, folding tends to result in hydrophobic regions of the protein ending up in the interior, while more hydrophilic regions are on the outside. A variety of noncovalent interactions stabilize the fold, dominated by hydrogen bonding and the clustering of hydrophobic groups. Secondary structures pack together in a variety of ways, such as the formation of β -sheets from either parallel or antiparallel β -strands. The atoms pack together very efficiently in most natural proteins.

There appears to be a limited number of ways in which secondary structures fold into domains. There are several instances where proteins that seem to be completely unrelated in terms of sequence are found to have the same fold and some researchers estimate that there may be only a few thousand different folds in nature. Currently, there are more than 35,000 known protein structures, and these are classified into approximately 2000 fold families. The fact that so many proteins fold into a similar structure even if their sequences are not very similar means that we can use bioinformatics tools to model structures of various proteins on similar folds as will be described in Section 13.2.

The stable folded state of a protein represents a state of low energy

A protein chain starts to fold as soon as it has been synthesized and thermodynamic considerations mean that the final fold it adopts is a state of low **free energy**. This is discussed in more detail in Section 13.1 and Appendix B. Folded proteins are

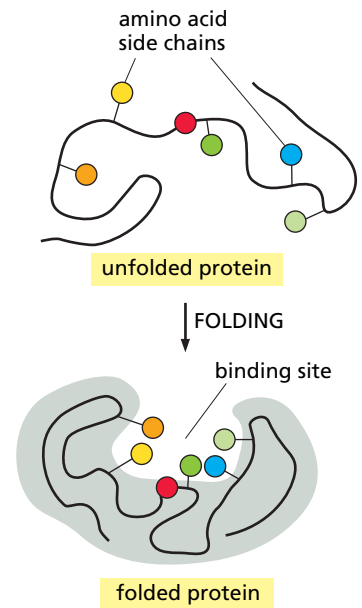
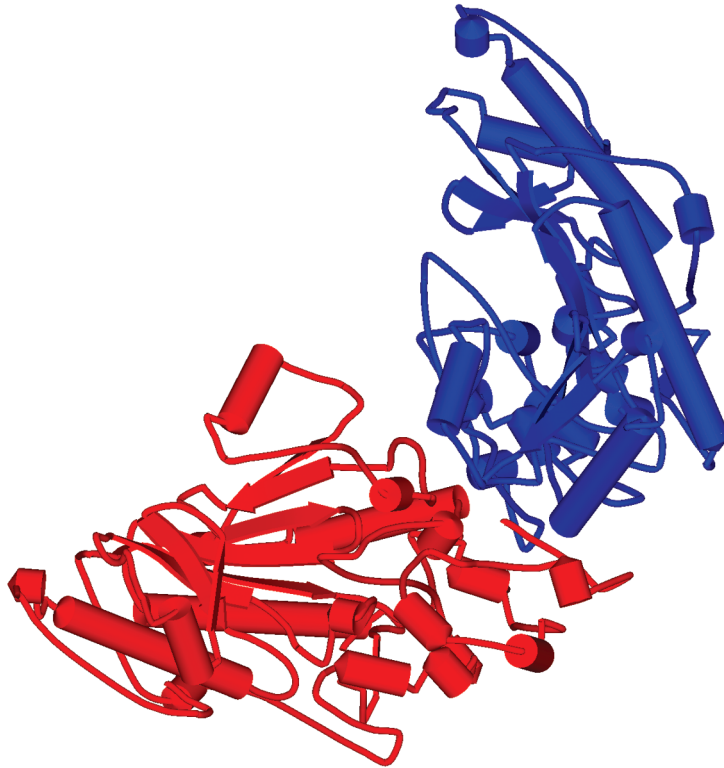


Figure 2.14

Distant residues can come close in the folded structure. A schematic which shows that when a polypeptide chain (primary structure) folds into a tertiary structure, residues that are far apart from each other in the sequence can come close together to form a functional unit, in this case a binding/catalytic site. (From B. Alberts et al., *Molecular Biology of the Cell*, 4th ed. New York: Garland Science, 2002.)

Figure 2.15

The quaternary structure of muscle creatine kinase. This structure is formed by the complex of two identical chains and structures. Therefore it is called a homodimer.



generally stable in the conditions in which they have to operate, but in a wider sense, proteins are unstable thermodynamically. Most proteins start to unfold above about 60°C, as the noncovalent bonds that hold them together are broken by thermal energy; unfolded proteins are said to be **denatured**. As it becomes denatured, a protein loses its function. Only specialized microorganisms are capable of living at temperatures this high.

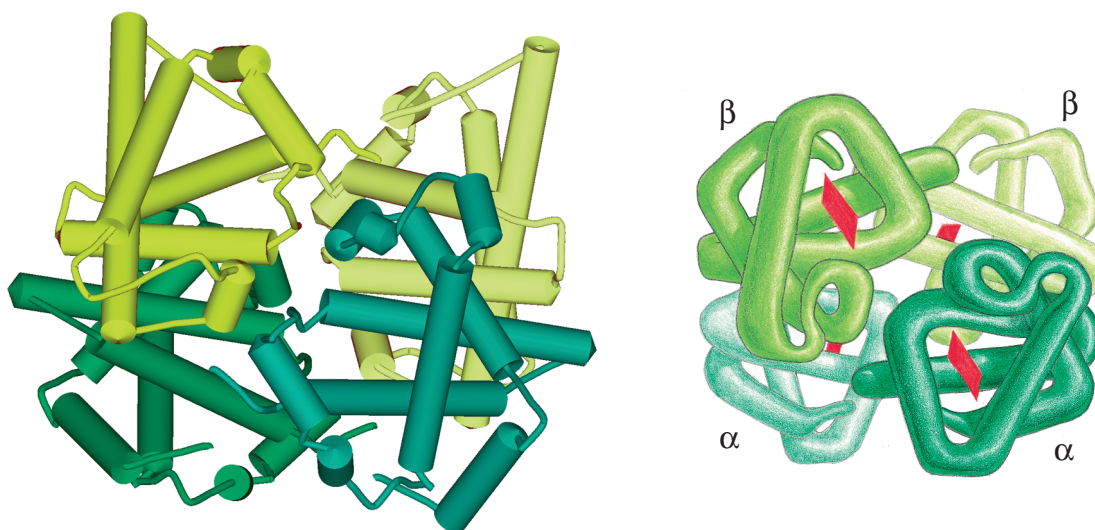
The stability of a protein is a fine balance between a number of effects. Predominant among these is the hydrophobic nature of many side chains, and so structure that depends on hydrophobic interactions is more stable away from contact with water. As a result there is a tendency for large hydrophobic groups to cluster inside the protein, with most of the polar groups on the protein surface. This is an oversimplification, however, as many nonpolar groups do reside on the surface, occasionally even having an important role in binding nonpolar ligands.

Hydrogen bonds also play an important role in the structure of the protein, although it is not as simple as the hydrophobic effect. The folded structure will form internal hydrogen bonds, both between the main chain atoms and some of the side chain atoms. However, in an unfolded state the atoms form hydrogen bonds with water. Therefore all these water-mediated hydrogen bonds are lost when the protein folds, which is not conducive to folding, but the release of all the otherwise hydrogen-bonded water creates more entropy in the solvent and this contributes to the folding effect.

Additionally, the residues in a protein pack together very effectively to occupy space, and there is some evidence that those proteins that have unoccupied space in the interior are less stable.

Many proteins are formed of multiple subunits

Individual folded polypeptides can interact with each other to form protein complexes or quaternary structures. Oligomeric proteins contain more than one



polypeptide chain. The individual chains are referred to as subunits or monomers of the oligomer. The assembly of these monomers is the quaternary structure. A protein consisting of two subunits is called a dimer, three a trimer, four a tetramer, and so on. The subunits can be the same protein sequence, such as in the dimer of muscle creatine kinase (see Figure 2.15), or made up of different polypeptide chains, such as the tetrameric hemoglobin, which has two α and two β subunits (see Figure 2.16). A quaternary structure associates in a similar way as the secondary or tertiary structure using noncovalent interactions. However, when quaternary structures assemble they often have hydrophobic groups that interact and exclude water. Often quaternary structure forms only when the complex is active, and frequently there is a sophisticated control of the activity involving several subunits. It is much more difficult to predict or model the structure of a quaternary structure compared to a single domain protein.

Figure 2.16

The quaternary structure of bovine deoxyhemoglobin which is a heterotetramer. In other words it is made up of different chains and structures from four folds. Two folds are the same therefore it consists of two homodimers. The heme groups are shown in red. (From B. Alberts et al., *Molecular Biology of the Cell*, 4th ed. New York: Garland Science, 2002.)

Summary

This chapter introduced the general principles associated with protein structure. Proteins are linear polymers composed of amino acids. They have a complex structure that includes some more regular features. Large proteins consist of several domains, each of which will have some regular secondary structures. The structures form a scaffold on which the active groups are located, ready to bind ligands, or carry information, and other roles in order for the protein to function properly.

One of the main aims of bioinformatics is to predict and analyze the structure of proteins and the relationship of the structure to the function. Many programs have been designed to predict protein secondary structure from the primary sequence information. These types of studies are described in Chapters 11 and 12 of this book. Further methods of modeling have been proposed that predict the three-dimensional structure of proteins as well as the quaternary complexes and these are described in Chapter 13. Some aspects of structure function analysis are dealt with in Chapter 14. To perform accurate predictions and to obtain more information about the protein sequence under study, it has to be aligned to other proteins to find homologs. This type of research is dealt with in detail, both for protein and nucleotide sequences, in Chapters 4 to 8 of this book.

Further Reading

Alberts B, Johnson A, Lewis J, Raff M, Roberts K & Walter P (2008) *Molecular Biology of the Cell*, 5th ed. New York: Garland Science.

Branden C & Tooze J (1999) *Introduction to Protein Structure*, 2nd ed. New York: Garland Science.

Lesk AM (2000) *Introduction to Protein Architecture*. Oxford: Oxford University Press.

Mezei M (1998) Chameleon sequences in the PDB. *Protein Eng* 11, 411–414.

Schulz GE & Schirmer RH (1984) *Principles of Protein Structure*. New York: Springer Verlag.

3

DEALING WITH DATABASES

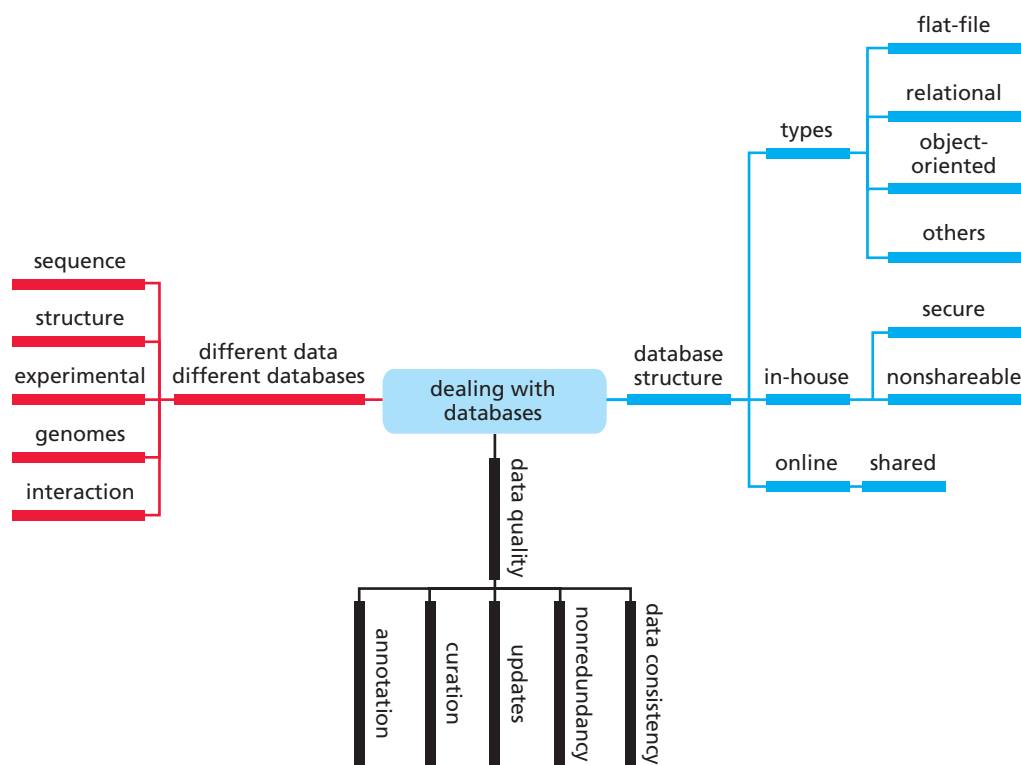
When you have read Chapter 3, you should be able to:

- Explain why databases are the backbone of bioinformatics research.
- Discuss how flat files were the first type of database, and why they are still used today.
- Show that relational databases are better for searching across tables.
- Outline the many other types of database structure that exist.
- Explain why databases contain both data and annotations of data.
- Discuss how many different types of database exist.
- Explain why the quality of data is important.
- Explain why checking the data and human curation are necessary.

The first databases were simply collections of data. For example old cards-in-a-box catalogs were noncomputerized databases. However, a database is more than just a collection of data. The modern database indeed stores data, but it also contains quite complex technology (such as ORACLE) to store the data in a structured manner and it is a complex model of tables and accompanying connections. A database is a sophisticated arrangement of storage, methods of storing, and architecture. It can be likened to a tax collection office, which not only contains the data about people's tax payments, but has a physical means of storing it, and methods by which the data can be **input**, accessed, and analyzed. In this context, the term architecture refers not only to the organized structure in which the data are kept but also to the building itself which provides the resources needed by the staff (also part of the architecture!) to process the data.

It is just over 50 years ago that the first protein sequence, that of bovine insulin, was determined by Frederick Sanger. Ten years later there were already attempts to collect all known sequences in a single database as an aid to the analysis of relationships between similar sequences. At the same time, programs for extracting and analyzing these sequences were written and the field of bioinformatics began, although it did not receive this name for some years.

The number of documented nucleotide sequences now numbers in the hundreds of thousands, and there are over a hundred thousand protein sequences too. This explosion in the number of sequences has made the use of electronic databases for storage and analysis essential. There has been a parallel increase in the quantity of data in other areas of biomedical research, such as molecular structures, and through the use



Mind Map 3.1

The schematic representation of topics important in understanding general aspects of databases.

The mind map highlights the points that there are many different databases used by bioinformaticians, that the database structure can vary, and also that data quality is very important.

of new experimental techniques such as microarrays and gene expression measurements. The need for databases has similarly increased in these areas. The existence of many different databases in closely related areas makes it useful to include cross references between related entries in different databases. As a result, today many of these databases can be regarded as linked together into a large network of information covering a broad range of biomedical and chemical research.

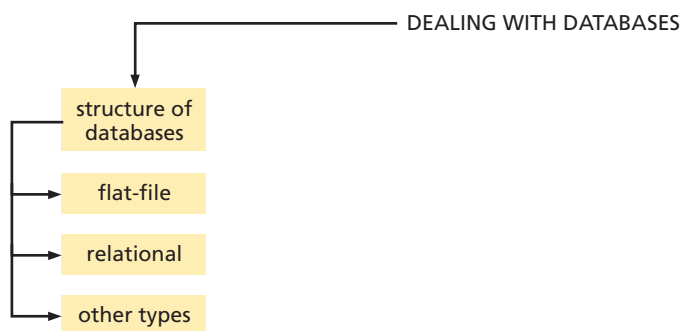
There are many different ways in which databases can be designed, both in terms of the ways the information is stored and the ways it can be retrieved and analyzed. There is no need to have a detailed technical understanding of these aspects in order to use databases, but we will describe some of the basic concepts as they can help in making effective use of these data sources.

Although there are many different types of databases, we will give an overview of the types most commonly used in bioinformatics research. Only a small fraction of the complete set of databases will be mentioned here, but the reader interested in discovering more about those not covered will be directed to an extensive list.

It is important to be able to have confidence in the accuracy of the data extracted from these sources. For this, certain aspects of database maintenance need to be understood before accessing any type of data for further analysis. Data quality issues are described in the last section of this chapter.

3.1 The Structure of Databases

A database is a repository of information that has a specific structure that enables the entering and extraction of data and in many cases also aids analysis of the data (see Flow Diagram 3.1). In general this database structure consists of files or tables, each containing numerous **records** and **fields**. Figure 3.1A shows an example of a very simple database table, in this case a single page with a contact list, with three

**Flow Diagram 3.1**

The key concept introduced in this section is that there are several distinct forms of electronic databases, each with particular advantages and disadvantages.

records each storing the details of one individual. There are three fields—Name, Telephone and Address—for each record.

A more complicated example would be a database of gene sequences stored in paper form in a filing cabinet, with gene data for each species stored in a separate file. Each file would contain many pages, each holding the information about a single gene. The information given about each gene will be in several distinct parts, such as the name of the gene, the gene sequence, or the name of the protein encoded by the gene. Each of these different pieces of information can occur in all genes, so that often the page used is printed with a standard form, with each section of the form, called a field, used to record one of the types of information. When databases are stored in electronic form their structure has many similarities to the paper form. Often a single computer file stores the entire database, and is the equivalent of the filing cabinet. Electronic database files consist of tables, which are the equivalent of the individual files in the cabinet. Thus, a gene sequence database might contain a separate table for each organism. Each gene would be listed in a separate row of the table, called a record, the electronic equivalent of the page. Each record will consist of several different pieces of information given in different columns of the table, and called fields. An example of the beginning of a record for gene TCP1-beta of *Saccharomyces cerevisiae* is shown in Figure 3.1B. This illustrates that the GenBank flat-file format is readable by humans as well as computers, with the field names shown here in blue. The complete record is very long, and so only the top section is shown.

(A)

CONTACT LIST		
NAME	TELEPHONE	ADDRESS
S. Claus	0203 450	The North Pole, Lapland
M. Mouse	0202 453	Disneyworld, Florida
A. Moonman	0104 459	Craterland, The Moon

(B) GenBank Flat-File Format

LOCUS	SCU49845	5028 bp	DNA
DEFINITION	Saccharomyces cerevisiae TCP1-beta gene, partial cds, and Axl2p (AXL2) and Rev7p (REV7) genes, complete cds.		
ACCESSION	U49845		
VERSION	U49845.1 GI:1293613		
KEYWORDS	.		
SOURCE	Saccharomyces cerevisiae (baker's yeast)		
ORGANISM	Saccharomyces cerevisiae Eukaryota; Fungi; Ascomycota; Saccharomycotina; Saccharomycetes; Saccharomycetales; Saccharomycetaceae; Saccharomyces.		

Figure 3.1

Two examples of flat-file database structures. (A) shows a contact list as a flat-file database in which a record holds the contact information for an individual, and consists of a number of fields (in this case three), such as name, telephone number, and address. (B) shows that a flat-file format can be very useful and is still used today, especially with text-handling computer languages. This is an example of a flat-file format obtained from the GenBank sequence databank. It is a very small part of the complete record, and the words in blue are the field names.

There are various types of electronic databases which differ in their structure. A structurally simple example of a database is the flat-file format, while a much more complex and therefore more versatile database structure is the relational form. Both of these will be discussed below in more detail. More modern database management structures include **object-oriented databases**, **data warehouses**, and **distributed databases**. These are also briefly described below. Note that a computerized database needs software that is used to control the database; this software is referred to as a database management system or DBMS for short.

Flat-file databases store data as text files

A flat-file database is the simplest form of a database, where collections of data, such as nucleotide and amino acid sequence, are stored as either a large single text file or as a collection of different text files. The file contains records, generally one record per line. The data files are flat, as in a sheet of paper, in contrast to more complex and versatile models such as a **relational database** discussed next.

A common example of a flat-file database is a simple name and address list. In this example the record will consist of a small and fixed number of fields such as a name, address, and contact number. Flat-file databases can be created by hand, for example when you write on a piece of paper the name, address, and phone number of your friends you will have created a flat-file database. However, although useful, this has its limitations; as the number of friends grows and as friends change addresses the database has to be updated, which would end up with the piece of paper being filled up and looking very messy with many crossed-out fields. Therefore using computers and computer programs to manipulate the data in a flat-file format is useful and was first implemented a long time ago. In the nineteenth century Herman Hollerith decided that any resident of the United States could be encoded by a string of exactly 80 digits and letters consisting of name, age, and other information, filled with blank spaces to make everyone's entry the same length. This led to the first-ever computerized census, in 1890, encoded on punch cards that were read by machines (see Figure 3.2).

Nowadays, electronic flat-file databases can be organized and maintained by sophisticated management software that provide a mechanism by which data can

Figure 3.2

First computer databases. This computer was designed in the late 19th century, and first used in the 1890 United States census. Hollerith developed an integrating tabulator housing separate adding machines—the upright units—that could simultaneously add totals recorded in separate areas, or fields, of a punched card. (Courtesy of Science Photo Library.)



be easily input and can be retrieved using detailed search queries. However, in general, the data available to biologists involves so many different aspects that it can be more effectively stored using more sophisticated database structures. Flat files are still used, especially to distribute the data (see Figure 3.1B), because many of the more complex database structures depend on specific, often expensive software, whereas the flat files can be read and analyzed by many alternative programs according to the user's preferences.

Relational databases are widely used for storing biological information

Probably the commonest type of database used for biological information is the relational database. A relational database stores the data within a number of tables. Each table consists of records and fields (rows and columns) as described above and in general these will be different for each table. However, each table will be linked to at least one other by a shared field called a **key**. It is these keys that distinguish a relational database from a flat-file database. It should be noted that not all fields can be keys, as the data for a key must be unique in each record. A relational database can have several different fields acting as keys, even in the same table, and the structure of the database is often drawn to show the keys linking different tables together.

For example, in Figure 3.3 a table called *protab1* is shown where each row contains a code for a protein, the name of the protein, the length of its sequence, and which species it comes from. In this case *protab1* is linked to other tables in a relational database by the protein code field, which is the key. This enables the extraction of related data from many tables so that, for example, using the protein code we can extract the name of a protein from *protab1* and the sequence of that protein from *protab2*, a table that only stores the protein code and protein sequence. In this way, the information held in different tables can be combined.

In relational database applications a set of operators is provided that allows for easy manipulation and analysis of the data. Some of these operators are mathematical in nature (addition, multiplication, and so on) and some are concerned with data handling, such as selecting specific subtypes of data from different tables. Frequently, the operations performed on the data in the tables produce results that are displayed in new tables. In the protein sequence example just described, the results table would contain the protein code, protein name, and protein sequence. In relational database management systems, the operators are written in query-specific languages that are based on relational algebra. **Structured Query Language (SQL)** is

protab1			
Protein-code	Protein-name	Length	Species-origin
P1001	Hemoglobin	145	Bovine
P1002	Hemoglobin	136	Ovine
P1003	Eye Lens Protein	234	Human
....			

protab2	
Protein-code	Protein-sequence
P1001	MDRTHGFDLKLSPRTVNQWLMLALFFGHS...
P1002	MDKTSHGFEIKLLTPKKLQQWLMIAIYFGHT...
P1003	SRTHEEGKLMQWPPRPLYIALFTEPPYP...
....	

Figure 3.3

Two example tables representing the relational database model. In this structure the records are connected by an identifier, often referred to as the primary key. There can be a number of subidentifiers or secondary keys. These connect various types of data from as many tables as necessary. Any field can be a key as long as it is unique.

commonly used and the examples below show how data could be extracted from the tables in Figure 3.3. The first query is a simple extraction of data from a single table.

Query 1

```
SELECT protein-code, protein-name  
FROM protab1  
WHERE species-origin = 'Bovine';
```

The SQL commands are in capital letters (this is a common standard when writing SQL), and all this simple bit of SQL code means is: select all protein codes and protein names from the table called “protab1” where the species of origin is bovine. The second query is slightly more complex and extracts data from two different tables using the key as the common factor.

Query 2

```
SELECT protab1.protein-name, protab2.protein-sequence  
FROM protab1, protab2  
WHERE protab1.protein-code = protab2.protein-code  
AND protab1.protein-code = 'P1002';
```

In this example we are telling the program to search and extract all protein names from protab1 and all sequences from protab2 where protein-code is p1002 in both tables.

These are just simple examples of basic queries. The queries can become extremely complicated when one deals with complex database structures containing many connections between many different tables. A detailed explanation of how relational databases and query languages work is outside the scope of this book; the interested reader should consult Further Reading on p. 67.

XML has the flexibility to define bespoke data classifications

Recent years have seen a significant effort to develop **eXtensible Markup Language (XML)** as a general tool for the storage of data and information. XML, a very powerful system for marking up (annotating) data, is one of many markup languages, including hypertext markup language (HTML) and XHTML, which are commonly used to write Web pages. In fact HTML and XHTML are subsets of XML, although they are considerably less powerful.

The key feature of these languages is the use of identifiers called tabs, which can enclose sections of data, for example <title>Understanding Bioinformatics</title>. While languages such as HTML can only use a restricted, small set of tabs, most of which are concerned with the presentation format (for example <bold> which indicates use of bold font), XML has mechanisms that allow arbitrary tags to be used. It is possible to construct and disseminate sets of tags appropriate for specialized data and for particular applications. For example, a <publisher> tag could be defined and used to identify book publishers. Recent versions of many word processing applications also use special sets of tags, allowing them to generate XML files. The ability to design appropriate specialized ways to classify data is one of the features that has made XML an attractive alternative to the databases described above. A particularly useful feature of XML is that it uses a plain file format, which makes XML files very portable and accessible. An increasing number of bioinformatics databases are being made available in XML format, although often their master copy is maintained using other solutions such as relational databases.

Having data marked up in XML or any other markup language is only useful if there are methods available to access and interpret the data. HTML and XHTML are interpreted by Web browsers, which identify the tags and use them as instructions for formatting and displaying the data. Interpreting programs have been written for

some specialized XML tag sets, for example the word processing applications already mentioned. In other cases, especially those where the tags identify the nature of the data, it is more appropriate to use general programs to extract the data, and to create bespoke applications if further analysis is required.

The extraction of data from an XML file is very similar to database querying. For example, although the underlying technical details might be very different, if an XML file and a database both held lists of the same books, the result of extracting/querying all book titles would be the same. There is an equivalent of SQL for XML files, called Xquery, so that extracting information is equally easy from either source. The XML language does not automatically have the complexity of database applications, so that XML files may not necessarily be the method of choice. However, they have the advantage of great flexibility, so that in certain situations they are likely to be preferred.

Many other database structures are used for biological data

There are other database models that are also used for biological systems, such as the object-oriented model. Object-oriented databases do not necessarily need SQL or any such special type of database programming language, and can use languages such as Java or C++. The key feature of object-oriented databases is their user interface. They have an apparent structure that consists of objects that correspond closely to the objects and concepts which occur in the field of interest. For example, the database structure might have cell objects with components and properties reflecting those of biological cells. The user of such a database can frame their queries in more familiar terms than is often possible using relational databases as described above. However, the actual structure used to store the information in an object-oriented database may still involve tables and have little, if any, real differences when compared to a relational database.

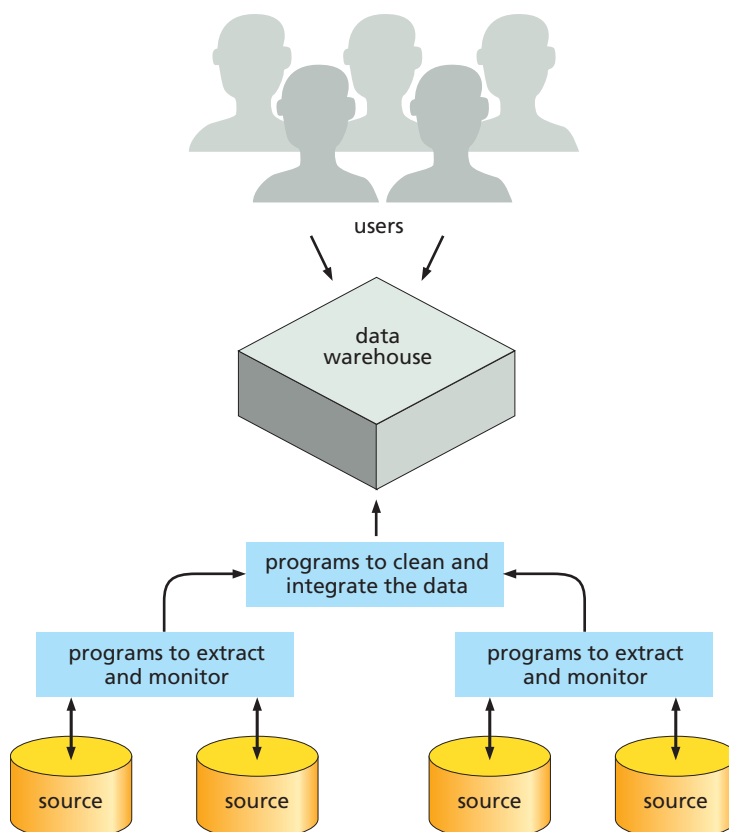


Figure 3.4

A schematic of data warehousing. The data come from different sources and are extracted automatically by special programs that also usually monitor the original source for updates. The data have to then be cleaned, checked, and put into a similar format so that integration can take place. Finally, the data are stored in a single database and are accessible to users.

Another form is the distributed database, in which although under the control of a central database management system, the actual database part (tables) may be stored in more than one computer located either in the same room or located anywhere but connected through the network. An example of a distributed database is the Reciprocal Net, which is a distributed database for crystallographers to store information about molecular structures. The central database management is at Indiana University but there are 19 other participating sites, ranging from those in the United States to those in the United Kingdom as well as Australia.

Data warehousing has been developed to overcome the problem of data heterogeneity (see next section) and data management between databases. The data warehouse basically stores information that has been integrated into one database (see Figure 3.4). It is, in effect, the opposite of the distributed database model.

Most large biological database sites, such as the Macromolecular Structure Database (MSD) at the European Bioinformatics Institute (EMBL-EBI), are made up of a mixture of the above models. MSD uses the relational model as well as having the data warehouse model for its structural information, integrating information about secondary structure, active sites, ligands, as well as information from external databases.

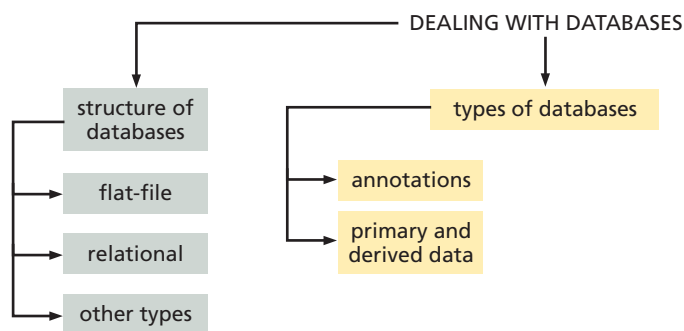
Databases can be accessed locally or online and often link to each other

Access to a database can be local, usually through user interface programs written on-site, for example in Java, or external through Web-based interfaces. A database stored locally has the advantages of faster access, more flexibility in designing specific queries, and security. However, it is necessary not only to have enough disk space to keep a database but also to invest time and money for the management and maintenance of the database. Therefore the preferred choice for most users is to access external databases via the Internet. A lot of effort has gone into providing Web-based forms and interfaces to biological and related databases and database resources, as can be seen by exploring the major bioinformatics database serving sites at the National Center for Biotechnology Information (NCBI) and EMBL-EBI.

Often many biological and related databases contain records that are relevant to the topic of interest. For example, there may be relevant entries for a protein in many different databases including those for protein sequence, protein structure, and gene sequence data. If each of these databases had to be separately queried it could take a long time to learn all that is known about the system, and some information may be missed if all databases are not searched. However, almost all databases in the biomedical area have some **links** to relevant entries in other databases, making it much easier to collate all the information. No database can be assumed to have links to all other relevant databases, so some care is still needed to obtain all the information, but much time can be saved. Note that some major sites such as NCBI and EBI augment these links so that not all accessible versions of a particular database necessarily have the same set of links. These links have made the collection of publicly available databases one of the most powerful resources in biomedical research.

3.2 Types of Database

There is a very large number of databases covering a wide range of scientific data with much duplication, both real and apparent. Just discovering what exists can be a difficult task, because many of the databases are created by small, independent teams and hosted on their own Web servers. In this section we will give a brief survey of the existing databases (see Flow Diagram 3.2).

**Flow Diagram 3.2**

The key concept introduced in this section is that most bioinformatics databases contain some results from analysis of the primary data, greatly increasing their usefulness.

An important feature of many databases is that they do not only store the data that are supposedly their reason for being created. For example, entries in the major protein and nucleotide sequence database often have a large amount of relevant non-sequence information. For clarity, in the remainder of this chapter we will use the term **data** to refer to the key information in the database entry: for example, the sequence, structure, or expression levels together with the minimal information required to identify the data, for example the name of the gene (and the organism) whose sequence is contained in the entry. All of the additional information in the database entry will be referred to as the **annotation**, and can include links to related entries in other databases, interpretation of the data, and relevant research citations. We will begin by briefly looking at some aspects of database annotation.

There's more to databases than just data

The minimal content of a database entry would include just the data and the data's identity (for example, protein name and source organism) and the author/submitter responsible for the entry. However, usually much more information is given, referred to as annotation, including published papers reporting the data, other known facts or interpretations (for example, enzyme reactions and substrates, or gene structures), and lists of highly relevant entries in other databases (for example, entries about a sequence motif, if one is present). Thus, even though the majority of databases concentrate almost wholly on one particular aspect (for example, DNA sequence, protein structure, and human genetic disorders) they contain information about a wide range of related aspects.

When the annotation lists related entries in other databases these are usually given as **URL** pointers, called links, with which one can very quickly surf through many databases. This makes the set of databases an extremely useful information resource often far more powerful than standard literature reviews. A successful database search will reveal many different aspects of the area of interest, and provide raw data and its interpretation as well as important research papers.

In addition to simply providing information, some of the databases present their data in an interactive graphical display. This enables the data to be browsed akin to thumbing the pages of a book. Sometimes this is the best way to become familiar with general features of the data, as well as being a more intuitive way to search through some types of data. In addition, some databases also provide programs for the online analysis of their data.

Primary and derived data

A distinction is sometimes made between databases of so-called primary data and those that contain secondary data derived from these primary sources. In some cases the primary data include the raw experimental results, for example scans of

gene expression arrays and two-dimensional (2-D) proteomic gels. In many cases the primary data are the initial experimental interpretation, for example nucleotide sequences. Although many protein sequences have been deduced from nucleotide sequences, the main protein sequence databases such as SWISS-PROT are often regarded as primary data sources. In general, except for experimental errors, these data are regarded as very reliable. (Note that this does not extend to the further annotation that might be present, especially in sequence databases.)

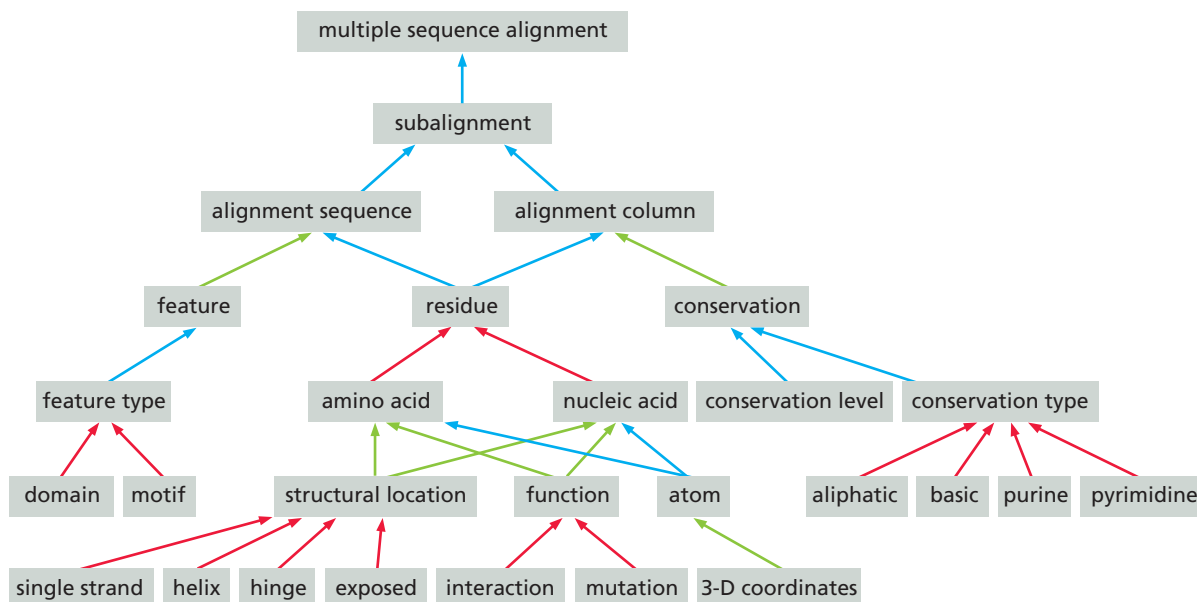
Examples of secondary databases are those that contain collections of conserved protein sequence motifs, or comparisons of multiple sequences that give measures of sequence similarity and relatedness. Because they can only be based on the data existing at the time, some of the entries in this kind of database may be proved incorrect once more primary data have been collected. For example, new protein sequence information may mean that a sequence hitherto regarded as a complete conserved motif turns out not to be conserved at all, or to be only part of a motif. If such secondary databases are not rederived at regular intervals, their data must be treated with a degree of suspicion.

Figure 3.5

Schematic diagram of a section of the MAO ontology architecture. The gray boxes show terms and the arrows represent relationships between them. Red arrows indicate “is a” relationship; for example, a domain is a feature type. Green arrows indicate “is attribute,” for example, function is an attribute of amino acid. Blue arrows indicate “part of,” thus an atom is part of an amino acid. (From J.D. Thompson et al., MAO: a multiple alignment ontology for nucleic acid and protein sequences, N.A.R. 33 (13):4164–4171, 2005, by permission of Oxford University Press.)

How we define and connect things is important: Ontologies

The organization of the complex data found in databases, as well as merging data from various places, is difficult when data are described using different terminology for each database. Therefore formal and explicit specifications of the terms used and the relationships between them have been defined; these are called **ontologies**. The specific terms defined in ontologies can be reused between different biological fields; this simplifies the sharing of data and the sharing of knowledge about the data, and facilitates analysis by computer programs, such as the GOTM program that is used to analyze and visualize sets of genes using the Gene Ontology. This widely used Gene Ontology (GO) is a collaborative project across many laboratories to provide a controlled vocabulary that describes gene and gene-associated information (but not gene products) for all organisms. The building blocks of any ontology are terms. For example in GO, each GO term has a number and a name, such as *signal transduction*. Each term is also assigned to one of three subontologies in GO: molecular function, cellular component, and biological process. Sharing a vocabulary such as GO between databases is a step toward unification of databases. Similarly there is MGED (Microarray Gene Expression Data), whose terms are developed to describe gene



expression experiments, sequence ontology project (SOP), which aims to define terms that describe features of a nucleotide or a protein sequence, and a multiple alignment ontology (MAO) for sequences. Figure 3.5 shows a graphical representation of a part of the MAO ontology structure. In MAO the sequence is placed into a context of the evolutionary family. It provides standard descriptions of most multiple sequence alignments and their methods, as well as integration of other relevant data such as structural or functional information. This facilitates inclusion of information based on conservation and evolution in the final analysis.

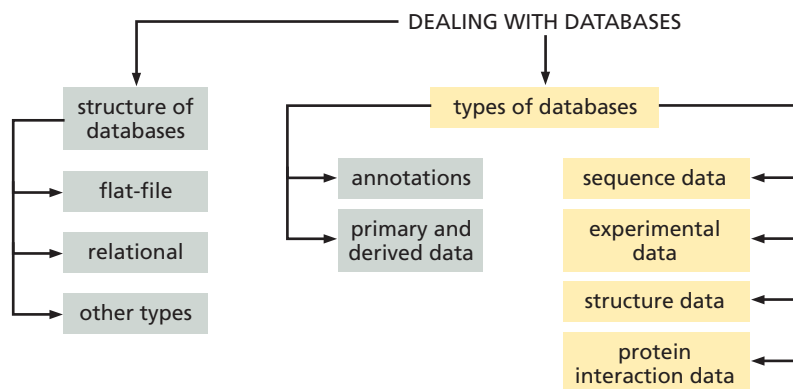
3.3 Looking for Databases

Beyond a thorough literature and Internet search, there is a key site to use when looking for relevant databases. Every year the first new issue in the journal *Nucleic Acids Research* (NAR) is devoted to papers reporting new and updated databases. In addition to the articles presented, there is a list of the URLs of databases that have been reported in these annual issues of NAR. This list, called the Molecular Biology Database Collection and numbering 858 entries in 2006, is available from the NAR home page. To give a feel for the range of material available in the biomedical databases, in 2006 the Molecular Biology Database Collection was divided into 14 categories (see Figure 3.6). Some of these categories will be described in more detail below as they are closely related to the bioinformatics analyses that are dealt with in the rest of this book (see Flow Diagram 3.3). Because of their importance as primary data sources, we will discuss sequence databases in some more detail.

There are a number of centers that have been funded to provide access to a large number of major databases in an integrated environment, to facilitate their use by the research community. The major database centers include the NCBI (<http://www.ncbi.nlm.nih.gov/>), the EBI (<http://www.ebi.ac.uk>), and the Sanger Institute (<http://www.sanger.ac.uk>). Each has links to over 100 different databases and is a very useful starting point for sequence, structure, and genome analysis. However, although these include most of the major databases, they are only a small fraction of the total available, and often a more specialist database can provide better information in the case of specific queries. In particular, a number of high-quality genome-specific databases have been produced, which would probably be the best starting point for learning about a gene or protein of that species.

Sequence databases

Nucleotide sequence related databases (8% of the 858 listed in the Molecular Biology Database Collection) include major international collaborations, such as GenBank and the EMBL-EBI Nucleotide Sequence database, as well as resources



Flow Diagram 3.3

The key concept introduced in this section is that bioinformatics databases can be classified according to the type of data they contain.